

Siemens PTI Report R136-19

MISO DPP 2018 April West Area Phase 1 Study

Prepared for

MISO

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Contents

Legal Notice.....	ix
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Executive Summary	xi
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1.1	Project List	xi
1.2	Reactive Power Requirements for Non-Synchronous Generation (FERC Order 827)	xiv
1.3	Total Network Upgrades for all Projects	xiv
1.4	Per Project Summary	xviii
1.4.1	J951 Summary	xviii
1.4.2	J952 Summary	xviii
1.4.3	J953 Summary	xviii
1.4.4	J954 Summary	xix
1.4.5	J958 Summary	xix
1.4.6	J959 Summary	xix
1.4.7	J963 Summary	xix
1.4.8	J966 Summary	xix
1.4.9	J967 Summary	xx
1.4.10	J971 Summary	xx
1.4.11	J975 Summary	xx
1.4.12	J977 Summary	xxi
1.4.13	J981 Summary	xxi
1.4.14	J982 Summary	xxi
1.4.15	J997 Summary	xxii
1.4.16	J998 Summary	xxii
1.4.17	J999 Summary	xxii
1.4.18	J1001 Summary	xxii
1.4.19	J1006 Summary	xxiii
1.4.20	J1008 Summary	xxiii
1.4.21	J1024 Summary	xxiii
1.4.22	J1032 Summary	xxiv
1.4.23	J1038 Summary	xxiv
1.4.24	J1040 Summary	xxiv

1.4.25	J1041 Summary	xxv
1.4.26	J1045 Summary	xxv
1.4.27	J1050 Summary	xxvi
1.4.28	J1054 Summary	xxvi
1.4.29	J1057 Summary	xxvi
1.4.30	J1061 Summary	xxvii
1.4.31	J1072 Summary	xxvii
1.4.32	J1084 Summary	xxvii
1.4.33	J1086 Summary	xxviii
1.4.34	J1092 Summary	xxviii
1.4.35	J1098 Summary	xxviii
1.4.36	J1105 Summary	xxix
1.4.37	J1106 Summary	xxix
1.4.38	J1108 Summary	xxix
1.4.39	J1109 Summary	xxx
1.4.40	J1110 Summary	xxx
1.4.41	J1114 Summary	xxxi
1.4.42	J1122 Summary	xxxi
1.4.43	J1124 Summary	xxxi
1.4.44	J1128 Summary	xxxii
1.4.45	J1131 Summary	xxxii
1.4.46	J1132 Summary	xxxii
1.4.47	J1135 Summary	xxxii
1.4.48	J1140 Summary	xxxiii
1.4.49	J1143 Summary	xxxiii
1.4.50	J1149 Summary	xxxiii
1.4.51	J1164 Summary	xxxiv
1.4.52	J1169 Summary	xxxiv
1.4.53	J1170 Summary	xxxiv
1.4.54	J1174 Summary	xxxv
1.4.55	J1175 Summary	xxxv
1.4.56	J1177 Summary	xxxvi
1.4.57	J1179 Summary	xxxvi
1.4.58	J1181 Summary	xxxvi
1.4.59	J1182 Summary	xxxvi

1.4.60	J1187 Summary	xxxvii
1.5	Study Compliance with NERC FAC-002-2 Standard	xxxviii
Introduction		1
Section 2 – Model Development and Study Criteria.....		2-1
2.1	Model Development.....	2-1
2.1.1	Benchmark Cases	2-1
2.1.2	Study Cases	2-1
2.2	Contingency Criteria	2-2
2.3	Monitored Elements.....	2-3
2.4	Performance Criteria.....	2-4
2.5	Reactive Power Requirements for Non-Synchronous Generation (FERC Order 827)	2-5
Section 3 – Summer Peak Steady-State Analysis		3-1
3.1	Study Procedure.....	3-1
3.1.1	Computer Programs	3-1
3.1.2	Study Methodology.....	3-1
3.2	Contingency Analysis Results for Summer Peak in Discharging Condition	3-1
3.2.1	System Intact Conditions.....	3-1
3.2.2	Post Contingency Conditions	3-1
3.3	Contingency Analysis Results for Summer Peak in Charging Condition.....	3-2
3.3.1	System Intact Conditions.....	3-2
3.3.2	Post Contingency Conditions	3-2
3.4	Summer Peak Worst Thermal Constraints.....	3-2
3.5	Network Upgrades Identified in MISO ERIS Analysis for 2024 Summer Peak Scenario	3-12
Section 4 – Summer Shoulder Steady-State Analysis		4-1
4.1	Study Procedure.....	4-1
4.2	Step 1 – Stage-1 ACCC Analysis.....	4-1
4.2.1	Stage-1 Voltage Violations	4-1
4.2.2	Stage-1 Thermal Violations	4-2
4.3	Step 2 – Base Case NUs Justification Analysis	4-2
4.4	Step 3 – Stage-2 ACCC Analysis.....	4-3
4.4.1	Stage-2 ACCC Analysis Results for Summer Shoulder in Discharging Condition	4-3

4.4.2	Stage-2 ACCC Analysis Results for Summer Shoulder in Charging Condition	4-4
4.4.3	Summer Shoulder Worst Thermal Constraints in the Stage-2 ACCC.....	4-4
4.5	Network Upgrades Identified in MISO ERIS Analysis for 2024 Summer Shoulder Scenario	4-29
Section 5 – Local Planning Criteria Analysis		5-1
5.1	GRE Local Planning Criteria Analysis	5-1
5.1.1	Additional Network Upgrades Identified in J1106 GRE LPC Analysis	5-1
5.1.2	Additional Network Upgrades Identified in J1140 GRE LPC Analysis	5-2
5.1.3	Additional Network Upgrades Identified in CCS GRE LPC Analysis	5-2
5.2	OTP Local Planning Criteria Analysis.....	5-4
5.2.1	Additional Network Upgrades Identified in OTP LPC Analysis	5-4
5.3	MDU Local Planning Criteria Analysis.....	5-5
5.3.1	Additional Network Upgrades Identified in MDU LPC Analysis	5-5
5.4	MRES Local Planning Criteria Analysis	5-7
5.4.1	Additional Network Upgrades Identified in MRES LPC Analysis	5-7
Section 6 – MWEX Voltage Stability Study		6-1
Section 7 – Deliverability Study.....		7-1
7.1	Project Description.....	7-1
7.2	Introduction	7-1
7.3	Study Methodology.....	7-1
7.4	Determining the MW restriction	7-1
7.5	2024 Deliverability Study Result.....	7-2
7.5.1	J953	7-2
7.5.2	J954	7-2
7.5.3	J958	7-2
7.5.4	J959	7-2
7.5.5	J963	7-2
7.5.6	J966	7-3
7.5.7	J967	7-3
7.5.8	J971	7-3

7.5.9	J975	7-4
7.5.10	J977	7-4
7.5.11	J981	7-4
7.5.12	J982	7-5
7.5.13	J997	7-5
7.5.14	J998	7-5
7.5.15	J999	7-6
7.5.16	J1001	7-6
7.5.17	J1006	7-6
7.5.18	J1008	7-6
7.5.19	J1024	7-7
7.5.20	J1032	7-7
7.5.21	J1040	7-7
7.5.22	J1041	7-8
7.5.23	J1045	7-8
7.5.24	J1050	7-8
7.5.25	J1054	7-9
7.5.26	J1057	7-9
7.5.27	J1061	7-9
7.5.28	J1072	7-9
7.5.29	J1084	7-9
7.5.30	J1086	7-10
7.5.31	J1092	7-10
7.5.32	J1098	7-10
7.5.33	J1105	7-10
7.5.34	J1106	7-11
7.5.35	J1108	7-11
7.5.36	J1109	7-11
7.5.37	J1110	7-12
7.5.38	J1114	7-12
7.5.39	J1122	7-12
7.5.40	J1124	7-13
7.5.41	J1128	7-13
7.5.42	J1131	7-13
7.5.43	J1132	7-14

7.5.44	J1135	7-14
7.5.45	J1140	7-14
7.5.46	J1143	7-15
7.5.47	J1149	7-15
7.5.48	J1164	7-15
7.5.49	J1169	7-15
7.5.50	J1170	7-16
7.5.51	J1174	7-16
7.5.52	J1175	7-17
7.5.53	J1177	7-17
7.5.54	J1179	7-17
7.5.55	J1181	7-17
7.5.56	J1182	7-18
7.5.57	J1187	7-18
Section 8 – Shared Network Upgrades Analysis		8-1
Section 9 – Cost Allocation		9-1
9.1	Cost Assumptions for Network Upgrades	9-1
9.2	ERIS Network Upgrades Proposed for DPP West Area Projects	9-1
9.3	Cost Allocation Methodology	9-9
9.4	Cost Allocation	9-10
Appendix A – Model Development		A-1
A.1	DPP 2018 April Generation Projects	A-1
A.2	DPP 2017 August West Area Phase 1 Network Upgrades	A-14
A.3	Model Review Comments	A-16
A.4	MISO Classic as the Study Sink	A-18
A.5	PJM Market as PJM Projects Sink	A-19
A.6	SPP Market as SPP Projects Sink	A-20
A.7	Contingency Files used in Steady-State Analysis	A-21
Appendix B – Model Data		B-1
B.1	Power Flow Model Data	B-1
B.2	2024 Slider Diagrams	B-3
Appendix C – Reactive Power Requirement Analysis Results (FERC Order 827)		C-1

Appendix D – 2024 Summer Peak Contingency Analysis Results	D-1
D.1 Constraints in 2024 Summer Peak (SPK) Discharging Condition	D-1
D.2 Constraints in 2024 Summer Peak (SPK) Charging Condition.....	D-3
Appendix E – 2024 Summer Shoulder Contingency Analysis Results	E-1
E.1 Stage-1 Contingency Analysis Results	E-1
E.2 Base Case Network Upgrades Justification Results	E-3
E.3 Stage-2 Contingency Analysis with Base Case NUs	E-5
Appendix F – Local Planning Criteria Analysis Results	F-1
F.1 GRE Local Planning Criteria Analysis Results	F-1
F.2 OTP LPC Analysis.....	F-7
F.3 MDU LPC Analysis.....	F-9
F.4 MRES LPC Analysis.....	F-11
Appendix G – MWEX Voltage Study Details	G-1
Appendix H – 2024 Cost Allocation Results	H-1
H.1 Distribution Factor (DF) and MW Contribution Results for Cost Allocation in 2024	H-1
H.2 Cost Allocation Details.....	H-3

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Executive Summary

This report presents results of a System Impact Study (SIS) performed to evaluate the interconnection of the DPP 2018 April Phase 1 West Area Group (DPP West Area) generating facilities. The results for 2024 scenario are summarized below.

1.1 Project List

The DPP West Area study group has sixty (60) generation projects with a combined nameplate rating of 9073 MW. The DPP West Area generating facilities are listed in Table ES-1. The modeling details and projects' slider diagrams are shown in Appendix B.

Table ES-1: Generating Facilities in DPP 2018 April West Area Group

MISO Project #	Service Type	TO	County	State	Point of Interconnection	Fuel Type	ERIS Output	NRIS Output	SH MW	SPK MW	Stability MW
J951	ERIS	OTP	Kittson	MN	Donaldson - Warsaw 115 kV	Wind	80	0	80.0	12.5	80.0
J952	ERIS	MDU	Corson	SD	McIntosh Junction 115 kV	Wind	54	0	54.0	8.4	54.0
J953	External NRIS	ITCM	Johnson	IA	AMIL.IOW_AFRYT	Diesel	1.83	1.83	0.0	1.8	1.8
J954	External NRIS	ITCM	Johnson	IA	AMIL.IOW_AFRYT	Solar	1.4	1.4	0.7	1.4	1.4
J958	NRIS	Xcel	Lincoln	MN	Buffalo Ridge 115 kV	Wind	100	100	100.0	15.6	100.0
J959	NRIS	SMMPA	Fayette	IA	Windsor 161 kV	Wind	150	150	150.0	23.4	150.0
J963	NRIS	ITCM	Cedar	IA	Bennett - Graham 69 kV	Diesel	10.2	10.2	0.0	10.2	10.2
J966	NRIS	ATXI	Putnam, Sullivan	MO	Zachary 345 kV	Wind	400	400	400.0	62.4	400.0
J967	NRIS	Xcel	Mower	MN	Adams 345 kV	Wind	150	150	150.0	23.4	150.0
J971	NRIS	Xcel	Nobles	MN	Nobles Co - Fenton 115 kV	Wind	250	250	250.0	39.0	250.0
J975	NRIS	OTP	Cass	ND	Buffalo 115 kV	Wind	150	150	150.0	23.4	150.0
J977	NRIS	ITCM	Jackson, Martin	MN	Jackson North 161 kV	Wind	300	300	300.0	46.8	300.0
J981	NRIS	MEC	Washington	IA	Sub T 345 kV	Wind	200	200	200.0	31.2	200.0
J982	NRIS	MEC	Dickinson, Emmet	IA	Obrien County - Kossuth 345 kV (33 mi from Obrien)	Wind	300	300	300.0	46.8	300.0
J997	NRIS	MDU	LaMoure	ND	Ellendale 345 230 kV	Solar	200	200	100.0	200.0	200.0

MISO Project #	Service Type	TO	County	State	Point of Interconnection	Fuel Type	ERIS Output	NRIS Output	SH MW	SPK MW	Stability MW
J998	NRIS	MEC	Dallas	IA	Adel - Waukee 69 kV	Solar	50	50	25.0	50.0	50.0
J999	NRIS	ITCM	Linn	IA	Fairfax 161 kV	Solar	100	100	50.0	100.0	100.0
J1001	NRIS	Xcel	Lincoln	MN	Buffalo Ridge 115 kV	Solar	40	40	20.0	40.0	40.0
J1006	NRIS	Xcel	Brookings	SD	Brookings - Yankee 115 kV	Solar	100	100	50.0	100.0	100.0
J1008	NRIS	ITCM	Faribault	MN	Blue Earth IPL - Blue Earth 69 kV	Solar	50	50	25.0	50.0	50.0
J1024	NRIS	MEC	Nodaway	MO	J611 - Clarinda 161 kV	Wind	200	200	200.0	31.2	200.0
J1032	NRIS	Xcel	Wabasha	MN	Briggs Rd. - N. Rochester 345 kV	Battery	50	50	50.0	50.0	50.0
J1038	ERIS	OTP	Grant	MN	Grant Co. - E. Fergus Falls - Hoot Lake 115 kV	Solar	32.5	0	16.3	32.5	32.5
J1040	NRIS	MDU	McIntosh	ND	Wishek Junction 230 kV	Wind	250	250	250.0	39.0	250.0
J1041	NRIS	GRE	Jackson	MN	Brewster-Heron Lake 161 kV	Battery	20	20	20.0	20.0	20.0
J1045	NRIS	Xcel	Murray	MN	Fenton - Chanarambie 115 kV	Battery	20	20	20.0	20.0	20.0
J1050	NRIS	ITCM	Boone, Hamilton	IA	Doud Tap 161 kV	Wind	225	225	225.0	35.1	225.0
J1054	NRIS	Xcel	Goodhue	MN	Cannon Falls 115 kV	Battery	30	30	30.0	30.0	30.0
J1057	NRIS	GRE	Jackson	MN	Brewster-Heron Lake 161 kV (0.2 mi from Brewster)	Solar	125	125	62.5	125.0	125.0
J1061	NRIS	Xcel	Wabasha	MN	Briggs Rd - N. Rochester 345 kV	Solar	300	300	150.0	300.0	300.0
J1072	NRIS	Xcel	Mower	MN	Adams 345 kV	Solar	150	150	75.0	150.0	150.0
J1084	NRIS	ITCM	Clinton	IA	Rock Creek 345 kV	Solar	150	150	75.0	150.0	150.0
J1086	NRIS	ITCM	Nobles	MN	Elk - Brewster 161 kV	Solar	120	120	60.0	120.0	120.0
J1092	NRIS	Xcel	Saint Croix	WI	Three Lakes 115 kV	Solar	100	100	50.0	100.0	100.0
J1098	NRIS	Xcel	Jackson	MN	Lakefield 345 kV	Solar	40	40	20.0	40.0	40.0
J1105	NRIS	Xcel	Dakota	MN	Hampton Corners 345 kV	Solar	200	200	100.0	200.0	200.0
J1106	NRIS	GRE	Redwood	MN	Lyon County - Cedar Mountain 345 kV (12 mi from Lyon Co)	Wind	414	414	414.0	64.6	414.0
J1108	NRIS	MRES	Douglas	MN	Alexandria 345 kV	Solar	150	150	75.0	150.0	150.0
J1109	NRIS	Xcel	Cass	ND	Bison 345 kV	Wind	207	207	207.0	32.3	207.0

MISO Project #	Service Type	TO	County	State	Point of Interconnection	Fuel Type	ERIS Output	NRIS Output	SH MW	SPK MW	Stability MW
J1110	NRIS	SMPMPA	Mower	MN	North Austin 161 kV	Solar	100	100	50.0	100.0	100.0
J1114	NRIS	MEC	Plymouth	IA	J506 (Raun - Highland 345 kV)	Wind	200	200	200.0	31.2	200.0
J1122	NRIS	MEC	Pottawattamie	IA	Council Bluffs - Fallow Avenue 345 kV (23.56 mi from Overland)	Wind	200	200	200.0	31.2	200.0
J1124	NRIS	SMPMPA	Olmsted	MN	Byron 345 kV	Solar	100	100	50.0	100.0	100.0
J1128	NRIS	SMPMPA	Freeborn	MN	Hayward - Murphy Creek 161 kV	Solar	150	150	75.0	150.0	150.0
J1131	NRIS	MEC	Scott	IA	Sub 56 161 kV	Solar	100	100	50.0	100.0	100.0
J1132	NRIS	ITCM	Union	IA	Creston East 69 kV	Solar	50	50	25.0	50.0	50.0
J1135	NRIS	ITCM	Des Moines	IA	Agency Street 69 kV	Solar	50	50	25.0	50.0	50.0
J1140	NRIS	MP	Benton	MN	Platte River 115 kV	Solar	100	100	50.0	100.0	100.0
J1143	NRIS	MP	Morrison	MN	Blanchard - Bellevu 115 kV (1000 feet from Blanchard)	Solar	100	100	50.0	100.0	100.0
J1149	NRIS	Xcel	Yellow Medicine	MN	Hazel Creek 230 kV	Solar	200	200	100.0	200.0	200.0
J1164	NRIS	ITCM	Rock	MN	Magnolia 161 kV	Solar	200	200	100.0	200.0	200.0
J1169	NRIS	Xcel	McCook	SD	Grant 115 kV	Solar	50	50	25.0	50.0	50.0
J1170	NRIS	Xcel	Cass	ND	Bison 345 kV	Solar	200	200	100.0	200.0	200.0
J1174	NRIS	ITCM	Worth	IA	Bison - Colby 345 kV (16 mi from Colby)	Solar	300	300	150.0	300.0	300.0
J1175	NRIS	ITCM	Worth	IA	Bison - Colby 345 kV (16 mi from Colby)	Wind	300	300	300.0	46.8	300.0
J1177	NRIS	ATXI	Schuyler	MO	Mark Twain 345 kV	Wind	300	300	300.0	46.8	300.0
J1179	NRIS	Xcel	Chisago	MN	Rush City - Red Rock 230 kV	Solar	300	300	150.0	300.0	300.0
J1181	NRIS	ITCM	Chickasaw	IA	Hazleton - Mitchell county 345 kV	Wind	200	200	200.0	31.2	200.0
J1182	NRIS	ATXI	Adair	MO	Zachary 345 kV	Solar	250	250	125.0	250.0	250.0
J1187	NRIS	GRE	Mercer	ND	Stanton 230 kV	Wind	151.8	151.8	151.8	23.7	151.8

1.2 Reactive Power Requirements for Non-Synchronous Generation (FERC Order 827)

For non-synchronous generation projects in the DPP 2018 April West Area study group, if they do not have signed Generator Interconnection Agreement (GIA) or Provisional GIA (PGIA) by September 21, 2016, they are required to provide dynamic reactive power within the range of 0.95 leading to 0.95 lagging at the high-side of the generator substation.

All non-synchronous generation projects in this study group are required to meet the reactive power requirements per FERC Order 827.

The reactive power requirement analysis results are summarized as following:

- J951, J958, J959, J966, J971, J981, J997, J998, J999, J1006, J1024, J1061, J1072, J1084, J1086, J1092, J1105, J1106, J1109, J1110, J1124, J1128, J1174, J1177, J1179, J1182, and J1187 do not meet the reactive power requirements per FERC Order 827.
- All other non-synchronous generation projects can meet the reactive power requirements per FERC Order 827.

1.3 Total Network Upgrades for all Projects

The cost allocation of Network Upgrades for the study group reflects responsibilities for mitigating system impacts based on Interconnection Customer-elected level of Network Resource Interconnection Service as of the System Impact Study report date. The total cost of network upgrades in the interconnection plan required for each generation project is listed in Table ES-2. The costs for Network Upgrades are planning level estimates and subject to be revised in the facility studies.

Table ES-2: Total Cost of Network Upgrades for DPP 2018 April West Area Generation Projects

Project Num	ERIS Network Upgrades (\$)							NRIS Network Upgrades (\$)	Interconnection Substation TO NUs (\$)	TO's Interconnection Facilities (TOIF)	SNU (\$)	Total Network Upgrade Cost (Exclude TOIF & Affected System) (\$)	M2 Received (\$)	M3 (\$)
	Base Case NUs	MWEX Voltage Stability	MISO Thermal & Voltage	MRES LPC	GRE LPC	MDU LPC	OTP LPC							
J951	\$0	\$0	\$500,000	\$0	\$0	\$0	\$0	\$0	\$3,500,000	NA	\$0	\$4,000,000	\$320,000	\$80,000
J952	\$1,164,349	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$5,000,000	NA	\$0	\$6,164,349	\$216,000	\$400,435
J953	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$8,000	\$0
J954	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$5,600	\$0
J958	\$7,764,609	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$3,500,000	NA	\$0	\$11,264,609	\$400,000	\$726,461
J959	\$0	\$0	\$312,985	\$0	\$0	\$0	\$0	\$0	\$3,800,000	NA	\$0	\$4,112,985	\$600,000	\$0
J963	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$23,200	\$0
J966	\$0	\$0	\$58,827,888	\$0	\$0	\$0	\$0	\$0	\$1,500,000	NA	\$0	\$60,327,888	\$1,600,000	\$4,432,789
J967	\$0	\$0	\$619,266	\$0	\$0	\$0	\$0	\$0	\$6,500,000	NA	\$0	\$7,119,266	\$600,000	\$111,927
J971	\$20,371,260	\$0	\$9,508,753	\$0	\$0	\$0	\$0	\$1,386,121	\$9,000,000	NA	\$0	\$40,266,134	\$1,000,000	\$3,026,613
J975	\$0	\$0	\$11,985,480	\$0	\$0	\$0	\$0	\$8,102,379	\$1,500,000	NA	\$0	\$21,587,859	\$600,000	\$1,558,786
J977	\$25,203,605	\$0	\$28,168,744	\$0	\$0	\$0	\$0	\$0	\$1,500,000	NA	\$0	\$54,872,349	\$1,200,000	\$4,287,235
J981	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$5,000,000	NA	\$0	\$5,000,000	\$800,000	\$0
J982	\$20,389,799	\$0	\$155,512	\$0	\$0	\$0	\$0	\$0	\$15,000,000	NA	\$0	\$35,545,311	\$1,200,000	\$2,354,531
J997	\$1,507,630	\$0	\$182,517	\$0	\$0	\$2,653,590	\$0	\$54,675	\$3,200,000	NA	\$0	\$7,598,412	\$800,000	\$0
J998	\$556,306	\$0	\$0	\$0	\$0	\$0	\$0	\$5,403	\$7,500,000	NA	\$0	\$8,061,709	\$200,000	\$606,171
J999	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$1,500,000	NA	\$0	\$1,500,000	\$400,000	\$0
J1001	\$1,552,922	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$3,500,000	NA	\$0	\$5,052,922	\$160,000	\$345,292
J1006	\$3,473,881	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$9,000,000	NA	\$0	\$12,473,881	\$400,000	\$847,388
J1008	\$1,490,232	\$0	\$20,263,000	\$0	\$0	\$0	\$0	\$0	\$3,000,000	NA	\$0	\$24,753,232	\$200,000	\$2,275,323
J1024	\$4,509,771	\$0	\$700,000	\$0	\$0	\$0	\$0	\$7,391,120	\$10,500,000	NA	\$0	\$23,100,890	\$800,000	\$1,510,089
J1032	\$0	\$0	\$5,937,403	\$0	\$0	\$0	\$0	\$0	\$6,500,000	NA	\$0	\$12,437,403	\$200,000	\$1,043,740
J1038	\$0	\$0	\$10,226,635	\$0	\$0	\$0	\$0	\$0	\$3,500,000	NA	\$0	\$13,726,635	\$130,000	\$1,242,664
J1040	\$2,924,848	\$0	\$3,317,483	\$0	\$0	\$68,596,410	\$0	\$0	\$2,400,000	NA	\$0	\$77,238,741	\$1,000,000	\$6,723,874
J1041	\$1,589,201	\$0	\$4,054,761	\$0	\$0	\$0	\$0	\$0	\$2,500,000	NA	\$0	\$8,143,961	\$60,000	\$754,396
J1045	\$1,639,455	\$0	\$0	\$0	\$0	\$0	\$0	\$113,879	\$250,000	NA	\$0	\$2,003,334	\$80,000	\$120,333
J1050	\$5,106,862	\$0	\$0	\$0	\$0	\$0	\$0	\$17,966	\$1,250,000	NA	\$0	\$6,374,828	\$900,000	\$0
J1054	\$0	\$0	\$2,518,524	\$0	\$0	\$0	\$0	\$0	\$3,500,000	NA	\$0	\$6,018,524	\$120,000	\$481,852
J1057	\$4,966,252	\$0	\$22,336,901	\$0	\$0	\$0	\$0	\$0	\$2,500,000	NA	\$0	\$29,803,154	\$500,000	\$2,480,315

Executive Summary

Project Num	ERIS Network Upgrades (\$)							NRIS Network Upgrades (\$)	Interconnection Substation TO NUs (\$)	TO's Interconnection Facilities (TOIF)	SNU (\$)	Total Network Upgrade Cost (Exclude TOIF & Affected System) (\$)	M2 Received (\$)	M3 (\$)
	Base Case NUs	MWEX Voltage Stability	MISO Thermal & Voltage	MRES LPC	GRE LPC	MDU LPC	OTP LPC							
J1061	\$0	\$0	\$16,508,089	\$0	\$0	\$0	\$0	\$0	\$6,500,000	NA	\$0	\$23,008,089	\$1,200,000	\$1,100,809
J1072	\$0	\$0	\$309,633	\$0	\$0	\$0	\$0	\$0	\$6,500,000	NA	\$0	\$6,809,633	\$600,000	\$80,963
J1084	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$2,000,000	NA	\$0	\$2,000,000	\$600,000	\$0
J1086	\$4,734,746	\$0	\$21,370,519	\$0	\$0	\$0	\$0	\$0	\$5,000,000	NA	\$0	\$31,105,265	\$480,000	\$2,630,527
J1092	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$3,500,000	NA	\$0	\$3,500,000	\$400,000	\$0
J1098	\$2,200,353	\$0	\$0	\$0	\$0	\$0	\$0	\$1,405,315	\$250,000	NA	\$0	\$3,855,668	\$160,000	\$225,567
J1105	\$0	\$0	\$8,881,476	\$0	\$0	\$0	\$0	\$0	\$6,500,000	NA	\$0	\$15,381,476	\$800,000	\$738,148
J1106	\$51,101,678	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$8,600,000	NA	\$0	\$59,701,678	\$1,656,000	\$4,314,168
J1108	\$0	\$0	\$88,375,665	\$30,000	\$0	\$0	\$0	\$0	\$2,500,000	NA	\$0	\$90,905,665	\$600,000	\$8,490,566
J1109	\$0	\$0	\$21,244,070	\$0	\$0	\$0	\$0	\$6,124,777	\$6,500,000	NA	\$0	\$33,868,847	\$800,000	\$2,586,885
J1110	\$0	\$0	\$62,763	\$0	\$0	\$0	\$0	\$0	\$3,350,000	NA	\$0	\$3,412,763	\$400,000	\$0
J1114	\$12,521,943	\$0	\$144,488	\$0	\$0	\$0	\$0	\$8,594,685	\$3,000,000	NA	\$0	\$24,261,116	\$800,000	\$1,626,112
J1122	\$5,362,554	\$0	\$0	\$0	\$0	\$0	\$0	\$140,128	\$16,000,000	NA	\$0	\$21,502,682	\$800,000	\$1,350,268
J1124	\$0	\$0	\$103,735	\$0	\$0	\$0	\$0	\$0	\$3,500,000	NA	\$0	\$3,603,735	\$400,000	\$0
J1128	\$0	\$0	\$13,975,891	\$0	\$0	\$0	\$0	\$6,150	\$5,400,000	NA	\$0	\$19,382,041	\$600,000	\$1,338,204
J1131	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$2,500,000	NA	\$0	\$2,500,000	\$400,000	\$0
J1132	\$559,728	\$0	\$0	\$0	\$0	\$0	\$0	\$3,745,383	\$1,000,000	NA	\$0	\$5,305,112	\$200,000	\$330,511
J1135	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$500,000	NA	\$0	\$500,000	\$200,000	\$0
J1140	\$0	\$0	\$8,599,332	\$0	\$100,000	\$0	\$0	\$0	\$5,000,000	NA	\$0	\$13,699,332	\$400,000	\$969,933
J1143	\$0	\$0	\$10,062,842	\$0	\$0	\$0	\$0	\$0	\$5,000,000	NA	\$0	\$15,062,842	\$400,000	\$1,106,284
J1149	\$5,742,456	\$0	\$3,500,000	\$0	\$0	\$0	\$0	\$0	\$5,000,000	NA	\$0	\$14,242,456	\$800,000	\$624,246
J1164	\$7,198,463	\$0	\$33,550,322	\$0	\$0	\$0	\$0	\$0	\$6,000,000	NA	\$0	\$46,748,785	\$800,000	\$3,874,879
J1169	\$1,430,338	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$3,500,000	NA	\$0	\$4,930,338	\$200,000	\$293,034
J1170	\$0	\$0	\$11,963,991	\$0	\$0	\$0	\$0	\$5,917,659	\$6,500,000	NA	\$0	\$24,381,651	\$800,000	\$1,638,165
J1174	\$5,009,747	\$0	\$6,933,333	\$0	\$0	\$0	\$0	\$21,925	\$10,000,000	NA	\$0	\$21,965,005	\$1,200,000	\$996,501
J1175	\$10,019,494	\$0	\$13,866,667	\$0	\$0	\$0	\$0	\$21,925	\$2,000,000	NA	\$0	\$25,908,086	\$1,200,000	\$1,390,809
J1177	\$260,969	\$0	\$28,257,100	\$0	\$0	\$0	\$0	\$0	\$1,500,000	NA	\$0	\$30,018,068	\$1,200,000	\$1,801,807
J1179	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$11,000,000	NA	\$0	\$11,000,000	\$1,200,000	\$0
J1181	\$0	\$0	\$4,290,234	\$0	\$0	\$0	\$0	\$0	\$10,000,000	NA	\$0	\$14,290,234	\$800,000	\$629,023
J1182	\$0	\$0	\$5,515,013	\$0	\$0	\$0	\$0	\$0	\$1,500,000	NA	\$0	\$7,015,013	\$2,000,000	\$0

Project Num	ERIS Network Upgrades (\$)							NRIS Network Upgrades (\$)	Interconnection Substation TO NUs (\$)	TO's Interconnection Facilities (TOIF)	SNU (\$)	Total Network Upgrade Cost (Exclude TOIF & Affected System) (\$)	M2 Received (\$)	M3 (\$)
	Base Case NUs	MWEX Voltage Stability	MISO Thermal & Voltage	MRES LPC	GRE LPC	MDU LPC	OTP LPC							
J1187	\$475,814	\$0	\$371,985	\$0	\$0	\$0	\$0	\$20,300,509	\$1,200,000	NA	\$0	\$22,348,307	\$607,200	\$1,627,631
Total (\$)	\$210,829,263	\$0	\$477,503,000	\$30,000	\$100,000	\$71,250,000	\$0	\$63,350,000	\$267,700,000	\$0	\$0	\$1,090,762,263	\$37,226,000	\$75,175,253

The study was performed under the direction of MISO by Siemens PTI and an ad hoc study group. The ad hoc study group was formed to review the study scope, methodology, models and results. The ad hoc study group consisted of representatives from the interconnection customers and the following utility companies – Ameren, American Transmission Company, Basin Electric Power, Cedar Falls Utilities, Central Iowa Power Cooperative, City of Springfield (IL) Water Light & Power, Columbia (MO) Water and Light, Commonwealth Edison, Corn Belt Power Cooperative, Dairyland Power, Great River Energy, ITC Midwest, Lincoln Electric System, Manitoba Hydro, MidAmerican Energy Company, MISO, Minnesota Power, Minnkota Power, Missouri River Energy Services, Montana-Dakota Utilities Co., Muscatine Power & Water, Nebraska Public Power District, Northwestern Public Service, Omaha Public Power District, Otter Tail Power, PJM, Southern Illinois Power Cooperative, Southern Minnesota Municipal Power Agency, SPP, Western Area Power Administration, and Xcel Energy.

1.4 Per Project Summary

This section provides estimated cost of Network Upgrades on a per project basis for the 2024 scenario. The shared cost of Network Upgrades for all the generation projects are listed below.

The Interconnection Customers are required to mitigate the constraints observed from the 2024 scenario.

1.4.1 J951 Summary

Network Upgrade	Cost	J951	NUs Type
Donalds-J951 POI 115 kV	\$500,000	\$500,000	MISO SH
Total Cost Per Project for Actual ERIS Elections for each Project		\$500,000	

1.4.2 J952 Summary

Network Upgrade	Cost	J952	NUs Type
Hazel Creek-Scott Co. 345 kV	\$210,829,263	\$1,164,349	Base Case NU Hazel Creek-Scott Co. 345 kV
Total Cost Per Project for Actual ERIS Elections for each Project		\$1,164,349	

1.4.3 J953 Summary

Network Upgrade	Cost	J953	NUs Type
No Network Upgrades		\$0	
Total Cost Per Project for Actual NRIS Elections for each Project		\$0	

1.4.4 J954 Summary

Network Upgrade	Cost	J954	NUs Type
No Network Upgrades		\$0	
Total Cost Per Project for Actual NRIS Elections for each Project		\$0	

1.4.5 J958 Summary

Network Upgrade	Cost	J958	NUs Type
Hazel Creek-Scott Co. 345 kV	\$210,829,263	\$7,764,609	Base Case NU Hazel Creek-Scott Co. 345 kV
Total Cost Per Project for Actual NRIS Elections for each Project		\$7,764,609	

1.4.6 J959 Summary

Network Upgrade	Cost	J959	NUs Type
Hazleton-Arnold 345 kV	\$1,230,000	\$312,985	MISO SH
Total Cost Per Project for Actual NRIS Elections for each Project		\$312,985	

1.4.7 J963 Summary

Network Upgrade	Cost	J963	NUs Type
No Network Upgrades		\$0	
Total Cost Per Project for Actual NRIS Elections for each Project		\$0	

1.4.8 J966 Summary

Network Upgrade	Cost	J966	NUs Type
Adair-Thomas Hill 161 kV	\$28,600,000	\$28,600,000	MISO SH
Montgomery-Spencer Crk 345 kV	\$35,000,000	\$17,648,040	MISO SH
2nd Maywood-Herleman 345 kV line	\$21,000,000	\$12,579,848	MISO SH
Total Cost Per Project for Actual NRIS Elections for each Project		\$58,827,888	

1.4.9 J967 Summary

Network Upgrade	Cost	J967	NUs Type
J1181 POI-Hazleton 345 kV	\$1,590,000	\$347,195	MISO SH
Hazleton-Arnold 345 kV	\$1,230,000	\$272,072	MISO SH
Wabaco-Rochester 161 kV	\$11,000,000	\$0 ¹	MISO SH
Total Cost Per Project for Actual NRIS Elections for each Project		\$619,266	

Note 1: J967 will assume \$3,367,472 cost responsibility if the Wabaco-Rochester 161 kV rebuild is no longer an approved MTEP Appendix A project.

1.4.10 J971 Summary

Network Upgrade	Cost	J971	NUs Type
Hazel Creek-Scott Co. 345 kV	\$210,829,263	\$20,371,260	Base Case NU Hazel Creek-Scott Co. 345 kV
Lakefield Tap-Lakefield 345 kV	\$37,600,000	\$9,508,753	MISO SH
Upgrade Fenton 115/69 kV Transformer	\$1,500,000	\$1,386,121	NRIS
Total Cost Per Project for Actual NRIS Elections for each Project		\$31,266,134	

1.4.11 J975 Summary

Network Upgrade	Cost	J975	NUs Type
Sheyenne-Maple River 230 kV	\$14,400,000	\$3,901,290	MISO SH
CSLTNET-Mapleton 115 kV	\$7,000,000	\$2,561,685	MISO SH
CSLTNET-Buffalo 115 kV	\$9,900,000	\$3,622,955	MISO SH
Maple River 345-230-13.8 kV xfmr #1	\$4,600,000	\$949,870	MISO SH & PK, NRIS
Maple River 345-230-13.8 kV xfmr #2	\$4,600,000	\$949,680	MISO SH & PK, NRIS
2nd Coyote 345/115 kV Transformer	\$7,000,000	\$2,611,349	NRIS
New Square Butte-Mandan 230 kV Line	\$31,000,000	\$5,190,143	NRIS
Leland Olds-Stanton 230 kV Rebuild	\$2,400,000	\$300,887	NRIS
Total Cost Per Project for Actual NRIS Elections for each Project		\$20,087,859	

1.4.12 J977 Summary

Network Upgrade	Cost	J977	NUs Type
Hazel Creek-Scott Co. 345 kV	\$210,829,263	\$25,203,605	Base Case NU Hazel Creek-Scott Co. 345 kV
Lakefield Tap-Lakefield 345 kV	\$37,600,000	\$17,568,744	MISO SH
Rutland-Fox Lake 161 kV	\$9,800,000	\$9,800,000	MISO SH
Rutland-Huntley 161 kV	\$800,000	\$800,000	MISO SH
Total Cost Per Project for Actual NRIS Elections for each Project		\$53,372,349	

1.4.13 J981 Summary

Network Upgrade	Cost	J981	NUs Type
No Network Upgrades		\$0	
Total Cost Per Project for Actual NRIS Elections for each Project		\$0	

1.4.14 J982 Summary

Network Upgrade	Cost	J982	NUs Type
Hazel Creek-Scott Co. 345 kV	\$210,829,263	\$20,389,799	Base Case NU Hazel Creek-Scott Co. 345 kV
J748 POI-Raun 345 kV	\$300,000	\$155,512	MISO SH
Total Cost Per Project for Actual NRIS Elections for each Project		\$20,545,311	

1.4.15 J997 Summary

Network Upgrade	Cost	J997	NUs Type
Hazel Creek-Scott Co. 345 kV	\$210,829,263	\$1,507,630	Base Case NU Hazel Creek-Scott Co. 345 kV
Hankinson-Forman 230 kV	\$500,000	\$182,517	MISO SH
Ellendale-Aberdeen 115 kV	\$250,000	\$91,980	MDU LPC
Ellendale 345-230-34.5 xfmr	\$7,000,000	\$2,561,610	MDU LPC
East Bismark Terminal Upgrades	\$100,000	\$54,675	NRIS
Total Cost Per Project for Actual NRIS Elections for each Project		\$4,398,412	

1.4.16 J998 Summary

Network Upgrade	Cost	J998	NUs Type
Hazel Creek-Scott Co. 345 kV	\$210,829,263	\$556,306	Base Case NU Hazel Creek-Scott Co. 345 kV
Council Bluffs-S3456 345 kV Terminal Equipment Upgrade	\$200,000	\$5,403	NRIS
Total Cost Per Project for Actual NRIS Elections for each Project		\$561,709	

1.4.17 J999 Summary

Network Upgrade	Cost	J999	NUs Type
No Network Upgrades		\$0	
Total Cost Per Project for Actual NRIS Elections for each Project		\$0	

1.4.18 J1001 Summary

Network Upgrade	Cost	J1001	NUs Type
Hazel Creek-Scott Co. 345 kV	\$210,829,263	\$1,552,922	Base Case NU Hazel Creek-Scott Co. 345 kV
Total Cost Per Project for Actual NRIS Elections for each Project		\$1,552,922	

1.4.19 J1006 Summary

Network Upgrade	Cost	J1006	NUs Type
Hazel Creek-Scott Co. 345 kV	\$210,829,263	\$3,473,881	Base Case NU Hazel Creek-Scott Co. 345 kV
Total Cost Per Project for Actual NRIS Elections for each Project		\$3,473,881	

1.4.20 J1008 Summary

Network Upgrade	Cost	J1008	NUs Type
Hazel Creek-Scott Co. 345 kV	\$210,829,263	\$1,490,232	Base Case NU Hazel Creek-Scott Co. 345 kV
J1008 POI-Blue Earth 69 kV	\$3,800,000	\$3,800,000	MISO PK
J1008 POI-Blue Earth Tap 69 kV	\$363,000	\$363,000	MISO PK
Huntley-Blue Earth Tap 69 kV	\$5,300,000	\$5,300,000	MISO PK
Elmore-Blue Earth Tap 69 kV	\$3,800,000	\$3,800,000	MISO PK
Elmore-Lakota CBPC 69 kV	\$5,500,000	\$5,500,000	MISO PK
Lakota-Lakota CBPC 69 kV	\$1,500,000	\$1,500,000	MISO PK
Total Cost Per Project for Actual NRIS Elections for each Project		\$21,753,232	

1.4.21 J1024 Summary

Network Upgrade	Cost	J1024	NUs Type
Hazel Creek-Scott Co. 345 kV	\$210,829,263	\$4,509,771	Base Case NU Hazel Creek-Scott Co. 345 kV
J1024 POI-Clarinda 161 kV	\$700,000	\$700,000	MISO SH
Council Bluffs-S3456 345 kV Terminal Equipment Upgrade	\$200,000	\$31,099	NRIS
Reconductor MCKSBRG-WNTRST	\$10,000,000	\$6,370,342	NRIS
WNTRST-NORWLK Structure Replacements	\$300,000	\$189,678	NRIS
Adams-Creston Structure Replacements	\$800,000	\$800,000	NRIS
Total Cost Per Project for Actual NRIS Elections for each Project		\$12,600,890	

1.4.22 J1032 Summary

Network Upgrade	Cost	J1032	NUs Type
J1181 POI-Hazleton 345 kV	\$1,590,000	\$47,168	MISO SH
Briggs Rd 345-161-34.5 kV xfmr #9	\$4,000,000	\$1,000,000	MISO SH
Adams-Beaver Creek 161 kV	\$15,100,000	\$3,775,000	MISO SH
Wabaco-Rochester 161 kV	\$11,000,000	\$0 ¹	MISO SH
Wabaco-Alma 161 kV	\$6,300,000	\$1,115,235	MISO SH
Total Cost Per Project for Actual NRIS Elections for each Project		\$5,937,403	

Note 1: J1032 will assume \$934,287 cost responsibility if the Wabaco-Rochester 161 kV rebuild is no longer an approved MTEP Appendix A project.

1.4.23 J1038 Summary

Network Upgrade	Cost	J1038	NUs Type
Inman-Wing River 230 kV	\$13,100,000	\$5,732,161	MISO SH
Inman-Henning 230 kV	\$2,800,000	\$2,800,000	MISO SH
Silver Lake-Henning 230 kV	\$1,700,000	\$349,818	MISO SH
Silver Lake-Fergus Falls 230 kV	\$460,000	\$94,657	MISO SH
Hoot Lake-J1038 POI 115 kV	\$1,250,000	\$1,250,000	MISO SH
Total Cost Per Project for Actual ERIS Elections for each Project		\$10,226,635	

1.4.24 J1040 Summary

Network Upgrade	Cost	J1040	NUs Type
Hazel Creek-Scott Co. 345 kV	\$210,829,263	\$2,924,848	Base Case NU Hazel Creek-Scott Co. 345 kV
Hankinson-Forman 230 kV	\$500,000	\$317,483	MISO SH
Ellendale-Aberdeen 115 kV	\$250,000	\$158,020	MDU LPC
Fox Tail-Tatanka North 230 kV	\$2,000,000	\$2,000,000	MISO SH
Merricourt-Tatanka North 230 kV	\$1,000,000	\$1,000,000	MISO SH
J580 POI-Mandan 230 kV	\$12,000,000	\$12,000,000	MDU LPC
Fox Tail-Tatanka North 230 kV (MDU LPC)	\$1,500,000	\$1,500,000	MDU LPC

Network Upgrade	Cost	J1040	NUs Type
Stein-NE Bismark 115 kV	\$250,000	\$250,000	MDU LPC
Heskett-Mandan 230 kV	\$250,000	\$250,000	MDU LPC
Fox Tail-Ellendale 230 kV	\$20,000,000	\$20,000,000	MDU LPC
Merricourt-Ellendale 230 kV	\$30,000,000	\$30,000,000	MDU LPC
Ellendale 345-230-34.5 xfmr	\$7,000,000	\$4,438,390	MDU LPC
Total Cost Per Project for Actual NRIS Elections for each Project		\$74,838,741	

1.4.25 J1041 Summary

Network Upgrade	Cost	J1041	NUs Type
Hazel Creek-Scott Co. 345 kV	\$210,829,263	\$1,589,201	Base Case NU Hazel Creek-Scott Co. 345 kV
J1057 POI-Heron Lake 161 kV	\$16,200,000	\$721,910	MISO PK
J1086 POI-Elk 161 kV	\$770,000	\$58,113	MISO PK
Lakefield Tap-Lakefield 345 kV	\$37,600,000	\$961,786	MISO SH
Split Rock-Rock Co. 161 kV	\$21,900,000	\$941,940	MISO SH & PK
Rock Co.-Magnolia 161 kV	\$24,600,000	\$1,058,065	MISO PK
Split Rock 161-115-36.2 kV xfmr #6	\$2,000,000	\$86,022	MISO SH & PK
Rebuild the Rushmore-Wolf Tap-Rushmore 69 kV line, replace three 3-way line switches	\$3,300,000	\$134,612	MISO PK
Magnolia-Elk 161 kV	\$2,000,000	\$91,399	MISO PK
Heron Lake-Lakefield 161 kV	\$20,000	\$915	MISO SH & PK
Total Cost Per Project for Actual NRIS Elections for each Project		\$5,643,961	

1.4.26 J1045 Summary

Network Upgrade	Cost	J1045	NUs Type
Hazel Creek-Scott Co. 345 kV	\$210,829,263	\$1,639,455	Base Case NU Hazel Creek-Scott Co. 345 kV
Upgrade Fenton 115/69 kV Transformer	\$1,500,000	\$113,879	NRIS
Total Cost Per Project for Actual NRIS Elections for each Project		\$1,753,334	

1.4.27 J1050 Summary

Network Upgrade	Cost	J1050	NUs Type
Hazel Creek-Scott Co. 345 kV	\$210,829,263	\$5,106,862	Base Case NU Hazel Creek-Scott Co. 345 kV
Council Bluffs-S3456 345 kV Terminal Equipment Upgrade	\$200,000	\$17,966	NRIS
Total Cost Per Project for Actual NRIS Elections for each Project		\$5,124,828	

1.4.28 J1054 Summary

Network Upgrade	Cost	J1054	NUs Type
Chub Lake 345-115-34.5 kV xfmr	\$11,400,000	\$2,518,524	MISO SH
Total Cost Per Project for Actual NRIS Elections for each Project		\$2,518,524	

1.4.29 J1057 Summary

Network Upgrade	Cost	J1057	NUs Type
Hazel Creek-Scott Co. 345 kV	\$210,829,263	\$4,966,252	Base Case NU Hazel Creek-Scott Co. 345 kV
J1057 POI-Heron Lake 161 kV	\$16,200,000	\$4,511,938	MISO PK
J1086 POI-Elk 161 kV	\$770,000	\$363,208	MISO PK
Lakefield Tap-Lakefield 345 kV	\$37,600,000	\$3,005,808	MISO SH
Split Rock-Rock Co. 161 kV	\$21,900,000	\$5,887,125	MISO SH & PK
Rock Co.-Magnolia 161 kV	\$24,600,000	\$6,612,903	MISO PK
Split Rock 161-115-36.2 kV xfmr #6	\$2,000,000	\$537,634	MISO SH & PK
Rebuild the Rushmore-Wolf Tap-Rushmore 69 kV line, replace three 3-way line switches	\$3,300,000	\$841,323	MISO PK
Magnolia-Elk 161 kV	\$2,000,000	\$571,246	MISO PK
Heron Lake-Lakefield 161 kV	\$20,000	\$5,717	MISO SH & PK
Total Cost Per Project for Actual NRIS Elections for each Project		\$27,303,154	

1.4.30 J1061 Summary

Network Upgrade	Cost	J1061	NUs Type
J1181 POI-Hazleton 345 kV	\$1,590,000	\$141,541	MISO SH
Briggs Rd 345-161-34.5 kV xfmr #9	\$4,000,000	\$3,000,000	MISO SH
Adams-Beaver Creek 161 kV	\$15,100,000	\$11,325,000	MISO SH
Wabaco-Rochester 161 kV	\$11,000,000	\$0 ¹	MISO SH
Wabaco-Alma 161 kV	\$6,300,000	\$2,041,548	MISO SH
Total Cost Per Project for Actual NRIS Elections for each Project		\$16,508,089	

Note 1: J1061 will assume \$1,797,820 cost responsibility if the Wabaco-Rochester 161 kV rebuild is no longer an approved MTEP Appendix A project.

1.4.31 J1072 Summary

Network Upgrade	Cost	J1072	NUs Type
J1181 POI-Hazleton 345 kV	\$1,590,000	\$173,597	MISO SH
Hazleton-Arnold 345 kV	\$1,230,000	\$136,036	MISO SH
Wabaco-Rochester 161 kV	\$11,000,000	\$0 ¹	MISO SH
Total Cost Per Project for Actual NRIS Elections for each Project		\$309,633	

Note 1: J1072 will assume \$1,002,595 cost responsibility if the Wabaco-Rochester 161 kV rebuild is no longer an approved MTEP Appendix A project.

1.4.32 J1084 Summary

Network Upgrade	Cost	J1084	NUs Type
No Network Upgrades		\$0	
Total Cost Per Project for Actual NRIS Elections for each Project		\$0	

1.4.33 J1086 Summary

Network Upgrade	Cost	J1086	NUs Type
Hazel Creek-Scott Co. 345 kV	\$210,829,263	\$4,734,746	Base Case NU Hazel Creek-Scott Co. 345 kV
J1057 POI-Heron Lake 161 kV	\$16,200,000	\$4,306,750	MISO PK
J1086 POI-Elk 161 kV	\$770,000	\$348,679	MISO PK
Lakefield Tap-Lakefield 345 kV	\$37,600,000	\$2,837,380	MISO SH
Split Rock-Rock Co. 161 kV	\$21,900,000	\$5,651,640	MISO SH & PK
Rock Co.-Magnolia 161 kV	\$24,600,000	\$6,348,387	MISO PK
Split Rock 161-115-36.2 kV xfmr #6	\$2,000,000	\$516,129	MISO SH & PK
Rebuild the Rushmore-Wolf Tap-Rushmore 69 kV line, replace three 3-way line switches	\$3,300,000	\$807,670	MISO PK
Magnolia-Elk 161 kV	\$2,000,000	\$548,396	MISO PK
Heron Lake-Lakefield 161 kV	\$20,000	\$5,488	MISO SH & PK
Total Cost Per Project for Actual NRIS Elections for each Project		\$26,105,265	

1.4.34 J1092 Summary

Network Upgrade	Cost	J1092	NUs Type
No Network Upgrades		\$0	
Total Cost Per Project for Actual NRIS Elections for each Project		\$0	

1.4.35 J1098 Summary

Network Upgrade	Cost	J1098	NUs Type
Hazel Creek-Scott Co. 345 kV	\$210,829,263	\$2,200,353	Base Case NU Hazel Creek-Scott Co. 345 kV
Second Webster 345/115 kV Transformer	\$10,000,000	\$1,405,315	NRIS
Total Cost Per Project for Actual NRIS Elections for each Project		\$3,605,668	

1.4.36 J1105 Summary

Network Upgrade	Cost	J1105	NUs Type
Chub Lake 345-115-34.5 kV xfmr	\$11,400,000	\$8,881,476	MISO SH
Total Cost Per Project for Actual NRIS Elections for each Project		\$8,881,476	

1.4.37 J1106 Summary

Network Upgrade	Cost	J1106	NUs Type
Hazel Creek-Scott Co. 345 kV	\$210,829,263	\$51,101,678	Base Case NU Hazel Creek-Scott Co. 345 kV
Total Cost Per Project for Actual NRIS Elections for each Project		\$51,101,678	

1.4.38 J1108 Summary

Network Upgrade	Cost	J1108	NUs Type
Rebuild Blanchard - J1143 POI - Bellevue - Little Falls 115 kV (8.34 mi)	\$4,170,000	\$128,519	MISO SH & PK
Douglas Co 115-69 kV xfmr #2	\$1,200,000	\$1,200,000	MISO SH
Douglas Co-Hudson 115 kV	\$10,390,000	\$10,390,000	MISO SH
Douglas Co-Little Sauk 115 kV	\$14,850,000	\$14,850,000	MISO SH
Quarry 345-115-34.5 kV xfmr #9	\$4,000,000	\$4,000,000	MISO SH
Douglas Co-West Union 69 kV	\$3,240,000	\$3,240,000	MISO SH
Blackoak-Grove 69 kV	\$120,000	\$120,000	MISO SH
Riverton-Mud Lake 230 kV	\$4,100,000	\$4,100,000	MISO SH
Pepin Lake-Long Prairie 115 kV	\$6,385,000	\$6,385,000	MISO SH
Pepin Lake-Blanchard 115 kV	\$7,115,000	\$7,115,000	MISO SH
Inman-Wing River 230 kV	\$13,100,000	\$7,367,839	MISO SH
Hubbard-Erie Jct 230 kV	\$2,265,000	\$2,265,000	MISO SH
West St. Cloud-Little Sauk Tap 115 kV	\$2,100,000	\$113,020	MISO SH
Wing River-Riverton 230 kV	\$13,720,000	\$13,720,000	MISO SH
Little Sauk Tap-Long Prairie 115 kV	\$4,670,000	\$4,670,000	MISO SH
Lesauk Tap-Fishill 115 kV	\$4,500,000	\$227,947	MISO SH
St. Stephens-Fishill 115 kV	\$5,100,000	\$258,340	MISO SH

Network Upgrade	Cost	J1108	NUs Type
West Union-Sauk Center Muni 69 kV	\$4,800,000	\$4,800,000	MISO SH
Kandota Tap-Sauk Center Muni 69 kV	\$540,000	\$540,000	MISO SH
Kandota Tap-Grove 69 kV	\$2,885,000	\$2,885,000	MISO SH
Alexandria MRES-Alexandria 115 kV	\$30,000	\$30,000	MRES LPC
Total Cost Per Project for Actual NRIS Elections for each Project		\$88,405,665	

1.4.39 J1109 Summary

Network Upgrade	Cost	J1109	NUs Type
Sheyenne-Lake Park 230 kV	\$3,150,000	\$2,123,941	MISO SH
Sheyenne-Maple River 230 kV	\$14,400,000	\$7,078,935	MISO SH
Silver Lake-Henning 230 kV	\$1,700,000	\$910,384	MISO SH
Silver Lake-Fergus Falls 230 kV	\$460,000	\$246,339	MISO SH
CSLTNET-Mapleton 115 kV	\$7,000,000	\$2,992,610	MISO SH
CSLTNET-Buffalo 115 kV	\$9,900,000	\$4,232,405	MISO SH
Audubon-Lake Park 230 kV	\$200,000	\$134,853	MISO SH
Maple River 345-230-13.8 kV xfmr #1	\$4,600,000	\$1,762,487	MISO SH & PK, NRIS
Maple River 345-230-13.8 kV xfmr #2	\$4,600,000	\$1,762,116	MISO SH & PK, NRIS
New Square Butte-Mandan 230 kV Line	\$31,000,000	\$5,771,006	NRIS
Leland Olds-Stanton 230 kV Rebuild	\$2,400,000	\$353,771	NRIS
Total Cost Per Project for Actual NRIS Elections for each Project		\$27,368,847	

1.4.40 J1110 Summary

Network Upgrade	Cost	J1110	NUs Type
J1181 POI-Hazleton 345 kV	\$1,590,000	\$62,763	MISO SH
Total Cost Per Project for Actual NRIS Elections for each Project		\$62,763	

1.4.41 J1114 Summary

Network Upgrade	Cost	J1114	NUs Type
Hazel Creek-Scott Co. 345 kV	\$210,829,263	\$12,521,943	Base Case NU Hazel Creek-Scott Co. 345 kV
J748 POI-Raun 345 kV	\$300,000	\$144,488	MISO SH
Second Webster 345/115 kV Transformer	\$10,000,000	\$8,594,685	NRIS
Total Cost Per Project for Actual NRIS Elections for each Project		\$21,261,116	

1.4.42 J1122 Summary

Network Upgrade	Cost	J1122	NUs Type
Hazel Creek-Scott Co. 345 kV	\$210,829,263	\$5,362,554	Base Case NU Hazel Creek-Scott Co. 345 kV
Council Bluffs-S3456 345 kV Terminal Equipment Upgrade	\$200,000	\$140,128	NRIS
Total Cost Per Project for Actual NRIS Elections for each Project		\$5,502,682	

1.4.43 J1124 Summary

Network Upgrade	Cost	J1124	NUs Type
J1181 POI-Hazleton 345 kV	\$1,590,000	\$103,735	MISO SH
Wabaco-Rochester 161 kV	\$11,000,000	\$0 ¹	MISO SH
Total Cost Per Project for Actual NRIS Elections for each Project		\$103,735	

Note 1: J1124 will assume \$836,821 cost responsibility if the Wabaco-Rochester 161 kV rebuild is no longer an approved MTEP Appendix A project.

1.4.44 J1128 Summary

Network Upgrade	Cost	J1128	NUs Type
J1181 POI-Hazleton 345 kV	\$1,590,000	\$75,891	MISO SH
Ellendale-County Line 69 kV	\$4,200,000	\$4,200,000	MISO SH
Hayward-County Line 69 kV	\$9,700,000	\$9,700,000	MISO SH
Emery Jumper Replacement	\$50,000	\$6,150	NRIS
Total Cost Per Project for Actual NRIS Elections for each Project	\$823,062,263	\$13,982,041	

1.4.45 J1131 Summary

Network Upgrade	Cost	J1131	NUs Type
No Network Upgrades		\$0	
Total Cost Per Project for Actual NRIS Elections for each Project		\$0	

1.4.46 J1132 Summary

Network Upgrade	Cost	J1132	NUs Type
Hazel Creek-Scott Co. 345 kV	\$210,829,263	\$559,728	Base Case NU Hazel Creek-Scott Co. 345 kV
Council Bluffs-S3456 345 kV Terminal Equipment Upgrade	\$200,000	\$5,403	NRIS
Reconductor MCKSBRG-WNTRST	\$10,000,000	\$3,629,658	NRIS
WNTRST-NORWLK Structure Replacements	\$300,000	\$110,322	NRIS
Total Cost Per Project for Actual NRIS Elections for each Project		\$4,305,112	

1.4.47 J1135 Summary

Network Upgrade	Cost	J1135	NUs Type
No Network Upgrades		\$0	
Total Cost Per Project for Actual NRIS Elections for each Project		\$0	

1.4.48 J1140 Summary

Network Upgrade	Cost	J1140	NUs Type
West St. Cloud-Westwood 115 kV	\$900,000	\$900,000	MISO SH
West St. Cloud-Little Sauk Tap 115 kV	\$2,100,000	\$1,238,388	MISO SH
Increase conductor clearance on Little Falls - Dewing - Mud Lake 115 kV (28.23 mi)	\$1,600,000	\$869,495	MISO PK
Lesauk Tap-Fishill 115 kV	\$4,500,000	\$2,620,992	MISO SH
St. Stephens-Fishill 115 kV	\$5,100,000	\$2,970,457	MISO SH
Benton 230 kV bus bar	\$100,000	\$100,000	CCS GRE LPC
Total Cost Per Project for Actual NRIS Elections for each Project		\$8,699,332	

1.4.49 J1143 Summary

Network Upgrade	Cost	J1143	NUs Type
Rebuild Blanchard - J1143 POI - Bellevue - Little Falls 115 kV (8.34 mi)	\$4,170,000	\$4,041,481	MISO SH & PK
West St. Cloud-Little Sauk Tap 115 kV	\$2,100,000	\$748,592	MISO SH
Increase conductor clearance on Little Falls - Dewing - Mud Lake 115 kV (28.23 mi)	\$1,600,000	\$730,505	MISO PK
Lesauk Tap-Fishill 115 kV	\$4,500,000	\$1,651,061	MISO SH
St. Stephens-Fishill 115 kV	\$5,100,000	\$1,871,203	MISO SH
Aldrich-Verndale 115 kV	\$1,020,000	\$1,020,000	MISO SH
Total Cost Per Project for Actual NRIS Elections for each Project		\$10,062,842	

1.4.50 J1149 Summary

Network Upgrade	Cost	J1149	NUs Type
Hazel Creek-Scott Co. 345 kV	\$210,829,263	\$5,742,456	Base Case NU Hazel Creek-Scott Co. 345 kV
Hazel Crk 345-230-13.8 kV xfmr #9	\$3,500,000	\$3,500,000	MISO SH
Total Cost Per Project for Actual NRIS Elections for each Project		\$9,242,456	

1.4.51 J1164 Summary

Network Upgrade	Cost	J1164	NUs Type
Hazel Creek-Scott Co. 345 kV	\$210,829,263	\$7,198,463	Base Case NU Hazel Creek-Scott Co. 345 kV
J1057 POI-Heron Lake 161 kV	\$16,200,000	\$6,659,402	MISO PK
Lakefield Tap-Lakefield 345 kV	\$37,600,000	\$3,717,530	MISO SH
Split Rock-Rock Co. 161 kV	\$21,900,000	\$9,419,295	MISO SH & PK
Rock Co.-Magnolia 161 kV	\$24,600,000	\$10,580,645	MISO PK
Split Rock 161-115-36.2 kV xfmr #6	\$2,000,000	\$860,215	MISO SH & PK
Rebuild the Rushmore-Wolf Tap-Rushmore 69 kV line, replace three 3-way line switches	\$3,300,000	\$1,516,396	MISO PK
Magnolia-Elk 161 kV	\$2,000,000	\$788,959	MISO PK
Heron Lake-Lakefield 161 kV	\$20,000	\$7,881	MISO SH & PK
Total Cost Per Project for Actual NRIS Elections for each Project		\$40,748,785	

1.4.52 J1169 Summary

Network Upgrade	Cost	J1169	NUs Type
Hazel Creek-Scott Co. 345 kV	\$210,829,263	\$1,430,338	Base Case NU Hazel Creek-Scott Co. 345 kV
Total Cost Per Project for Actual NRIS Elections for each Project		\$1,430,338	

1.4.53 J1170 Summary

Network Upgrade	Cost	J1170	NUs Type
Sheyenne-Lake Park 230 kV	\$3,150,000	\$1,026,059	MISO SH
Sheyenne-Maple River 230 kV	\$14,400,000	\$3,419,775	MISO SH
Silver Lake-Henning 230 kV	\$1,700,000	\$439,799	MISO SH
Silver Lake-Fergus Falls 230 kV	\$460,000	\$119,004	MISO SH
CSLTNET-Mapleton 115 kV	\$7,000,000	\$1,445,705	MISO SH
CSLTNET-Buffalo 115 kV	\$9,900,000	\$2,044,640	MISO SH
Audubon-Lake Park 230 kV	\$200,000	\$65,147	MISO SH
Maple River 345-230-13.8 kV xfmr #1	\$4,600,000	\$1,702,110	MISO SH & PK, NRIS
Maple River 345-230-13.8 kV xfmr #2	\$4,600,000	\$1,701,753	MISO SH & PK, NRIS

Network Upgrade	Cost	J1170	NUs Type
New Square Butte-Mandan 230 kV Line	\$31,000,000	\$5,575,851	NRIS
Leland Olds-Stanton 230 kV Rebuild	\$2,400,000	\$341,808	NRIS
Total Cost Per Project for Actual NRIS Elections for each Project		\$17,881,651	

1.4.54 J1174 Summary

Network Upgrade	Cost	J1174	NUs Type
Hazel Creek-Scott Co. 345 kV	\$210,829,263	\$5,009,747	Base Case NU Hazel Creek-Scott Co. 345 kV
Lime Crk-Colby 161 kV	\$4,800,000	\$1,600,000	MISO SH
Colby 345-161 kV xfmr	\$8,000,000	\$2,666,667	MISO SH
Webster-Wright 161 kV	\$8,000,000	\$2,666,667	MISO SH
Emery Jumper Replacement	\$50,000	\$21,925	NRIS
Total Cost Per Project for Actual NRIS Elections for each Project		\$11,965,005	

1.4.55 J1175 Summary

Network Upgrade	Cost	J1175	NUs Type
Hazel Creek-Scott Co. 345 kV	\$210,829,263	\$10,019,494	Base Case NU Hazel Creek-Scott Co. 345 kV
Lime Crk-Colby 161 kV	\$4,800,000	\$3,200,000	MISO SH
Colby 345-161 kV xfmr	\$8,000,000	\$5,333,333	MISO SH
Webster-Wright 161 kV	\$8,000,000	\$5,333,333	MISO SH
Emery Jumper Replacement	\$50,000	\$21,925	NRIS
Total Cost Per Project for Actual NRIS Elections for each Project		\$23,908,086	

1.4.56 J1177 Summary

Network Upgrade	Cost	J1177	NUs Type
Hazel Creek-Scott Co. 345 kV	\$210,829,263	\$260,969	Base Case NU Hazel Creek-Scott Co. 345 kV
Montgomery-Spencer Crk 345 kV	\$35,000,000	\$11,836,947	MISO SH
2nd Maywood-Herleman 345 kV line	\$21,000,000	\$8,420,152	MISO SH
Ottumwa 345-161 kV xfmr	\$8,000,000	\$8,000,000	MISO SH
Total Cost Per Project for Actual NRIS Elections for each Project		\$28,518,068	

1.4.57 J1179 Summary

Network Upgrade	Cost	J1179	NUs Type
No Network Upgrades		\$0	
Total Cost Per Project for Actual NRIS Elections for each Project		\$0	

1.4.58 J1181 Summary

Network Upgrade	Cost	J1181	NUs Type
J1181 POI-Hazleton 345 kV	\$1,590,000	\$638,110	MISO SH
Hazleton-Arnold 345 kV	\$1,230,000	\$508,907	MISO SH
Wabaco-Rochester 161 kV	\$11,000,000	\$0 ¹	MISO SH
Wabaco-Alma 161 kV	\$6,300,000	\$3,143,217	MISO SH
Total Cost Per Project for Actual NRIS Elections for each Project		\$4,290,234	

Note 1: J1181 will assume \$3,061,004 cost responsibility if the Wabaco-Rochester 161 kV rebuild is no longer an approved MTEP Appendix A project.

1.4.59 J1182 Summary

Network Upgrade	Cost	J1182	NUs Type
Montgomery-Spencer Crk 345 kV	\$35,000,000	\$5,515,013	MISO SH
Total Cost Per Project for Actual NRIS Elections for each Project		\$5,515,013	

1.4.60 J1187 Summary

Network Upgrade	Cost	J1187	NUs Type
Hazel Creek-Scott Co. 345 kV	\$210,829,263	\$475,814	Base Case NU Hazel Creek-Scott Co. 345 kV
Maple River 345-230-13.8 kV xfmr #1	\$4,600,000	\$185,533	MISO SH & PK, NRIS
Maple River 345-230-13.8 kV xfmr #2	\$4,600,000	\$186,451	MISO SH & PK, NRIS
2nd Coyote 345/115 kV Transformer	\$7,000,000	\$4,388,651	NRIS
New Square Butte-Mandan 230 kV Line	\$31,000,000	\$14,463,000	NRIS
Leland Olds-Stanton 230 kV Rebuild	\$2,400,000	\$1,403,533	NRIS
East Bismark Terminal Upgrades	\$100,000	\$45,325	NRIS
Total Cost Per Project for Actual NRIS Elections for each Project		\$21,148,307	

1.5 Study Compliance with NERC FAC-002-2 Standard

This DPP 2018 April West Area study was completed in compliance with NERC FAC-002-2:

R1.1: The reliability impact of the new interconnection, or materially modified existing interconnection, on affected system(s).

Section 3 covers summer peak steady-state analysis results which include thermal and voltage constraints impacted by the DPP West Area generating facilities. Thermal and voltage upgrades required to interconnect the new generating facilities are also identified.

Section 4 covers summer shoulder steady-state analysis results which include thermal and voltage constraints impacted by the DPP West Area generating facilities. Thermal and voltage upgrades required to interconnect the new generating facilities are also identified.

Section 5.1 covers reliability impact of the generating facilities per GRE Local planning Criteria (LPC). Network Upgrades required to interconnect the new generating facilities are also identified.

Section 5.2 covers reliability impact of the generating facilities per OTP Local planning Criteria (LPC). Network Upgrades required to interconnect the new generating facilities are also identified.

Section 5.3 covers reliability impact of the generating facilities per MDU Local planning Criteria (LPC). Network Upgrades required to interconnect the new generating facilities are also identified.

Section 5.4 covers reliability impact of the generating facilities per MRES Local planning Criteria (LPC). Network Upgrades required to interconnect the new generating facilities are also identified.

Section 6 covers voltage stability (PV) analysis on the MWEX System Operating Limit (SOL). Network Upgrades required for MWEX voltage stability are identified.

Section 7 covers Deliverability reliability impact of the new NRIS generating facilities.

R1.2: Adherence to applicable NERC Reliability Standards; regional and Transmission Owner planning criteria; and Facility interconnection requirements.

Sections 2.2-2.4 cover NERC Reliability Standard TPL-001-4.

Section 5.1 covers GRE Local Planning Criteria (LPC).

Section 5.2 covers OTP LPC.

Section 5.3 covers MDU LPC.

Section 5.4 covers MRES LPC.

Section 6 (voltage stability analysis) covers individual system planning criteria (ATC).

Section 7 covers MISO system planning criteria.

R1.3: Steady-state, short-circuit, and dynamics studies, as necessary, to evaluate system performance under both normal and contingency conditions.

Section 3 and Section 4 cover MISO steady-state assessment including NERC category P0 to P7 contingencies (TPL-001-4).

Section 5.1 covers GRE's LPC assessment including NERC category P0 to P7 contingencies (TPL-001-4).

Section 5.2 covers OTP's LPC assessment including NERC category P0 to P7 contingencies (TPL-001-4).

Section 5.3 covers MDU's LPC assessment including NERC category P0 to P7 contingencies (TPL-001-4).

Section 5.4 covers MRES's LPC assessment including NERC category P0 to P7 contingencies (TPL-001-4).

Section 6 covers steady-state voltage stability assessment.

Section 7 covers MISO deliverability study (steady-state assessment) including NERC category P0 to P1 contingencies (TPL-001-4).

R1.4: Study assumptions, system performance, alternatives considered, and coordinated recommendations. While these studies may be performed independently, the results shall be evaluated and coordinated by the entities involved.

Section 2.1, Section 2.2, Section 2.3, and Section 2.4 cover study assumptions and system performance criteria.

Jointly coordinated recommendations can be found in Section 5.1 (MISO and GRE), Section 5.2 (MISO and OTP), Section 5.3 (MISO and MDU), Section 5.4 (MISO and MRES), and Section 6 (MISO and ATC). Results in Section 3, 4, 6, and 7 have also been reviewed by PJM, SPP, and ATC.

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Introduction

Sixty (60) generation projects, listed in Table A-1 (Appendix A.1), have requested to interconnect to the MISO transmission network in the West Area and have advanced to the Definitive Planning Phase (DPP) 2018 April Phase 1 study (DPP West Area). J951, J952, and J1038 have requested Energy Resource Interconnection Service (ERIS); J953 and J954 have requested external Network Resource Interconnection Service (external NRIS); All other generating facilities have requested both ERIS and NRIS.

This report presents the study results of a System Impact Study (SIS) performed to evaluate the interconnection of the generating facilities in the DPP West Area Phase 1 study.

The study was performed under the direction of MISO by Siemens PTI and an ad hoc study group. The ad hoc study group was formed to review the study scope, methodology, models and results. The ad hoc study group consisted of representatives from the interconnection customers and the following utility companies – Ameren, American Transmission Company, Basin Electric Power, Cedar Falls Utilities, Central Iowa Power Cooperative, City of Springfield (IL) Water Light & Power, Columbia (MO) Water and Light, Commonwealth Edison, Corn Belt Power Cooperative, Dairyland Power, Great River Energy, ITC Midwest, Lincoln Electric System, Manitoba Hydro, MidAmerican Energy Company, MISO, Minnesota Power, Minnkota Power, Missouri River Energy Services, Montana-Dakota Utilities Co., Muscatine Power & Water, Nebraska Public Power District, Northwestern Public Service, Omaha Public Power District, Otter Tail Power, PJM, Southern Illinois Power Cooperative, Southern Minnesota Municipal Power Agency, SPP, Western Area Power Administration, and Xcel Energy.

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Model Development and Study Criteria

2.1 Model Development

2.1.1 Benchmark Cases

DPP 2018 April West area power flow benchmark cases representing 2024 summer shoulder and summer peak conditions were developed from the MTEP19 models with LBA dispatch.

The benchmark cases for DPP 2018 April study were created as follows:

- MISO Prior queued generation projects and their associated Network Upgrades (NU) were modeled. Appendix A.2 lists all DPP 2017 August West Area Phase 1 MISO Network Upgrades included in the models.
- DPP 2018 April generation projects in the West Area (DPP West Area, Table A-1), Central Area (Table A-3), Michigan Area (Table A-4), and ATC Area (Table A-5) were modeled with offline status.
- For MISO generation projects, their output was sunk to the MISO Classic (Appendix A.4, Table A-8), where generation was scaled uniformly;
- PJM generation projects were modeled and dispatched. The generation output was sunk to the PJM market (Appendix A.5, Table A-9), where generation was scaled uniformly.
- SPP generation projects were modeled and dispatched. The generation output was sunk to the SPP market (Appendix A.6, Table A-10), where generation was scaled uniformly. The Network Upgrades identified in the SPP DIS2016-001 and DIS2016-002 studies were also modeled.
- The Hickory Creek–Cardinal 345 kV project (MVP project 3127) was included in the 2024 models; the Hickory Creek–Cardinal 345 kV project has an in-service date of 12/31/2023.
- Models were further reviewed by the Ad Hoc study members (transmission owners and customers). Model corrections and changes were made based on the comments and feedback. These modeling changes are listed in Appendix A.3.
- Adjusted Square Butte DC to match the total output of the Bison (Bison 1 to 5) and Oliver County (Oliver County 1 and 2) wind farms.
- Adjusted CU DC to match the total output of Coal Creek generation units #1 and #2.
- MHEX interface transfer level is at 1530 MW in summer shoulder and 1800 MW in summer peak cases.

2.1.2 Study Cases

Summer peak study case was created by dispatching the DPP 2018 April generation projects at the specified summer peak level from the benchmark case.

Summer shoulder study case was created by dispatching the DPP 2018 April generation projects at the specified summer shoulder level from the benchmark case.

The MISO Classic was used for power balance, where generation was scaled uniformly.

Due to voltage collapse under system intact condition in both study and benchmark summer shoulder cases, two (2) fictitious large size SVCs in SPP area (Table 2-1) were added to the summer shoulder study and benchmark cases to achieve converged power flow thermal solutions.

Table 2-1: Fictitious SVCs Added Only in Summer Shoulder Case

Location	Bus #	SVC Mvar
Post Rock 345 kV	530583	500
Mingo 345 kV	531451	400

Both study and benchmark power flow cases were solved with transformer tap adjustment enabled, area interchange disabled, phase shifter adjustment enabled and switched shunt adjustment enabled.

The interface transfer levels in the 2024 study cases are summarized in Table 2-2.

Table 2-2: Interface Transfer Levels in 2024 Steady State Study Cases

Interface	2024 SH Case (MW)	2024 SPK Case (MW)
MHEX	1529	1800
MWEX	1353.4	448.0
Arrowhead PST	541.5	43.7
J732 POI – Stone Lake 345 kV	739.4	506.1

2.2 Contingency Criteria

A variety of contingencies were considered for steady-state analysis:

- NERC Category P0 with system intact (no contingencies)
- NERC Category P1 contingencies
 - NERC Category P1 contingencies, at buses with a nominal voltage of 69 kV and above, in the following areas: CWLD (area 333), AMMO (area 356), AMIL (area 357), CWLP (area 360), SIPC (area 361), WEC (area 295), WEC MI (area 296), XCEL (area 600), MP (area 608), SMMPA (area 613), GRE (area 615), OTP (area 620), ITCM (area 627), MPW (area 633), MEC (area 635), MDU (area 661), BEPC-MISO (area 663), MHEB (area 667), DPC (area 680), ALTE (area 694), WPS (area 696), MGE (area 697), UPPC (area 698), CE(area 222), NPPD (area 640), OPPD (area 645), LES (area 650),

- WAPA (area 652), BEPC-SPP (area 659), AECI (area 330), MIPU (area 540), KCPL (area 541), KACY (area 542), INDN (area 545).
- Multiple-element NERC Category P1 contingencies, in Dakotas, Illinois, Iowa, Minnesota, Missouri, and Wisconsin. These specified Category P1 contingencies are listed in Appendix A.7.
- NERC Category P2-P7 contingencies
 - Selected NERC Category P2-P7 contingencies provided by the Ad Hoc Study Group, in the study region of Dakotas, Illinois, Iowa, Minnesota, Missouri, and Wisconsin. These specified Category P2-P7 contingencies are listed in Appendix A.7.

For all contingency and post-disturbance analyses, cases were solved with transformer tap adjustment enabled, area interchange adjustment disabled, phase shifter adjustment disabled (fixed) and switched shunt adjustment enabled.

2.3 Monitored Elements

The study area is defined in Table 2-3. Facilities in the study area were monitored for system intact and contingency conditions. Under NERC category P0 conditions (system intact) branches were monitored for loading above the normal (PSS[®]E rate A) rating. Under NERC category P1-P7 conditions, branches were monitored for loading as shown in the column labeled "Post-Disturbance Thermal Limits".

Table 2-3: Monitored Elements

Owner / Area	Monitored Facilities	Thermal Limits ¹		Voltage Limits ²	
		Pre-Disturbance	Post-Disturbance	Pre-Disturbance	Post-Disturbance
AECI	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
AMIL	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.075/0.90
AMMO	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.075/0.90
ATCLLC	69 kV and above	95% of Rate A	95% of Rate B	1.05/0.95	1.10/0.90
BEPC-MISO	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
BEPC-SPP	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
CBPC	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
CMPMA	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.07/0.90
CWLD	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
CWLP	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.075/0.90
CE	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
DPC	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
GMO	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
GRE	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.92/0.90
INDN	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
ITCM	69 kV and above	100% of Rate A	100% of Rate B	1.07/1.05/0.95	1.10/0.93

Owner / Area	Monitored Facilities	Thermal Limits ¹		Voltage Limits ²	
		Pre-Disturbance	Post-Disturbance	Pre-Disturbance	Post-Disturbance
KACY	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
KCPL	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
LES	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
MDU	57 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
MEC	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.96/0.95	1.05/0.96/0.95/0.94/0.93 ³
MHEB	69 kV and above	100% of Rate A	100% of Rate B	1.12/1.1/1.07/1.05/1.04/0.99/0.97/0.96/0.95	1.15/1.10/0.94/0.90
MP	69 kV and above	100% of Rate A	100% of Rate B	1.05/1.00	1.10/0.95
MPC	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.97	1.10/0.92
MPW	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.06/0.92
MRES	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.97	1.05/0.92
NPPD	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
OPPD	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
OTP	40 kV and above	100% of Rate A	100% of Rate B	1.07/1.05/0.97	1.10/0.92
PPI	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.075/0.90
RPU	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.92
SIPC	69 kV and above	100% of Rate A	100% of Rate B	1.07/0.95	1.09/0.91
SMMPA	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
WAPA	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
WPPI	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.1/0.9
XEL	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.05/0.92

Notes

- 1: PSS®E Rate A, Rate B or Rate C
- 2: Limits dependent on nominal bus voltage
- 3: For facilities in Cedar Falls Utilities or Ames Municipal Utilities, post-contingency voltage limits are 1.05/0.94 for >200 kV, and 1.05/0.93 for others.

2.4 Performance Criteria

A branch is considered as a thermal injection constraint if the branch is loaded above its applicable normal or emergency rating for the post-change case, and any of the following conditions are met:

- 1) the generator (NR/ER) has a larger than 20% DF on the overloaded facility under post contingent condition or 5% DF under system intact condition, or

- 2) the megawatt impact due to the generator is greater than or equal to 20% of the applicable rating (normal or emergency) of the overloaded facility, or
- 3) the overloaded facility or the overload-causing contingency is at generator's outlet, or
- 4) for any other constrained facility, where none of the study generators meet one of the above criteria in 1), 2), or 3), however, the cumulative megawatt impact of the group of study generators (NR/ER) is greater than 20% of the applicable rating, then only those study generators whose individual MW impact is greater than 5% of the applicable rating and has DF greater than 5% (OTDF or PTDF) will be responsible for mitigating the cumulative MW impact constraint.

A bus is considered a voltage constraint if both of the following conditions are met. All voltage constraints must be resolved before a project can receive interconnection service.

- 1) the bus voltage is outside of applicable normal or emergency limits for the post-change case, and
- 2) the change in bus voltage is greater than 0.01 per unit.

All DPP 2018 April West Area study generators must mitigate thermal injection constraints and voltage constraints in order to obtain unconditional Interconnection Service.

Further, all generators requesting Network Resource Interconnection Service (NRIS) must mitigate constraints found by using the deliverability algorithm, to meet the system performance criteria for NERC category P0-P1 events, if the constraint demonstrates an incremental flow caused by the generator equal to or greater than 5% of the generator's maximum dispatch level in each case.

2.5 Reactive Power Requirements for Non-Synchronous Generation (FERC Order 827)

For non-synchronous generation projects in the DPP 2018 April West Area study group, if they do not have signed Generator Interconnection Agreement (GIA) or Provisional GIA (PGIA) by September 21, 2016, they are required to provide dynamic reactive power within the range of 0.95 leading to 0.95 lagging at the high-side of the generator substation.

All non-synchronous generation projects in this study group are required to meet the reactive power requirements per FERC Order 827.

Collector system and shunt compensation of DPP West projects are modeled, which are listed in Appendix A.1, Table A-2. An analysis was performed to study the reactive power requirements (FERC Order 827) for the non-synchronous generation projects in the DPP 2018 April West study group. The analysis was performed as follows:

Step 1: Verify whether total dynamic reactive power (reactive power from generators and dynamic compensation devices) in the plant can meet the dynamic reactive power range of 0.95 leading to 0.95 lagging at the generator terminal bus. The verification in Step 1 was performed when generator data was submitted and modeled.

Step 2: Verify whether total reactive power (reactive power from generators, dynamic compensation devices, and static compensation devices) in the plant can meet the dynamic reactive power range of 0.95 leading to 0.95 lagging at the high-side of the generator substation. The testing procedure in Step 2 is described in the following:

- Lock the high-side of the generator substation at 1.0 pu voltage by adding a fictitious SVC. This is to ensure that the test result is not affected by the system condition.
- Lock the reactive power output of the generator to the maximum limit (Q_{max}). Make sure all shunt compensation devices within the substation are at the maximum capacitive output. Adjust transformer tap to ensure bus voltages within the substation are within 0.95 – 1.05 pu range. Measure real power and reactive power from the generator plant to the high-side of the generator substation. Calculate the power factor to verify if it meets the 0.95 lagging requirement.
- Lock the reactive power output of the generator to the minimum limit (Q_{min}). Make sure all shunt compensation devices within the substation are at the maximum inductive output. Adjust transformer tap to ensure bus voltages within the substation are within 0.95 – 1.05 pu range. Measure real power and reactive power from the generator plant to the high-side of the generator substation. Calculate the power factor to verify if it meets the 0.95 leading requirement.

Appendix C lists reactive power requirement analysis results for the DPP West generation projects. The results are summarized as following:

- J951, J958, J959, J966, J971, J981, J997, J998, J999, J1006, J1024, J1061, J1072, J1084, J1086, J1092, J1105, J1106, J1109, J1110, J1124, J1128, J1174, J1177, J1179, J1182, and J1187 do not meet the reactive power requirements per FERC Order 827.
- All other non-synchronous generation projects can meet the reactive power requirements per FERC Order 827.

Summer Peak Steady-State Analysis

Summer peak steady-state analysis was performed in summer peak scenario to identify thermal and voltage upgrades required interconnecting the generating facilities in the DPP 2018 April West Area group to the transmission system. Two conditions for discharging and charging of battery projects of J1032, J1041, J1045, and J1054 in the DPP West group were both studied.

3.1 Study Procedure

3.1.1 Computer Programs

Steady-state analyses were performed using PSS®E version 33.12.1 and PSS®MUST version 12.4.0.

3.1.2 Study Methodology

Summer peak power flow case was created in the procedure as described in Section 2.1. The summer peak case can converge under post-contingency conditions. Therefore, fictitious SVCs are not modeled. Nonlinear (AC) contingency analysis was performed on the benchmark and study cases, and the incremental impact of the DPP West Area generating facilities was evaluated by comparing the steady-state performance of the transmission system in the benchmark and study cases. Network upgrades were identified to mitigate any summer peak constraints.

3.2 Contingency Analysis Results for Summer Peak in Discharging Condition

The incremental impact of the proposed interconnection on individual facilities was evaluated by comparing flows and voltages between benchmark case (without DPP West Area projects) and study case (with DPP West Area projects). The four battery projects (J1032, J1041, J1045, and J1054) were in discharging condition.

3.2.1 System Intact Conditions

For NERC category P0 (system intact) conditions, thermal constraints are listed in Table D-1. No voltage constraints were identified (Table D-2).

3.2.2 Post Contingency Conditions

The results in this Section are for analysis of conditions following NERC Category P1-P7 contingencies. All category P1 contingencies were converged.

For P1 contingencies, thermal constraints are listed in Table D-3. No voltage constraints were identified (Table D-4).

Two category P2-P7 contingencies (Table D-7) were not converged, and their dc thermal results are listed in Table D-8. The contingency was not converged in the benchmark or study cases. No mitigation plan is required for the study projects for these contingencies.

For P2-P7 contingencies, thermal constraints are listed in Table D-5, and no voltage constraints were identified (Table D-6).

3.3 Contingency Analysis Results for Summer Peak in Charging Condition

The incremental impact of the proposed interconnection on individual facilities was evaluated by comparing flows and voltages between benchmark case (without DPP West Area projects) and study case (with DPP West Area projects). The four battery projects (J1032, J1041, J1045, and J1054) were in charging condition.

3.3.1 System Intact Conditions

For NERC category P0 (system intact) conditions, thermal constraints are listed in Table D-9. No voltage constraints were identified (Table D-10).

3.3.2 Post Contingency Conditions

The results in this Section are for analysis of conditions following NERC Category P1-P7 contingencies. All category P1 contingencies were converged.

For P1 contingencies, thermal constraints are listed in Table D-11. No voltage constraints were identified (Table D-12).

Two category P2-P7 contingencies (Table D-15) were not converged, and their dc thermal results are listed in Table D-16. The contingency was not converged in the benchmark or study cases. No mitigation plan is required for the study projects for these contingencies.

For P2-P7 contingencies, thermal constraints are listed in Table D-13, and no voltage constraints were identified (Table D-14).

3.4 Summer Peak Worst Thermal Constraints

Combining the results for both discharging and charging conditions, Table 3-1 lists worst thermal constraints and Network Upgrades in the 2024 summer peak scenario.

Table 3-1: 2024 Summer Peak Thermal Constraints and Network Upgrades, Maximum Screened Loading

Generator	Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Scenario	Mitigation
				(MVA)	(%)				
J1008	J1008 POI-Blue Earth 69 kV	45.0	ITCM	49.8	110.8	CEII Redacted	P1	PK Discharge	Rebuild 5.25 miles
J1008	J1008 POI-Blue Earth Tap 69 kV	45.0	ITCM	49.7	110.4	CEII Redacted	P1	PK Discharge	Rebuild 0.5 miles
J1008	J1008 POI-Blue Earth Tap 69 kV	45.0	ITCM	49.4	109.8	CEII Redacted	P2-P7	PK Discharge	Rebuild 0.5 miles
J1057,J1086,J1164, J1041	J1057 POI-Heron Lake 161 kV	167.0	ITCM	226.5	135.6	CEII Redacted	P0	PK Discharge	MTEP 15 Project 8106 mitigates, 296 MVA SN/SE rating
J1057,J1086,J1164, J1041	J1057 POI-Heron Lake 161 kV	167.0	ITCM	345.2	206.7	CEII Redacted	P1	PK Discharge	MTEP 15 Project 8106 and rebuild 14.7 miles

Generator	Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Scenario	Mitigation
				(MVA)	(%)				
J1057,J1086,J1164, J1041	J1057 POI-Heron Lake 161 kV	167.0	ITCM	345.2	206.7	CEII Redacted	P2-P7	PK Discharge	MTEP 15 Project 8106 and rebuild 14.7 miles
J1057,J1086,J1041	J1086 POI-Elk 161 kV	200.0	ITCM	251.1	125.6	CEII Redacted	P1	PK Discharge	Replace jumpers and rebuild approx 0.7 miles
J1057,J1086,J1041	J1086 POI-Elk 161 kV	200.0	ITCM	250.0	125.0	CEII Redacted	P2-P7	PK Discharge	Replace jumpers and rebuild approx 0.7 miles
J1143	J1143 POI-Blanch Rd. 115 kV	90.0	MP	97.7	108.6	CEII Redacted	P1	PK Discharge	Rebuild Blanchard - J1143 POI - Bellevue - Little Falls 115 kV (8.34 mi)
J1143	J1143 POI-Blanch Rd. 115 kV	90.0	MP	97.7	108.6	CEII Redacted	P2-P7	PK Discharge	Rebuild Blanchard - J1143 POI - Bellevue - Little Falls 115 kV (8.34 mi)
J1057,J1086,J1164, J1041	Split Rock-Rock Co. 161 kV	224.2	XEL	329.1	146.8	CEII Redacted	P1	PK Discharge	Uprate with 795 ACSS

Generator	Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Scenario	Mitigation
				(MVA)	(%)				
J1057,J1086,J1164, J1041	Split Rock-Rock Co. 161 kV	224.2	XEL	328.5	146.5	CEII Redacted	P2-P7	PK Discharge	Uprate with 795 ACSS
J1057,J1086,J1164, J1041	Rock Co.-Magnolia 161 kV	171.0	XEL ITCM	319.7	186.9	CEII Redacted	P1	PK Discharge	XEL: Uprate with 795 ACSS ITCM: Rebuild 11.4 miles, \$12.54M
J1057,J1086,J1164, J1041	Rock Co.-Magnolia 161 kV	171.0	XEL ITCM	319.0	186.6	CEII Redacted	P2-P7	PK Discharge	XEL: Uprate with 795 ACSS ITCM: Rebuild 11.4 miles, \$12.54M
J1057,J1086,J1164, J1041	Split Rock 161-115-36.2 kV xfmr #6	187.0	XEL	325.9	174.3	CEII Redacted	P1	PK Discharge	Upgrade with 336 MVA

Generator	Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Scenario	Mitigation
				(MVA)	(%)				
J1057,J1086,J1164, J1041	Split Rock 161-115-36.2 kV xfmr #6	187.0	XEL	325.3	173.9	CEII Redacted	P2-P7	PK Discharge	Upgrade with 336 MVA
J1140,J1143	Dewing-Little Falls 115 kV	78.0	GRE MP	83.4	106.9	CEII Redacted	P1	PK Discharge	Increase conductor clearance on Little Falls - Dewing - Mud Lake 115 kV (28.23 mi)
J1140,J1143	Dewing-Mud Lake 115 kV	78.0	GRE MP	79.6	102.1	CEII Redacted	P1	PK Discharge	Increase conductor clearance on Little Falls - Dewing - Mud Lake 115 kV (28.23 mi)
J1143	J1143 POI-Bellvu 115 kV	90.0	GRE MP	98.6	109.6	CEII Redacted	P1	PK Charge	Rebuild Blanchard - J1143 POI - Bellevue - Little Falls 115 kV (8.34 mi)
J1143	J1143 POI-Bellvu 115 kV	90.0	GRE MP	98.6	109.5	CEII Redacted	P2-P7	PK Discharge	Rebuild Blanchard - J1143 POI - Bellevue - Little Falls 115 kV (8.34 mi)
J1143	Bellevu-Little Falls 115 kV	90.0	GRE MP	96.3	107.0	CEII Redacted	P1	PK Charge	Rebuild Blanchard - J1143 POI - Bellevue - Little Falls 115 kV (8.34 mi)
J1143	Bellevu-Little Falls 115 kV	90.0	GRE MP	96.2	106.9	CEII Redacted	P2-P7	PK Discharge	Rebuild Blanchard - J1143 POI - Bellevue - Little Falls 115 kV (8.34 mi)

Generator	Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Scenario	Mitigation
				(MVA)	(%)				
J1041,J1086,J1164	J1057 POI-Brewster 161 kV	167.0	GRE ITCM	207.2	124.1	CEII Redacted	P1	PK Charge	N/A, per POI discussion Brewster would be on new terminal in new J1057 interconnection switch station
J1041,J1086,J1164	J1057 POI-Brewster 161 kV	167.0	GRE ITCM	207.2	124.0	CEII Redacted	P2-P7	PK Charge	N/A, per POI discussion Brewster would be on new terminal in new J1057 interconnection switch station
J1086,J1164	J1086 POI-Brewster 161 kV	200.0	GRE ITCM	212.7	106.3	CEII Redacted	P1	PK Charge	GRE: GRE equipment rating at Brewster is 296 MVA. \$0 ITCM: N/A, Rating 219 SN/SE MVA. \$0
J1086,J1164	J1086 POI-Brewster 161 kV	200.0	GRE ITCM	212.7	106.3	CEII Redacted	P2-P7	PK Charge	GRE: GRE equipment rating at Brewster is 296 MVA. \$0 ITCM: N/A, Rating 219 SN/SE MVA. \$0
J1057,J1086,J1164	Rushmore-Wolf Tap 69 kV	45.4	GRE XEL	52.6	115.8	CEII Redacted	P1	PK Discharge	GRE: Rebuild the Rushmore-Wolf Tap-Rushmore 69 kV line, replace three 3-way line switches.
J1057,J1086,J1164	Rushmore-Wolf Tap 69 kV	45.4	GRE XEL	52.5	115.5	CEII Redacted	P2-P7	PK Discharge	GRE: Rebuild the Rushmore-Wolf Tap-Rushmore 69 kV line, replace three 3-way line switches.
J1108	West Union-Sauk Center Muni 69 kV	38.0	GRE XEL	43.0	113.3	CEII Redacted	P1	PK Charge	XEL: line is 477 ACSR. \$0 GRE: switches and sub are rated 143.4 MVA. \$0

Generator	Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Scenario	Mitigation
				(MVA)	(%)				
J1108	West Union-Sauk Center Muni 69 kV	38.0	GRE XEL	39.3	103.4	CEII Redacted	P2-P7	PK Charge	XEL: line is 477 ACSR. \$0 GRE: switches and sub are rated 143.4 MVA. \$0
J975,J1109,J1170	Maple River 345-230-13.8 kV xfmr #1	420.0	OTP MPC	472.1	112.4	CEII Redacted	P1	PK Charge	Replace the transformer with a larger transformer.
J975,J1109,J1170	Maple River 345-230-13.8 kV xfmr #1	420.0	OTP MPC	467.7	111.4	CEII Redacted	P2-P7	PK Charge	Replace the transformer with a larger transformer.
J975,J1109,J1170	Maple River 345-230-13.8 kV xfmr #2	420.0	OTP MPC	475.2	113.1	CEII Redacted	P1	PK Charge	Replace the transformer with a larger transformer.
J975,J1109,J1170	Maple River 345-230-13.8 kV xfmr #2	420.0	OTP MPC	470.8	112.1	CEII Redacted	P2-P7	PK Charge	Replace the transformer with a larger transformer.
J1041,J1057,J1086, J1164	Magnolia 161-69 kV xfmr	56.0	ITCM	63.7	113.7	CEII Redacted	P1	PK Discharge	Terminal limit rated 62/66 MVA SN/SE
J1057,J1086,J1164	Magnolia 161-69 kV xfmr	56.0	ITCM	63.5	113.4	CEII Redacted	P2-P7	PK Discharge	Terminal limit rated 62/66 MVA SN/SE

Generator	Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Scenario	Mitigation
				(MVA)	(%)				
J1008	Huntley-Blue Earth Tap 69 kV	45.0	ITCM	49.5	110.0	CEII Redacted	P1	PK Discharge	Rebuild 7.34 miles
J1008	Elmore-Blue Earth Tap 69 kV	48.0	ITCM	49.6	103.4	CEII Redacted	P1	PK Discharge	Rebuild 5.24 miles
J1008	Elmore-Blue Earth Tap 69 kV	48.0	ITCM	49.4	102.9	CEII Redacted	P2-P7	PK Discharge	Rebuild 5.24 miles
J1008	Elmore-Lakota CBPC 69 kV	48.0	ITCM CBPC	49.0	102.0	CEII Redacted	P1	PK Discharge	Rebuild 7.58 miles
J1008	Elmore-Lakota CBPC 69 kV	48.0	ITCM CBPC	48.7	101.5	CEII Redacted	P2-P7	PK Discharge	Rebuild 7.58 miles

Generator	Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Scenario	Mitigation
				(MVA)	(%)				
J1041,J1057,J1086,J1164	Rushmore Tap-Wolf Tap 69 kV	48.0	GRE XEL	52.0	108.4	CEII Redacted	P1	PK Discharge	Rebuild the Rushmore-wolf tap-Rushmore 69 kV line, replace three 3-way line switches.
J1057,J1086,J1164	Rushmore Tap-Wolf Tap 69 kV	48.0	GRE XEL	51.9	108.1	CEII Redacted	P2-P7	PK Discharge	Rebuild the Rushmore-wolf tap-Rushmore 69 kV line, replace three 3-way line switches.
J1008	Lakota-Lakota CBPC 69 kV	48.0	ITCM CBPC	48.6	101.3	CEII Redacted	P1	PK Discharge	Rebuild 2.02 miles
J1008	Lakota-Lakota CBPC 69 kV	48.0	ITCM CBPC	48.4	100.8	CEII Redacted	P2-P7	PK Discharge	Rebuild 2.02 miles

Generator	Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Scenario	Mitigation
				(MVA)	(%)				
J1041,J1057,J1086, J1164	Magnolia-Elk 161 kV	151.0	ITCM	198.3	131.3	CEII Redacted	P1	PK Discharge	Replace structures to mitigate sag limits, new rating 210 MVA SN/SE
J1041,J1057,J1086, J1164	Magnolia-Elk 161 kV	151.0	ITCM	197.1	130.5	CEII Redacted	P2-P7	PK Discharge	Replace structures to mitigate sag limits, new rating 210 MVA SN/SE
J1041,J1057,J1086, J1164	Heron Lake-Lakefield 161 kV	167.0	ITCM	264.4	158.3	CEII Redacted	P1	PK Discharge	MTEP 15 Project 8106 and replace wavetrap
J1041,J1057,J1086, J1164	Heron Lake-Lakefield 161 kV	167.0	ITCM	264.4	158.3	CEII Redacted	P2-P7	PK Discharge	MTEP 15 Project 8106 and replace wavetrap

3.5 Network Upgrades Identified in MISO ERS Analysis for 2024 Summer Peak Scenario

Based on the MISO 2024 summer peak steady state analyses, the MISO thermal NUs and cost are listed in Table 4-3. No reactive power NUs are required.

Table 3-2: Summer Peak Thermal NUs

Constraint	Owner	Mitigation	Cost (\$)
J1008 POI-Blue Earth 69 kV	ITCM	Rebuild 5.25 miles	\$3,800,000
J1008 POI-Blue Earth Tap 69 kV	ITCM	Rebuild 0.5 miles	\$363,000
J1057 POI-Heron Lake 161 kV	ITCM	MTEP 15 Project 8106 and rebuild 14.7 miles	\$16,200,000
J1086 POI-Elk 161 kV	ITCM	Replace jumpers and rebuild approx 0.7 miles	\$770,000
J1143 POI-Blanch Rd. 115 kV	MP	Rebuild Blanchard - J1143 POI - Bellevue - Little Falls 115 kV (8.34 mi)	\$4,170,000
Split Rock-Rock Co. 161 kV	XEL	Uprate with 795 ACSS	\$21,900,000
Rock Co.-Magnolia 161 kV	XEL ITCM	XEL: Uprate with 795 ACSS, \$24.6M ITCM: Rebuild 11.4 miles, \$7.47M	\$24,600,000
Split Rock 161-115-36.2 kV xfmr #6	XEL	Upgrade with 336 MVA	\$2,000,000
Dewing-Little Falls 115 kV	GRE MP	Increase conductor clearance on Little Falls - Dewing - Mud Lake 115 kV (28.23 mi)	\$1,600,000
Dewing-Mud Lake 115 kV	GRE MP	Increase conductor clearance on Little Falls - Dewing - Mud Lake 115 kV (28.23 mi)	Incl.
J1143 POI-Bellevu 115 kV	GRE MP	Rebuild Blanchard - J1143 POI - Bellevue - Little Falls 115 kV (8.34 mi)	Incl.
Bellevu-Little Falls 115 kV	GRE MP	Rebuild Blanchard - J1143 POI - Bellevue - Little Falls 115 kV (8.34 mi)	Incl.
J1057 POI-Brewster 161 kV	GRE ITCM	N/A, per POI discussion Brewster would be on new terminal in new J1057 interconnection switch station	\$0
J1086 POI-Brewster 161 kV	GRE ITCM	GRE: GRE equipment rating at Brewster is 296 MVA. \$0 ITCM: N/A, Rating 219 SN/SE MVA. \$0	\$0
Rushmore-Wolf Tap 69 kV	GRE XEL	GRE: Rebuild the Rushmore-Wolf Tap-Rushmore 69 kV line, replace three 3-way line switches.	\$3,300,000
West Union-Sauk Center Muni 69 kV	GRE XEL	XEL: line is 477 ACSR. \$0 GRE: switches and sub are rated 143.4 MVA. \$0	\$0
Maple River 345-230-13.8 kV xfmr #1	OTP MPC	Replace the #1 transformer with a larger transformer.	\$4,600,000

Constraint	Owner	Mitigation	Cost (\$)
Maple River 345-230-13.8 kV xfmr #2	OTP MPC	Replace the #2 transformer with a larger transformer.	\$4,600,000
Magnolia 161-69 kV xfmr	ITCM	Terminal limit rated 62/66 MVA SN/SE	\$0
Huntley-Blue Earth Tap 69 kV	ITCM	Rebuild 7.34 miles	\$5,300,000
Elmore-Blue Earth Tap 69 kV	ITCM	Rebuild 5.24 miles	\$3,800,000
Elmore-Lakota CBPC 69 kV	ITCM CBPC	Rebuild 7.58 miles	\$5,500,000
Rushmore Tap-Wolf Tap 69 kV	GRE XEL	Rebuild the Rushmore-wolf tap-Rushmore 69 kV line, replace three 3-way line switches.	Incl.
Lakota-Lakota CBPC 69 kV	ITCM CBPC	Rebuild 2.02 miles	\$1,500,000
Magnolia-Elk 161 kV	ITCM	Replace structures to mitigate sag limits, new rating 210 MVA SN/SE	\$2,000,000
Heron Lake-Lakefield 161 kV	ITCM	MTEP 15 Project 8106 and replace wavetrap	\$20,000

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Section

4

Summer Shoulder Steady-State Analysis

Summer shoulder steady-state analysis was performed in summer shoulder scenario to identify thermal and voltage upgrades required interconnecting the generating facilities in the DPP 2018 April West Area group to the transmission system. Two conditions for discharging and charging of battery projects of J1032, J1041, J1045, and J1054 in the DPP West group were both studied.

4.1 Study Procedure

Summer shoulder power flow case was created in the procedure as described in Section 2.1. Due to post-contingent voltage collapse and thermal overloads in the initial power flow case, steady-state analysis was performed in the following three-step procedure:

- Step 1: Non-linear (AC) contingency analysis (Stage-1 ACCC) was performed for single critical contingencies (contingencies in ≥ 200 kV system in MISO West area) to identify voltage collapse and thermal overloads.
- Step 2: Based on the identified voltage collapse and thermal overloads in the Stage-1 ACCC, project justification analysis was performed to determine Network Upgrades (NUs) required for interconnection of DPP West projects. These selected NUs are called Base Case NUs.
- Step 3: The Base Case NUs were added to the models. Stage-2 ACCC was performed to identify any remaining thermal and voltage constraints.

4.2 Step 1 – Stage-1 ACCC Analysis

AC contingency analysis was performed for single critical contingencies (contingencies in ≥ 200 kV system in MISO West area) to identify voltage collapse and thermal overloads. Analysis was performed in the 2024 summer shoulder discharging scenario using PSS®E.

4.2.1 Stage-1 Voltage Violations

All single critical contingencies (contingencies in ≥ 200 kV system in MISO West area) were converged. Voltage violations are listed in Table E-1. Potential voltage collapse (voltages below 0.9 per unit in >200 kV system) were identified in the following areas:

- Lyon Co-Cedar Mountain-Helena-Chub Lake 345 kV
- Helena-Sheas Lake 345 kV
- Riverview 345 kV
- Panther-McLeod-Blue Lake 230 kV
- Willmar-Paynesville 230 kV

- Lake Park-Audubon-Erie Jct.-Hubbard-Badoura 230 kV
- Fergus Falls-Silver Lake-Henning-Inman-Wing River 230 kV

4.2.2 Stage-1 Thermal Violations

Thermal violations are listed in Table E-2. The following 345 kV lines were heavily loaded to more than 1460 MVA. A typical 345 kV line has Surge Impedance Loading (SIL) around 400 MVA. A transmission line absorbs reactive power from system quadratically proportional to line current when line flow is more than its SIL.

- Wilmarth-Sheas Lake-Helena-Chub Lake 345 kV
- Scott Co-Blue Lake 345 kV
- Crandal-Field 345 kV

4.3 Step 2 – Base Case NUs Justification Analysis

Based on the identified potential voltage collapse in the Stage-1 ACCC and the identified heavily overloaded 345 kV lines under post-contingent conditions, various transmission Network Upgrades (NUs) were tested. The Hazel Creek-Scott County 345 kV line was justified as Base Case NU due to the following performance advantages:

- 1) The new line carries 727 MVA significant flow under system intact condition.
- 2) The new line can effectively mitigate the identified voltage collapse in the following areas:
 - Lyon Co-Cedar Mountain-Helena-Chub Lake 345 kV
 - Helena-Sheas Lake 345 kV
 - Riverview 345 kV
 - Panther-McLeod-Blue Lake 230 kV
 - Willmar-Paynesville 230 kV
 - Lake Park-Audubon-Erie Jct.-Hubbard-Badoura 230 kV
 - Fergus Falls-Silver Lake-Henning-Inman-Wing River 230 kV
- 3) The new line can mitigate or significantly reduce the following identified thermal overloads:
 - Wilmarth-Sheas Lake-Helena-Chub Lake 345 kV
 - Minn. Valley-Panther-McLeod 230 kV
 - Granite Falls-Willmar 230 kV
 - Lakefield-Crandal-Fieldon 345 kV

The Hazel Creek-Scott County 345 kV Base Case NU justification analysis was performed based on the Stage-1 ACCC results. Detailed justification results are in Appendix E.2.

In summary, the following Base Case NUs (Table 4-1) are justified and required:

Table 4-1: MISO Base Case NUs

NUs	Miles	Cost (\$)
Hazel Creek-Scott County 345 kV	115	\$210,829,263
Total		\$210,829,263

4.4 Step 3 – Stage-2 ACCC Analysis

The Hazel Creek-Scott County 345 kV Base Case NU was added to create Stage-2 models. AC contingency analysis was performed in the Stage-2 models to identify any remaining thermal and voltage constraints. Two conditions for discharging and charging of battery projects J1032, J1041, J1045, and J1054 in the DPP West group were both studied.

4.4.1 Stage-2 ACCC Analysis Results for Summer Shoulder in Discharging Condition

The discharging condition includes four battery projects (J1032, J1041, J1045, and J1054) in discharging condition.

4.4.1.1 System Intact Conditions

For NERC category P0 (system intact) conditions, thermal constraints are listed in Table E-5. No voltage constraints were identified (Table E-6).

4.4.1.2 Post Contingency Conditions

The results in this Section are for analysis of conditions following NERC Category P1-P7 contingencies. All NERC Category P1 contingencies were converged.

For P1 contingencies, thermal constraints are listed in Table E-7. No voltage constraints were identified (Table E-8).

There are three P2-P7 contingencies (Table E-11) not converged in the benchmark case but converged in the study case. These contingencies were not converged in the benchmark case due to voltage collapse in the areas of Coal Creek and Twin Cities. No mitigation plan is required for the study projects because these contingencies were converged in the study case.

Two category P2-P7 contingencies (Table E-11) were not converged in both the benchmark and study cases. No mitigation plan is required for the study projects for these non-converged contingencies.

For the non-converged contingencies in Table E-11, DC contingency analysis was performed to get the dc thermal results in the benchmark case. The dc thermal results for non-converged contingencies are listed in Table E-12.

For P2-P7 contingencies, thermal constraints are listed in Table E-9. No voltage constraints were identified (Table E-10).

4.4.2 Stage-2 ACCC Analysis Results for Summer Shoulder in Charging Condition

The charging condition includes four battery projects (J1032, J1041, J1045, and J1054) in charging condition.

4.4.2.1 System Intact Conditions

For NERC category P0 (system intact) conditions, thermal constraints are listed in Table E-13. No voltage constraints were identified (Table E-14).

4.4.2.2 Post Contingency Conditions

The results in this Section are for analysis of conditions following NERC Category P1-P7 contingencies. All NERC Category P1 contingencies were converged.

For P1 contingencies, thermal constraints are listed in Table E-15. No voltage constraints were identified (Table E-16).

There are three P2-P7 contingencies (Table E-19) not converged in the benchmark case but converged in the study case. These contingencies were not converged in the benchmark case due to voltage collapse in the areas of Coal Creek and Twin Cities. No mitigation plan is required for the study projects because these contingencies were converged in the study case.

Two category P2-P7 contingencies (Table E-19) were not converged in both the benchmark and study cases. No mitigation plan is required for the study projects for these non-converged contingencies.

For the non-converged contingencies in Table E-19, DC contingency analysis was performed to get the dc thermal results in the benchmark case. The dc thermal results for non-converged contingencies are listed in Table E-20.

For P2-P7 contingencies, thermal constraints are listed in Table E-17. No voltage constraints were identified (Table E-18).

4.4.3 Summer Shoulder Worst Thermal Constraints in the Stage-2 ACCC

Combining the results for both discharging and charging conditions, Table 4-2 lists worst thermal constraints and Network Upgrades identified in the Stage-2 ACCC for 2024 summer shoulder scenario.

Table 4-2: 2024 Summer Shoulder Thermal Constraints, Maximum Screened Loading, Stage-2 ACCC

Generator	Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Scenario	Mitigation
				(MVA)	(%)				
J1024	J1024 POI-J611 POI 161 kV	199.0	MEC	215.0	108.1	CEII Redacted	P1	SH Discharge	MTEP 18885 to reconductor line in 2020
J1024	J1024 POI-J611 POI 161 kV	199.0	MEC	226.5	113.8	CEII Redacted	P2-P7	SH Discharge	MTEP 18885 to reconductor line in 2020
J1024	J1024 POI-Clarinda 161 kV	199.0	MEC	275.9	138.6	CEII Redacted	P1	SH Charge	Substation terminal equipment upgrades
J1024	J1024 POI-Clarinda 161 kV	199.0	MEC	275.9	138.7	CEII Redacted	P2-P7	SH Discharge	Substation terminal equipment upgrades
J1041,J1057,J1086, J1164	J1057 POI-Heron Lake 161 kV	167.0	ITCM	231.1	138.4	CEII Redacted	P1	SH Discharge	MTEP 15 Project 8106
J1041,J1057,J1086, J1164	J1057 POI-Heron Lake 161 kV	167.0	ITCM	231.1	138.4	CEII Redacted	P2-P7	SH Discharge	MTEP 15 Project 8106
J1108	J1143 POI-Blanch Rd. 115 kV	90.0	MP	122.7	136.3	CEII Redacted	P1	SH Charge	Rebuild Blanchard - J1143 POI - Bellevue - Little Falls 115 kV (8.34 mi)

Summer Shoulder Steady-State Analysis

Generator	Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Scenario	Mitigation
				(MVA)	(%)				
J967,J1061&J1032, J1072,J1110,J1124, J1128,J1181	J1181 POI-Hazleton 345 kV	872.0	ITCM MEC	1045.9	119.9	CEII Redacted	P1	SH Charge	MEC: Structure replacements. New MEC Only rating expected to be 1154/1154 MVA. \$750K ITCM: Structure replacements. New ITCM rating 1151 SN/SE. \$840K
J967,J1072,J1110, J1124,J1128,J1181	J1181 POI-Hazleton 345 kV	872.0	ITCM MEC	1145.7	131.4	CEII Redacted	P2-P7	SH Discharge	MEC: Structure replacements. New MEC Only rating expected to be 1154/1154 MVA. \$750K ITCM: Structure replacements. New ITCM rating 1151 SN/SE. \$840K
J967,J1072,J1124	J1181 POI-Mitchell Co 345 kV	872.0	ITCM	952.2	109.2	CEII Redacted	P2-P7	SH Discharge	ITCM portion existing rating 1064 MVA SN/SE
J1040	J580 POI-J302 POI 230 kV	383.0	MDU	393.1	102.6	CEII Redacted	P1	SH Charge	Rebuild line to ACSS New Rating: 610 / 610 MVA
J1040	J580 POI-J302 POI 230 kV	383.0	MDU	502.2	131.1	CEII Redacted	P2-P7	SH Charge	Rebuild line to ACSS New Rating: 610 / 610 MVA
J1040	J580 POI-Heskett 230 kV	383.0	MDU	395.4	103.2	CEII Redacted	P1	SH Charge	J580 POI-Heskett 230 kV Rebuild
J1040	J580 POI-Heskett 230 kV	383.0	MDU	457.9	119.6	CEII Redacted	P2-P7	SH Charge	J580 POI-Heskett 230 kV Rebuild

Summer Shoulder Steady-State Analysis

Generator	Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Scenario	Mitigation
				(MVA)	(%)				
J1024	J611 POI-Maryville 161 kV	152.0	MEC GMO	254.9	167.7	CEII Redacted	P0	SH Discharge	MEC: Existing MEC only rating expected to be 410 MVA after DPP 2016 AUG West line reconductor network upgrade. GMO: NU is not required unless it is identified as constraint in affected system study.
J1024	J611 POI-Maryville 161 kV	171.0	MEC GMO	296.0	173.1	CEII Redacted	P1	SH Discharge	MEC: Existing MEC only rating expected to be 410 MVA after DPP 2016 AUG West line reconductor network upgrade. GMO: NU is not required unless it is identified as constraint in affected system study.
J1024	J611 POI-Maryville 161 kV	171.0	MEC GMO	304.3	178.0	CEII Redacted	P2-P7	SH Discharge	MEC: Existing MEC only rating expected to be 410 MVA after DPP 2016 AUG West line reconductor network upgrade. GMO: NU is not required unless it is identified as constraint in affected system study.
J982,J1114	J748 POI-Raun 345 kV	864.0	MEC	910.6	105.4	CEII Redacted	P1	SH Discharge	Structure replacements. New MEC Only rating expected to be 952/952 MVA.
J982,J1114	J748 POI-Raun 345 kV	864.0	MEC	941.6	109.0	CEII Redacted	P2-P7	SH Discharge	Structure replacements. New MEC Only rating expected to be 952/952 MVA.

Summer Shoulder Steady-State Analysis

Generator	Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Scenario	Mitigation
				(MVA)	(%)				
J966,J1177	Palmyra 345-161 kV xfmr	336.0	AECI	456.2	135.8	CEII Redacted	P1	SH Discharge	NU is not required unless it is identified as constraint in affected system study.
J966	Palmyra 345-161 kV xfmr	336.0	AECI	430.7	128.2	CEII Redacted	P2-P7	SH Discharge	NU is not required unless it is identified as constraint in affected system study.
J1024	Marysville 161-69 kV xfmr #2	50.0	AECI	50.8	101.6	CEII Redacted	P1	SH Discharge	NU is not required unless it is identified as constraint in affected system study.
J966,J1182	Novelty SW 161-69 kV xfmr	84.0	AECI	105.3	125.3	CEII Redacted	P1	SH Discharge	NU is not required unless it is identified as constraint in affected system study.
J966	Novelty SW 161-69 kV xfmr	84.0	AECI	106.4	126.7	CEII Redacted	P2-P7	SH Discharge	NU is not required unless it is identified as constraint in affected system study.
J966,J1177,J1182	South River-Emerson 161 kV	236.0	AECI	281.9	119.4	CEII Redacted	P1	SH Discharge	NU is not required unless it is identified as constraint in affected system study.
J966,J1177,J1182	South River-Emerson 161 kV	236.0	AECI	275.9	116.9	CEII Redacted	P2-P7	SH Discharge	NU is not required unless it is identified as constraint in affected system study.
J966,J1182	South River-Palmyra 161 kV	167.0	AECI	264.7	158.5	CEII Redacted	P1	SH Charge	NU is not required unless it is identified as constraint in affected system study.

Summer Shoulder Steady-State Analysis

Generator	Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Scenario	Mitigation
				(MVA)	(%)				
J966,J1177,J1182	South River-Palmyra 161 kV	167.0	AECI	259.2	155.2	CEII Redacted	P2-P7	SH Discharge	NU is not required unless it is identified as constraint in affected system study.
J966,J1182	Adair 161 kV bus 1-2 tie	366.0	AMRN	413.0	112.8	CEII Redacted	P1	SH Discharge	Invalid. Adair 161 kV bus tie breakers have been upgraded to 2000 A (558 MVA)
J966,J1182	Adair 161 kV bus 1-2 tie	366.0	AMRN	419.9	114.7	CEII Redacted	P2-P7	SH Discharge	Invalid. Adair 161 kV bus tie breakers have been upgraded to 2000 A (558 MVA)
J966,J1177,J1182	Adair 161 kV bus 2-3 tie	335.0	AMRN	555.9	165.9	CEII Redacted	P1	SH Discharge	Invalid. Adair 161 kV bus tie breakers have been upgraded to 2000 A (558 MVA)
J966,J1177,J1182	Adair 161 kV bus 2-3 tie	335.0	AMRN	562.0	167.8	CEII Redacted	P2-P7	SH Discharge	Invalid. Adair 161 kV bus tie breakers have been upgraded to 2000 A (558 MVA)
J966	Adair-Thomas Hill 161 kV	333.0	AMRN AECI	359.7	108.0	CEII Redacted	P1	SH Discharge	Rebuild to 3000 A
J966	Adair-Thomas Hill 161 kV	333.0	AMRN AECI	366.4	110.0	CEII Redacted	P2-P7	SH Discharge	Rebuild to 3000 A
J966,J1177,J1182	Adair-Novelty SW 161 kV	334.0	AMRN AECI	400.0	119.8	CEII Redacted	P1	SH Discharge	Limited by AECI equipment
J966,J1177,J1182	Adair-Novelty SW 161 kV	334.0	AMRN AECI	395.2	118.3	CEII Redacted	P2-P7	SH Discharge	Limited by AECI equipment

Summer Shoulder Steady-State Analysis

Generator	Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Scenario	Mitigation
				(MVA)	(%)				
J966,J1177,J1182	Montgomery-Spencer Crk 345 kV	1272.0	AMRN	1416.9	111.4	CEII Redacted	P2-P7	SH Discharge	Rebuild Montgomery-Spencer Creek-5503 345 kV line
J966	Adair 69 kV bus 1-2 tie	56.0	AMRN	57.9	103.5	CEII Redacted	P2-P7	SH Discharge	Adair 69 kV bus tie should be 72 MVA/600 A
J966,J1177	Herleman-Marblehead N 138 kV	287.0	AMRN	299.6	104.4	CEII Redacted	P1	SH Discharge	2nd Maywood-Herleman 345 kV line
J966,J1177	Marblehead N 161-138 kV xfmr	300.0	AMRN	328.5	109.5	CEII Redacted	P1	SH Discharge	2nd Maywood-Herleman 345 kV line
J966,J1177	Marblehead N-Palmyra 161 kV	278.0	AMRN AECI	330.6	118.9	CEII Redacted	P1	SH Discharge	2nd Maywood-Herleman 345 kV line
J966,J1177	Marblehead N-Palmyra 161 kV	278.0	AMRN AECI	290.3	104.4	CEII Redacted	P2-P7	SH Discharge	2nd Maywood-Herleman 345 kV line
J1024	Maryville-Midway 161 kV	171.0	GMO	173.2	101.3	CEII Redacted	P2-P7	SH Discharge	NU is not required unless it is identified as constraint in affected system study.
J951,J975,J1040, J1109,J1170,J1187	G16-017 Tap-Ft. Thompson 345 kV	717.0	WAPA	788.3	110.0	CEII Redacted	P0	SH Discharge	NU is not required unless it is identified as constraint in affected system study.
J997,J1040,J1108, J1109&J1170	G16-017 Tap-Ft. Thompson 345 kV	789.0	WAPA	819.1	103.8	CEII Redacted	P1	SH Discharge	NU is not required unless it is identified as constraint in affected

Summer Shoulder Steady-State Analysis

Generator	Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Scenario	Mitigation
				(MVA)	(%)				
									system study.
J1098	Wilmarth-Sheas Lake 345 kV	1515.8	XEL	1633.9	107.8	CEII Redacted	P2-P7	SH Charge	Wilmarth-Sheas Lake 345 kV Rebuild
J1006	Split Rock-White 345 kV	717.1	XEL WAPA	806.0	112.4	CEII Redacted	P1	SH Charge	XEL: Limiter is on WAPA facility. \$0 WAPA: NU is not required unless it is identified as constraint in affected system study.
J1006	Split Rock-White 345 kV	717.1	XEL WAPA	835.7	116.5	CEII Redacted	P2-P7	SH Charge	XEL: Limiter is on WAPA facility. \$0 WAPA: NU is not required unless it is identified as constraint in affected system study.
J958,J971,J977, J982,J1001,J1006, J1008,J1032,J1041, J1045,J1054,J1057, J1086,J1098,J1106, J1114,J1149,J1164, J1169	Blue Lake-Scott Co 345 kV	1378.0	XEL	1451.6	105.3	CEII Redacted	P0	SH Charge	Blue Lake-Scott County 345 kV Rebuild
J1106,J1174&J1175	Blue Lake-Scott Co 345 kV	1515.8	XEL	2110.6	139.2	CEII Redacted	P1	SH Charge	Blue Lake-Scott County 345 kV Rebuild
J1106	Blue Lake-Scott Co 345 kV	1515.8	XEL	2091.8	138.0	CEII Redacted	P2-P7	SH Charge	Blue Lake-Scott County 345 kV Rebuild
J977,J1041,J1057, J1086	Lakefield Tap-Lakefield 345 kV	1034.0	XEL ITCM	1075.6	104.0	CEII Redacted	P1	SH Charge	Rebuild 18.8 mi to 2x954 ACSS

Summer Shoulder Steady-State Analysis

Generator	Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Scenario	Mitigation
				(MVA)	(%)				
J971,J977,J1041, J1057,J1086,J1164	Lakefield Tap-Lakefield 345 kV	1034.0	XEL ITCM	1154.9	111.7	CEII Redacted	P2-P7	SH Charge	Rebuild 18.8 mi to 2x954 ACSS
J977,J1041,J1057, J1086,J1098	Field S-Crandal 345 kV	1332.6	XEL	1387.4	104.1	CEII Redacted	P1	SH Charge	Crandal-Fieldon 345 kV Rebuild
J971,J977,J1041, J1057,J1086,J1098	Field S-Crandal 345 kV	1332.6	XEL	1468.2	110.2	CEII Redacted	P2-P7	SH Charge	Crandal-Fieldon 345 kV Rebuild
J1032,J1061	Briggs Rd 345-161-34.5 kV xfmr #9	515.2	XEL	516.9	100.3	CEII Redacted	P1	SH Discharge	Add 2nd 448 MVA TR10
J1032,J1061	Briggs Rd 345-161-34.5 kV xfmr #9	515.2	XEL	515.5	100.1	CEII Redacted	P2-P7	SH Discharge	Add 2nd 448 MVA TR10
J971,J977,J982, J998,J1008,J1032, J1050,J1054,J1057, J1086,J1098,J1114, J1122,J1132,J1164,	Helena-Sheas Lk 345 kV	1195.1	XEL	1197.0	100.2	CEII Redacted	P0	SH Charge	Helena-Sheas Lake 345 kV Rebuild

Summer Shoulder Steady-State Analysis

Generator	Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Scenario	Mitigation
				(MVA)	(%)				
J1174,J1175									
J977,J982,J1008, J1098,J1114,J1128, J1174,J1175,J1177	Helena-Sheas Lk 345 kV	1195.1	XEL	1438.3	120.3	CEII Redacted	P1	SH Charge	Helena-Sheas Lake 345 kV Rebuild
J977,J982,J1008, J1098,J1114,J1174, J1175	Helena-Sheas Lk 345 kV	1195.1	XEL	1625.0	136.0	CEII Redacted	P2-P7	SH Charge	Helena-Sheas Lake 345 kV Rebuild
J1149	Hazel Crk 345-230-13.8 kV xfmr #9	514.7	XEL	593.4	115.3	CEII Redacted	P2-P7	SH Charge	Upgrade to 672 MVA transformer
J1041,J1057,J1086, J1164	Split Rock-Rock Co. 161 kV	224.2	XEL	225.8	100.7	CEII Redacted	P1	SH Discharge	Uprate with 795 ACSS
J1041,J1057,J1086, J1164	Split Rock-Rock Co. 161 kV	224.2	XEL	224.9	100.3	CEII Redacted	P2-P7	SH Discharge	Uprate with 795 ACSS
J1109&J1170	Sheyenne-Lake Park 230 kV	477.2	XEL OTP	490.2	102.7	CEII Redacted	P1	SH Charge	XEL: Uprate 3 miles of line owned by Xcel to 954 ACSR. \$3M OTP: line clearance mitigations. \$150K

Summer Shoulder Steady-State Analysis

Generator	Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Scenario	Mitigation
				(MVA)	(%)				
J975,J1109,J1170	Sheyenne-Maple River 230 kV	483.0	XEL MPC	491.2	101.7	CEII Redacted	P1	SH Charge	XEL: Rebuild 8.0 mi to 954 ACSR. \$14.4M MPC: Replace Bus Jumpers at Maple River. \$300K
J975,J1109,J1170	Sheyenne-Maple River 230 kV	483.0	XEL MPC	496.9	102.9	CEII Redacted	P2-P7	SH Charge	XEL: Rebuild 8.0 mi to 954 ACSR. \$14.4M MPC: Replace Bus Jumpers at Maple River. \$300K
J1108	Minn Valley Tap-Granite Falls 230 kV	259.0	XEL WAPA	270.0	104.3	CEII Redacted	P1	SH Charge	XEL: Summer normal and emergency ratings are 413 MVA WAPA: NU is not required unless it is identified as constraint in affected system study.
J1041,J1057,J1086, J1164	Split Rock 161-115-36.2 kV xfmr #6	187.0	XEL	226.7	121.2	CEII Redacted	P1	SH Discharge	Upgrade with 336 MVA
J1041,J1057,J1086, J1164	Split Rock 161-115-36.2 kV xfmr #6	187.0	XEL	225.5	120.6	CEII Redacted	P2-P7	SH Discharge	Upgrade with 336 MVA
J975,J1109,J1170	Sheyenne-Mapleton 115 kV	396.8	XEL OTP	452.6	114.1	CEII Redacted	P2-P7	SH Charge	Sheyenne-Mapleton 115 kV Rebuild

Summer Shoulder Steady-State Analysis

Generator	Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Scenario	Mitigation
				(MVA)	(%)				
J1108	Douglas Co 115-69 kV xfmr #2	80.5	XEL	88.6	110.0	CEII Redacted	P1	SH Charge	Replace TR1 to match TR2 with 80 MVA
J1108	Douglas Co-Hudson 115 kV	239.0	XEL GRE	312.3	130.7	CEII Redacted	P1	SH Charge	Rebuild 6.8 mi to 795 ACSS
J1108	Douglas Co-Little Sauk 115 kV	107.0	XEL GRE	162.2	151.6	CEII Redacted	P1	SH Charge	Rebuild 9.9 mi to 795 ACSS
J1108	Quarry 345-115-34.5 kV xfmr #9	515.2	XEL	629.5	122.2	CEII Redacted	P1	SH Charge	Add 2nd 448 MVA TR10
J1108	Quarry 345-115-34.5 kV xfmr #9	515.2	XEL	628.7	122.0	CEII Redacted	P2-P7	SH Charge	Add 2nd 448 MVA TR10
J1108	Douglas Co-West Union 69 kV	71.7	XEL GRE	112.3	156.6	CEII Redacted	P1	SH Charge	Rebuild 5.4 mi to 795 ACSR
J1108	Blackoak-Grove 69 kV	57.2	XEL GRE	100.1	174.9	CEII Redacted	P1	SH Charge	Rebuild 0.2 mi to 795 ACSR
J1108	Riverton-Mud Lake 230 kV	391.0	MP GRE	396.2	101.3	CEII Redacted	P1	SH Charge	MP: No MP limiting equipment, 414.0/414.0 SN/SE. \$0 GRE: Rebuild 8.6 total miles of line. \$4.1M
J1108	Pepin Lake-Long Prairie 115 kV	107.0	MP	126.8	118.5	CEII Redacted	P1	SH Charge	Rebuild Pepin Lake-Long Prairie 115 kV (12.77 miles)
J1108	Pepin Lake-Blanchard 115 kV	99.0	MP	122.4	123.7	CEII Redacted	P1	SH Charge	Rebuild Blanchard - Pepin Lake 115 kV (14.23 miles)

Summer Shoulder Steady-State Analysis

Generator	Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Scenario	Mitigation
				(MVA)	(%)				
J1128	Austin-Murphy 161 kV	331.3	SMMPA	378.7	114.3	CEII Redacted	P2-P7	SH Charge	Austin-Murphy Creek 161 kV Rebuild
J1008,J1128	Murphy-J1128 POI 161 kV	276.0	SMMPA	278.6	100.9	CEII Redacted	P0	SH Charge	Murphy Creek–J1128 POI 161 kV Rebuild
J977,J1128	Murphy-J1128 POI 161 kV	276.0	SMMPA	350.8	127.1	CEII Redacted	P1	SH Charge	Murphy Creek–J1128 POI 161 kV Rebuild
J1128	Murphy-J1128 POI 161 kV	276.0	SMMPA	399.2	144.6	CEII Redacted	P2-P7	SH Charge	Murphy Creek–J1128 POI 161 kV Rebuild
J977	Rutland-Fox Lake 161 kV	195.0	ITCM	252.7	129.6	CEII Redacted	P1	SH Discharge	Rebuild 8.9 miles

Summer Shoulder Steady-State Analysis

Generator	Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Scenario	Mitigation
				(MVA)	(%)				
J977	Rutland-Fox Lake 161 kV	195.0	ITCM	263.5	135.1	CEII Redacted	P2-P7	SH Discharge	Rebuild 8.9 miles
J977	Rutland-Huntley 161 kV	218.0	ITCM	220.2	101.0	CEII Redacted	P2-P7	SH Discharge	Rebuild 0.73 miles
J1108	Inman-Wing River 230 kV	318.6	GRE	410.6	128.9	CEII Redacted	P1	SH Charge	Rebuild line, replace jumpers, CT, hookstick switches
J1038	Inman-Wing River 230 kV	318.6	GRE	381.9	119.9	CEII Redacted	P2-P7	SH Charge	Rebuild line, replace jumpers, CT, hookstick switches
J1038	Inman-Henning 230 kV	343.7	GRE	406.1	118.1	CEII Redacted	P2-P7	SH Charge	Rebuild line, replace jumpers at terminals

Summer Shoulder Steady-State Analysis

Generator	Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Scenario	Mitigation
				(MVA)	(%)				
J1108	Hubbard-Erie Jct 230 kV	380.5	GRE OTP MP	397.1	104.4	CEII Redacted	P1	SH Charge	GRE: GRE equipment at Hubbard is rated 522.6 MVA. \$0 OTP: Sufficient for flows seen in study. \$0 MP: Reconductor on MP's segment. \$2.265M
J1140	West St. Cloud-Westwood 115 kV	396.8	GRE XEL	400.7	101.0	CEII Redacted	P1	SH Charge	Rebuild 0.6 mi to 2x795 ACSS
J1140	West St. Cloud-Westwood 115 kV	396.8	GRE XEL	400.1	100.8	CEII Redacted	P2-P7	SH Charge	Rebuild 0.6 mi to 2x795 ACSS
J1108,J1140,J1143	West St. Cloud-Little Sauk Tap 115 kV	146.6	GRE XEL	170.9	116.6	CEII Redacted	P1	SH Charge	GRE: XEL owns the circuit and equipmnet. \$0 XEL: Uprate line to 795 ACSS. \$2.1M
J1140,J1143	West St. Cloud-Little Sauk Tap 115 kV	146.6	GRE XEL	171.1	116.7	CEII Redacted	P2-P7	SH Charge	GRE: XEL owns the circuit and equipmnet. \$0 XEL: Uprate line to 795 ACSS. \$2.1M
J1108	Wing River-Riverton 230 kV	268.0	GRE MP	307.2	114.6	CEII Redacted	P1	SH Charge	GRE: Increase conductor clearance to 75C on GRE portion (13.3 miles). 327.0/359.0 SN/SE. \$4M MP: Increase conductor clearance to 75C on MP portion of Riverton-Wing River 230 kV (35.97 miles). 327.0/359.0 SN/SE. \$9.72M

Summer Shoulder Steady-State Analysis

Generator	Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Scenario	Mitigation
				(MVA)	(%)				
J1038,J1109&J1170	Silver Lake-Henning 230 kV	334.0	GRE MRES OTP	347.8	104.1	CEII Redacted	P1	SH Charge	GRE: Replace jumpers at Henning. \$50K OTP: line clearance mitigations. \$650k MRES: Phase raisers. \$1M
J1038	Silver Lake-Henning 230 kV	334.0	GRE MRES OTP	416.2	124.6	CEII Redacted	P2-P7	SH Charge	GRE: Replace jumpers at Henning. \$50K OTP: line clearance mitigations. \$650k MRES: Phase raisers. \$1M
J1038,J1109&J1170	Silver Lake-Fergus Falls 230 kV	337.0	GRE OTP MRES	355.9	105.6	CEII Redacted	P1	SH Charge	GRE: GRE equipment at Silver Lake is rated 796.7 MVA. \$0 OTP: Line clearance mitigations. \$400K MRES: Phase raisers. \$60K
J1038	Silver Lake-Fergus Falls 230 kV	337.0	GRE OTP MRES	424.9	126.1	CEII Redacted	P2-P7	SH Charge	GRE: GRE equipment at Silver Lake is rated 796.7 MVA. \$0 OTP: Line clearance mitigations. \$400K MRES: Phase raisers. \$60K
J1105,J1054	Chub Lake 345-115-34.5 kV xfmr	448.0	GRE	618.8	138.1	CEII Redacted	P2-P7	SH Charge	Add second 345/115 kV transformer at Chub Lake

Summer Shoulder Steady-State Analysis

Generator	Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Scenario	Mitigation
				(MVA)	(%)				
J1143	J1143 POI-Bellvu 115 kV	90.0	GRE MP	90.9	101.0	CEII Redacted	P0	SH Charge	Rebuild Blanchard - J1143 POI - Bellevue - Little Falls 115 kV (8.34 mi)
J1108,J1143	J1143 POI-Bellvu 115 kV	90.0	GRE MP	153.2	170.2	CEII Redacted	P1	SH Charge	Rebuild Blanchard - J1143 POI - Bellevue - Little Falls 115 kV (8.34 mi)
J1143	J1143 POI-Bellvu 115 kV	90.0	GRE MP	153.3	170.4	CEII Redacted	P2-P7	SH Charge	Rebuild Blanchard - J1143 POI - Bellevue - Little Falls 115 kV (8.34 mi)
J1108,J1143	Bellvu-Little Falls 115 kV	90.0	GRE MP	151.6	168.4	CEII Redacted	P1	SH Charge	Rebuild Blanchard - J1143 POI - Bellevue - Little Falls 115 kV (8.34 mi)
J1143	Bellvu-Little Falls 115 kV	90.0	GRE MP	151.7	168.6	CEII Redacted	P2-P7	SH Charge	Rebuild Blanchard - J1143 POI - Bellevue - Little Falls 115 kV (8.34 mi)

Summer Shoulder Steady-State Analysis

Generator	Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Scenario	Mitigation
				(MVA)	(%)				
J1108	Little Sauk Tap-Long Prairie 115 kV	107.0	GRE MP	160.0	149.6	CEII Redacted	P1	SH Charge	GRE: GRE tap switch (Little Sauk) is rated 239 MVA. \$0 MP: Rebuild MP segment of Long Prairie-GRE Little Sauk 115 kV (9.34 miles). 185.0/203.0 SN/SE. \$4.67M XEL: Don't own anything on that section of line
J1108,J1140,J1143	Lesauk Tap-Fishill 115 kV	128.3	GRE XEL	175.6	136.8	CEII Redacted	P1	SH Charge	GRE: XEL owns equipment and line. \$0 XEL: Uprate line to 795 ACSS. \$4.5M
J1140,J1143	Lesauk Tap-Fishill 115 kV	128.3	GRE XEL	175.7	137.0	CEII Redacted	P2-P7	SH Charge	GRE: XEL owns equipment and line. \$0 XEL: Uprate line to 795 ACSS. \$4.5M
J1108,J1140,J1143	St. Stephens-Fishill 115 kV	124.5	GRE XEL	179.8	144.5	CEII Redacted	P1	SH Charge	GRE: XEL owns equipment and line. \$0 XEL: Uprate line to 795 ACSS. \$5.1M
J1140,J1143	St. Stephens-Fishill 115 kV	124.5	GRE XEL	180.0	144.6	CEII Redacted	P2-P7	SH Charge	GRE: XEL owns equipment and line. \$0 XEL: Uprate line to 795 ACSS. \$5.1M

Summer Shoulder Steady-State Analysis

Generator	Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Scenario	Mitigation
				(MVA)	(%)				
J1108	West Union-Sauk Center Muni 69 kV	38.0	GRE XEL	110.8	291.5	CEII Redacted	P1	SH Charge	Rebuild 8.0 mi to 795 ACSR
J1108	Kandota Tap-Sauk Center Muni 69 kV	39.6	GRE XEL	103.4	261.2	CEII Redacted	P1	SH Charge	Rebuild 0.9 mi to 795 ACSR
J1108	Kandota Tap-Grove 69 kV	53.2	GRE XEL	101.4	190.6	CEII Redacted	P1	SH Charge	GRE: Replace Kandota 3-way switch (1200 amp). \$385K XEL: Rebuild to 795 ACSR, \$2.5M
J1143	Aldrich-Verndale 115 kV	77.0	GRE MP	78.5	101.9	CEII Redacted	P1	SH Charge	Increase conductor clearance to 65C on Dog Lake - Verndale 115 kV (16.99 miles)
J975,J1109,J1170	CSLTNET-Mapleton 115 kV	418.4	OTP	459.3	109.8	CEII Redacted	P2-P7	SH Charge	Rebuild of ~12 miles of 115 kV line and terminal equipment replacement.
J975,J1109,J1170	CSLTNET-Buffalo 115 kV	418.0	OTP	484.0	115.8	CEII Redacted	P2-P7	SH Charge	Rebuild of ~16 miles of 115 kV line and terminal equipment replacement.
J1038	Hoot Lake-J1038 POI 115 kV	104.0	OTP	162.4	156.2	CEII Redacted	P1	SH Discharge	Rebuild of ~2 miles of 115 kV line and terminal equipment replacement.
J1038	Hoot Lake-J1038 POI 115 kV	104.0	OTP	162.2	155.9	CEII Redacted	P2-P7	SH Discharge	Rebuild of ~2 miles of 115 kV line and terminal equipment replacement.

Summer Shoulder Steady-State Analysis

Generator	Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Scenario	Mitigation
				(MVA)	(%)				
J951	Donalds-J951 POI 115 kV	116.0	OTP	145.5	125.4	CEII Redacted	P1	SH Charge	Line clearance mitigations and terminal equipment replacement.
J951	Donalds-J951 POI 115 kV	116.0	OTP	136.6	117.7	CEII Redacted	P2-P7	SH Charge	Line clearance mitigations and terminal equipment replacement.
J1108,J1109&J1170	Erie Jct-Audubon 230 kV	380.5	OTP XEL	402.6	105.8	CEII Redacted	P1	SH Charge	OTP: Clearance mitigations. \$150K XEL: Clearance mitigations. \$300K
J997,J1040	Hankinson-Forman 230 kV	432.1	OTP	470.0	108.8	CEII Redacted	P1	SH Charge	Clearance mitigations.
J1109&J1170	Wahpeton-Fergus Falls 230 kV	379.1	OTP	425.3	112.2	CEII Redacted	P1	SH Charge	Wahpeton-Fergus Falls 230 kV Uprate
J1109&J1170	Audubon-Lake Park 230 kV	469.5	OTP	482.2	102.7	CEII Redacted	P1	SH Charge	Clearance mitigations.
J975,J1109,J1170	Maple River 345-230-13.8 kV xfmr #1	420.0	OTP MPC	494.1	117.6	CEII Redacted	P1	SH Charge	Replace the #1 transformer with a larger transformer.
J975,J1109,J1170	Maple River 345-230-13.8 kV xfmr #1	420.0	OTP MPC	493.1	117.4	CEII Redacted	P2-P7	SH Charge	Replace the #1 transformer with a larger transformer.
J975,J1109,J1170	Maple River 345-230-13.8 kV xfmr #2	420.0	OTP MPC	497.1	118.4	CEII Redacted	P1	SH Charge	Replace the #2 transformer with a larger transformer.

Summer Shoulder Steady-State Analysis

Generator	Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Scenario	Mitigation
				(MVA)	(%)				
J975,J1109,J1170	Maple River 345-230-13.8 kV xfmr #2	420.0	OTP MPC	496.1	118.1	CEII Redacted	P2-P7	SH Charge	Replace the #2 transformer with a larger transformer.
J1128	Ellendale-County Line 69 kV	48.0	ITCM	55.1	114.7	CEII Redacted	P1	SH Charge	Rebuild 5.79 miles
J1128	Hayward-County Line 69 kV	48.0	ITCM	57.1	118.9	CEII Redacted	P1	SH Charge	Rebuild 13.32 miles
J1132	Osceola-Osceola REC 69 kV	38.0	ITCM CIPCO	39.7	104.5	CEII Redacted	P0	SH Discharge	ITCM: ITCM rating 42/44 MVA SN/SE CIPCO: NU is not required unless identified in affected system study
J1041,J1057,J1086,J1164	Heron Lake-Lakefield 161 kV	167.0	ITCM	167.2	100.1	CEII Redacted	P0	SH Discharge	MTEP 15 Project 8106 and replace wavetrap
J1041,J1057,J1086,J1164	Heron Lake-Lakefield 161 kV	167.0	ITCM	244.4	146.4	CEII Redacted	P1	SH Discharge	MTEP 15 Project 8106 and replace wavetrap
J1041,J1057,J1086,J1164	Heron Lake-Lakefield 161 kV	167.0	ITCM	244.4	146.4	CEII Redacted	P2-P7	SH Discharge	MTEP 15 Project 8106 and replace wavetrap
J1174,J1175	Lime Crk-Colby 161 kV	438.0	ITCM	467.0	106.6	CEII Redacted	P2-P7	SH Discharge	Rebuild 4.39 miles

Summer Shoulder Steady-State Analysis

Generator	Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Scenario	Mitigation
				(MVA)	(%)				
J1177	Ottumwa 345-161 kV xfmr	534.0	ITCM	551.4	103.3	CEII Redacted	P1	SH Discharge	Add second 345/161 kV transformer
J1177	Ottumwa 345-161 kV xfmr	534.0	ITCM	572.2	107.2	CEII Redacted	P2-P7	SH Discharge	Add second 345/161 kV transformer
J1128	Adams-Hayward 161 kV	233.0	ITCM	263.3	113.0	CEII Redacted	P1	SH Charge	Adams-Hayward 161 kV Uprate
J1110,J1128	Adams-Hayward 161 kV	233.0	ITCM	263.3	113.0	CEII Redacted	P2-P7	SH Charge	Adams-Hayward 161 kV Uprate
J1061&J1032	Adams-Beaver Creek 161 kV	264.0	ITCM DPC	270.0	102.3	CEII Redacted	P1	SH Discharge	Rebuild line with 795 ACSS conductor
J1181	Hazleton-Arnold 345 kV	923.0	ITCM MEC	1053.3	114.1	CEII Redacted	P1	SH Discharge	MEC: Structure replacements. New MEC Only rating expected to be 1229/1229 MVA. \$750K ITCM: Structure replacements. New ITCM rating 1285 MVA/SN/SE. \$480K
J959,J967,J1072,J1181	Hazleton-Arnold 345 kV	923.0	ITCM MEC	1220.7	132.3	CEII Redacted	P2-P7	SH Discharge	MEC: Structure replacements. New MEC Only rating expected to be 1229/1229 MVA. \$750K ITCM: Structure replacements. New ITCM rating 1285 MVA/SN/SE.

Summer Shoulder Steady-State Analysis

Generator	Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Scenario	Mitigation
				(MVA)	(%)				
									\$480K
J1181	Hazleton-Hickory Crk 345 kV	1195.0	ITCM	1308.3	109.5	CEII Redacted	P1	SH Discharge	ITCM rating 1569 MVA SN/SE
J1181	Hazleton-Hickory Crk 345 kV	1195.0	ITCM	1324.9	110.9	CEII Redacted	P2-P7	SH Discharge	ITCM rating 1569 MVA SN/SE
J1174,J1175	Colby 345-161 kV xfmr	558.0	ITCM	644.1	115.4	CEII Redacted	P1	SH Discharge	Add second 345/161 kV transformer
J1174&J1175	Webster-Wright 161 kV	212.0	MEC	246.9	116.5	CEII Redacted	P1	SH Discharge	Reconductor line and substation terminal equipment upgrades. New rating predicted to be 315/335 MVA.
J1040	Ward-Bismark 230 kV	352.0	BEPC WAPA	376.1	106.8	CEII Redacted	P1	SH Charge	NU is not required unless it is identified as constraint in affected system study.
J1040	Ward-Bismark 230 kV	352.0	BEPC WAPA	412.2	117.1	CEII Redacted	P2-P7	SH Charge	NU is not required unless it is identified as constraint in affected system study.
J951,J975,J1040,J1109,J1170,J1187	Ft. Thompson 345 kV bus 1-3 TIE	717.0	WAPA	787.0	109.8	CEII Redacted	P0	SH Discharge	NU is not required unless it is identified as constraint in affected system study.
J997,J1040,J1108,J1109&J1170,J1187	Ft. Thompson 345 kV bus 1-3 TIE	717.0	WAPA	817.9	114.1	CEII Redacted	P1	SH Discharge	NU is not required unless it is identified as constraint in affected system study.

Summer Shoulder Steady-State Analysis

Generator	Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Scenario	Mitigation
				(MVA)	(%)				
J951,J975,J1109,J1170	Wilton-Winger 230 kV	288.0	MPC OTP	334.8	116.3	CEII Redacted	P0	SH Charge	OTP: Sufficient for the flows of 395.2 MVA. \$0 MPC: Raise structures to 90 C. \$350K
J951,J975,J997,J1040,J1108,J1109&J1170	Wilton-Winger 230 kV	288.0	MPC OTP	395.2	137.2	CEII Redacted	P1	SH Charge	OTP: Sufficient for the flows of 395.2 MVA. \$0 MPC: Raise structures to 90 C. \$350K
J951	Drayton 230-115-13.8 kV xfmr	161.0	MPC	169.9	105.6	CEII Redacted	P2-P7	SH Charge	MPC: Generator offloads line. \$0
J1040	Ellendale-Aberdeen 115 kV	128.0	MDU NWE	139.4	108.9	CEII Redacted	P1	SH Charge	MDU's rating for Ellendale is 142 MVA. NWPS owns line and equipment at Aberdeen.
J1040	Mandan-Ward 230 kV	391.0	MDU BEPC	400.0	102.3	CEII Redacted	P1	SH Charge	MDU's rating at Mandan is 956 MVA. MPC owns equipment at Mandan. WAPA owns the line. BEPC owns Ward.
J1040	Mandan-Ward 230 kV	391.0	MDU BEPC	436.2	111.5	CEII Redacted	P2-P7	SH Charge	MDU's rating at Mandan is 956 MVA. MPC owns equipment at Mandan. WAPA owns the line. BEPC owns Ward.
J1040	Fox Tail-Tatanka North 230 kV	478.0	MDU	501.8	105.0	CEII Redacted	P1	SH Charge	Replace Breakers. New Rating: 478/526 MVA

Summer Shoulder Steady-State Analysis

Generator	Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Scenario	Mitigation
				(MVA)	(%)				
J1040	Fox Tail-Tatanka North 230 kV	478.0	MDU	502.9	105.2	CEII Redacted	P2-P7	SH Charge	Replace Breakers. New Rating: 478/526 MVA
J1040	Merricourt-Tatanka North 230 kV	343.0	MDU	411.3	119.9	CEII Redacted	P1	SH Charge	Rebuild line to ACSS. New Rating: 478/478 MVA
J1040	Merricourt-Tatanka North 230 kV	343.0	MDU	412.3	120.2	CEII Redacted	P2-P7	SH Charge	Rebuild line to ACSS. New Rating: 478/478 MVA
J1040	Wishek-J302 POI 230 kV	343.0	MDU	354.9	103.5	CEII Redacted	P2-P7	SH Charge	J302 POI-Wishek 230 kV Rebuild
J1061&J1032, J1072&J967,J1124, J1181	Wabaco-Rochester 161 kV	221.1	DPC	401.0	181.4	CEII Redacted	P1	SH Discharge	Rebuild line with 795 ACSS conductor. \$11M. MTEP Appendix A project
J967,J1181	Wabaco-Rochester 161 kV	221.1	DPC	289.6	131.0	CEII Redacted	P2-P7	SH Charge	Rebuild line with 795 ACSS conductor. \$11M. MTEP Appendix A project
J1061&J1032,J1181	Wabaco-Alma 161 kV	291.0	DPC	378.9	130.2	CEII Redacted	P1	SH Discharge	Rebuild line with 795 ACSS conductor

4.5 Network Upgrades Identified in MISO ERIIS Analysis for 2024 Summer Shoulder Scenario

Based on the MISO 2024 summer shoulder steady state analyses, the MISO Base Case NU and cost are listed in Table 4-3, and additional thermal NUs and cost are listed in Table 4-4. No reactive power NUs are required.

Table 4-3: MISO Base Case NUs

NUs	Miles	Cost (\$)
Hazel Creek-Scott County 345 kV	115	\$210,829,263
Total		\$210,829,263

Table 4-4: Additional Thermal NUs

Constraint	Owner	Mitigation	Cost (\$)
J1024 POI-J611 POI 161 kV	MEC	MTEP 18885 to reconductor line in 2020	\$0
J1024 POI-Clarinda 161 kV	MEC	Substation terminal equipment upgrades	\$700,000
J1057 POI-Heron Lake 161 kV	ITCM	MTEP 15 Project 8106	\$0
J1143 POI-Blanch Rd. 115 kV	MP	Rebuild Blanchard - J1143 POI - Bellevue - Little Falls 115 kV (8.34 mi)	\$4,170,000 ¹
J1181 POI-Hazleton 345 kV	ITCM MEC	MEC: Structure replacements. New MEC Only rating expected to be 1154/1154 MVA. \$750K ITCM: Structure replacements. New ITCM rating 1151 SN/SE. \$840K	\$1,590,000
J1181 POI-Mitchell Co 345 kV	ITCM	ITCM portion existing rating 1064 MVA SN/SE	\$0
J580 POI-J302 POI 230 kV	MDU	Rebuild line to ACSS New Rating: 610 / 610 MVA	\$2,300,000 ²
J580 POI-Heskett 230 kV	MDU	J580 POI-Heskett 230 kV Rebuild	\$4,500,000 ²
J611 POI-Maryville 161 kV	MEC GMO	MEC: Existing MEC only rating expected to be 410 MVA after DPP 2016 AUG West line reconductor network upgrade. GMO: NU is not required unless it is identified as constraint in affected system study.	\$0
J748 POI-Raun 345 kV	MEC	Structure replacements. New MEC Only rating expected to be 952/952 MVA.	\$300,000
Palmyra 345-161 kV xfmr	AECI	NU is not required unless it is identified as constraint in affected system study.	\$0
Marysville 161-69 kV xfmr #2	AECI	NU is not required unless it is identified as constraint in affected system study.	\$0

Constraint	Owner	Mitigation	Cost (\$)
Novelty SW 161-69 kV xfmr	AECI	NU is not required unless it is identified as constraint in affected system study.	\$0
South River-Emerson 161 kV	AECI	NU is not required unless it is identified as constraint in affected system study.	\$0
South River-Palmyra 161 kV	AECI	NU is not required unless it is identified as constraint in affected system study.	\$0
Adair 161 kV bus 1-2 tie	AMRN	Invalid. Adair 161 kV bus tie breakers have been upgraded to 2000 A (558 MVA)	\$0
Adair 161 kV bus 2-3 tie	AMRN	Invalid. Adair 161 kV bus tie breakers have been upgraded to 2000 A (558 MVA)	\$0
Adair-Thomas Hill 161 kV	AMRN AECI	Rebuild to 3000 A	\$28,600,000
Adair-Novelty SW 161 kV	AMRN AECI	Limited by AECI equipment	\$0
Montgomery-Spencer Crk 345 kV	AMRN	Rebuild Montgomery-Spencer Creek-5503 345 kV line	\$35,000,000
Adair 69 kV bus 1-2 tie	AMRN	Adair 69 kV bus tie should be 72 MVA/600 A	\$0
Herleman-Marblehead N 138 kV	AMRN	2nd Maywood-Herleman 345 kV line	\$21,000,000
Marblehead N 161-138 kV xfmr	AMRN	2nd Maywood-Herleman 345 kV line	Incl.
Marblehead N-Palmyra 161 kV	AMRN AECI	2nd Maywood-Herleman 345 kV line	Incl.
Maryville-Midway 161 kV	GMO	NU is not required unless it is identified as constraint in affected system study.	\$0
G16-017 Tap-Ft. Thompson 345 kV	WAPA	NU is not required unless it is identified as constraint in affected system study.	\$0
Wilmarth-Sheas Lake 345 kV	XEL	Wilmarth-Sheas Lake 345 kV Rebuild	\$49,400,000 ²
Split Rock-White 345 kV	XEL WAPA	XEL: Limiter is on WAPA facility. \$0 WAPA: NU is not required unless it is identified as constraint in affected system study.	\$0
Blue Lake-Scott Co 345 kV	XEL	Blue Lake-Scott County 345 kV Rebuild	\$24,000,000 ²
Lakefield Tap-Lakefield 345 kV	XEL ITCM	Rebuild 18.8 mi to 2x954 ACSS	\$37,600,000
Field S-Crandal 345 kV	XEL	Crandal-Fieldon 345 kV Rebuild	\$40,000,000 ²
Briggs Rd 345-161-34.5 kV xfmr #9	XEL	Add 2nd 448 MVA TR10	\$4,000,000
Helena-Sheas Lk 345 kV	XEL	Helena-Sheas Lake 345 kV Rebuild	\$15,000,000 ²

Constraint	Owner	Mitigation	Cost (\$)
Hazel Crk 345-230-13.8 kV xfmr #9	XEL	Upgrade to 672 MVA transformer	\$3,500,000
Split Rock-Rock Co. 161 kV	XEL	Upgrade with 795 ACSS	\$21,900,000 ¹
Sheyenne-Lake Park 230 kV	XEL OTP	XEL: Upgrade 3 miles of line owned by Xcel to 954 ACSR. \$3M OTP: line clearance mitigations. \$150K	\$3,150,000
Sheyenne-Maple River 230 kV	XEL MPC	XEL: Rebuild 8.0 mi to 954 ACSR. \$14.4M MPC: Replace Bus Jumpers at Maple River. \$300K	\$14,400,000
Minn Valley Tap-Granite Falls 230 kV	XEL WAPA	XEL: Summer normal and emergency ratings are 413 MVA WAPA: NU is not required unless it is identified as constraint in affected system study.	\$0
Split Rock 161-115-36.2 kV xfmr #6	XEL	Upgrade with 336 MVA	\$2,000,000 ¹
Sheyenne-Mapleton 115 kV	XEL OTP	Sheyenne-Mapleton 115 kV Rebuild	\$22,200,000 ²
Douglas Co 115-69 kV xfmr #2	XEL	Replace TR1 to match TR2 with 80 MVA	\$1,200,000
Douglas Co-Hudson 115 kV	XEL GRE	Rebuild 6.8 mi to 795 ACSS	\$10,390,000
Douglas Co-Little Sauk 115 kV	XEL GRE	Rebuild 9.9 mi to 795 ACSS	\$14,850,000
Quarry 345-115-34.5 kV xfmr #9	XEL	Add 2nd 448 MVA TR10	\$4,000,000
Douglas Co-West Union 69 kV	XEL GRE	Rebuild 5.4 mi to 795 ACSR	\$3,240,000
Blackoak-Grove 69 kV	XEL GRE	Rebuild 0.2 mi to 795 ACSR	\$120,000
Riverton-Mud Lake 230 kV	MP GRE	MP: No MP limiting equipment, 414.0/414.0 SN/SE. \$0 GRE: Rebuild 8.6 total miles of line. \$4.1M	\$4,100,000
Pepin Lake-Long Prairie 115 kV	MP	Rebuild Pepin Lake-Long Prairie 115 kV (12.77 miles)	6,385,000
Pepin Lake-Blanchard 115 kV	MP	Rebuild Blanchard - Pepin Lake 115 kV (14.23 miles)	7,115,000
Austin-Murphy 161 kV	SMMPA	Austin-Murphy Creek 161 kV Rebuild	\$1,875,000 ²
Murphy-J1128 POI 161 kV	SMMPA	Murphy Creek-J1128 POI 161 kV Rebuild	\$5,812,500 ²
Rutland-Fox Lake 161 kV	ITCM	Rebuild 8.9 miles	\$9,800,000
Rutland-Huntley 161 kV	ITCM	Rebuild 0.73 miles	\$800,000
Inman-Wing River 230 kV	GRE	Rebuild line, replace jumpers, CT, hookstick switches	\$13,100,000
Inman-Henning 230 kV	GRE	Rebuild line, replace jumpers at terminals	\$2,800,000
Hubbard-Erie Jct 230 kV	GRE OTP	GRE: GRE equipment at Hubbard is rated 522.6 MVA. \$0 OTP: Sufficient for flows seen in study. \$0	\$2,265,000

Constraint	Owner	Mitigation	Cost (\$)
	MP	MP: Reconductor on MP's segment. \$2.265M	
West St. Cloud-Westwood 115 kV	GRE XEL	Rebuild 0.6 mi to 2x795 ACSS	\$900,000
West St. Cloud-Little Sauk Tap 115 kV	GRE XEL	GRE: XEL owns the circuit and equipmnet. \$0 XEL: Uprate line to 795 ACSS. \$2.1M	\$2,100,000
Wing River-Riverton 230 kV	GRE MP	GRE: Increase conductor clearance to 75C on GRE portion (13.3 miles). 327.0/359.0 SN/SE. \$4M MP: Increase conductor clearance to 75C on MP portion of Riverton-Wing River 230 kV (35.97 miles). 327.0/359.0 SN/SE. \$9.72M	\$13,720,000
Silver Lake-Henning 230 kV	GRE MRES OTP	GRE: Replace jumpers at Henning. \$50K OTP: line clearance mitigations. \$650k MRES: Phase raisers. \$1M	\$1,700,000
Silver Lake-Fergus Falls 230 kV	GRE OTP MRES	GRE: GRE equipment at Silver Lake is rated 796.7 MVA. \$0 OTP: Line clearance mitigations. \$400K MRES: Phase raisers. \$60K	\$460,000
Chub Lake 345-115-34.5 kV xfmr	GRE	Add second 345/115 kV transformer at Chub Lake	\$11,400,000
J1143 POI-Bellevu 115 kV	GRE MP	Rebuild Blanchard - J1143 POI - Bellevue - Little Falls 115 kV (8.34 mi)	Incl.
Bellevu-Little Falls 115 kV	GRE MP	Rebuild Blanchard - J1143 POI - Bellevue - Little Falls 115 kV (8.34 mi)	Incl.
Little Sauk Tap-Long Prairie 115 kV	GRE MP	GRE: GRE tap switch (Little Sauk) is rated 239 MVA. \$0 MP: Rebuild MP segment of Long Prairie-GRE Little Sauk 115 kV (9.34 miles). 185.0/203.0 SN/SE. \$4.67M XEL: Don't own anything on that section of line	\$4,670,000
Lesauk Tap-Fishill 115 kV	GRE XEL	GRE: XEL owns equipment and line. \$0 XEL: Uprate line to 795 ACSS. \$4.5M	\$4,500,000
St. Stephens-Fishill 115 kV	GRE XEL	GRE: XEL owns equipment and line. \$0 XEL: Uprate line to 795 ACSS. \$5.1M	\$5,100,000
West Union-Sauk Center Muni 69 kV	GRE XEL	Rebuild 8.0 mi to 795 ACSR	\$4,800,000
Kandota Tap-Sauk Center Muni 69 kV	GRE XEL	Rebuild 0.9 mi to 795 ACSR	\$540,000
Kandota Tap-Grove 69 kV	GRE XEL	GRE: Replace Kandota 3-way switch (1200 amp). \$385K XEL: Rebuild to 795 ACSR, \$2.5M	\$2,885,000
Aldrich-Verndale 115 kV	GRE MP	Increase conductor clearance to 65C on Dog Lake - Verndale 115 kV (16.99 miles)	\$1,020,000
CSLTNET-Mapleton 115 kV	OTP	Rebuild of ~12 miles of 115 kV line and terminal equipment replacement.	\$7,000,000
CSLTNET-Buffalo 115 kV	OTP	Rebuild of ~16 miles of 115 kV line and terminal equipment replacement.	\$9,900,000

Constraint	Owner	Mitigation	Cost (\$)
Hoot Lake-J1038 POI 115 kV	OTP	Rebuild of ~2 miles of 115 kV line and terminal equipment replacement.	\$1,250,000
Donalds-J951 POI 115 kV	OTP	Line clearance mitigations and terminal equipment replacement.	\$500,000
Erie Jct-Audubon 230 kV	OTP XEL	OTP: Clearance mitigations. \$150K XEL: Clearance mitigations. \$300K	\$450,000 ²
Hankinson-Forman 230 kV	OTP	Clearance mitigations.	\$500,000
Wahpeton-Fergus Falls 230 kV	OTP	Wahpeton-Fergus Falls 230 kV Uprate	\$850,000 ²
Audubon-Lake Park 230 kV	OTP	Clearance mitigations.	\$200,000
Maple River 345-230-13.8 kV xfmr #1	OTP MPC	Replace the #1 transformer with a larger transformer.	\$4,600,000 ¹
Maple River 345-230-13.8 kV xfmr #2	OTP MPC	Replace the #2 transformer with a larger transformer.	\$4,600,000 ¹
Ellendale-County Line 69 kV	ITCM	Rebuild 5.79 miles	\$4,200,000
Hayward-County Line 69 kV	ITCM	Rebuild 13.32 miles	\$9,700,000
Osceola-Osceola REC 69 kV	ITCM CIPCO	ITCM: ITCM rating 42/44 MVA SN/SE CIPCO: NU is not required unless identified in affected system study	\$0
Heron Lake-Lakefield 161 kV	ITCM	MTEP 15 Project 8106 and replace wavetrap	\$20,000
Lime Crk-Colby 161 kV	ITCM	Rebuild 4.39 miles	\$4,800,000
Ottumwa 345-161 kV xfmr	ITCM	Add second 345/161 kV transformer	\$8,000,000
Adams-Hayward 161 kV	ITCM	Adams-Hayward 161 kV Uprate	\$480,000 ²
Adams-Beaver Creek 161 kV	ITCM DPC	Rebuild line with 795 ACSS conductor	\$15,100,000
Hazleton-Arnold 345 kV	ITCM MEC	MEC: Structure replacements. New MEC Only rating expected to be 1229/1229 MVA. \$750K ITCM: Structure replacements. New ITCM rating 1285 MVA/SN/SE. \$480K	\$1,230,000
Hazleton-Hickory Crk 345 kV	ITCM	ITCM rating 1569 MVA SN/SE	\$0
Colby 345-161 kV xfmr	ITCM	Add second 345/161 kV transformer	\$8,000,000
Webster-Wright 161 kV	MEC	Reconductor line and substation terminal equipment upgrades. New rating predicted to be 315/335 MVA.	\$8,000,000
Ward-Bismark 230 kV	BEPC WAPA	NU is not required unless it is identified as constraint in affected system study.	\$0

Constraint	Owner	Mitigation	Cost (\$)
Ft. Thompson 345 kV bus 1-3 TIE	WAPA	NU is not required unless it is identified as constraint in affected system study.	\$0
Wilton-Winger 230 kV	MPC OTP	OTP: Sufficient for the flows of 395.2 MVA. \$0 MPC: Raise structures to 90 C. \$350K	\$0
Drayton 230-115-13.8 kV xfmr	MPC	MPC: Generator offloads line. \$0	\$0
Ellendale-Aberdeen 115 kV	MDU NWE	MDU's rating for Ellendale is 142 MVA. NWPS owns line and equipment at Aberdeen.	\$0
Mandan-Ward 230 kV	MDU BEPC	MDU's rating at Mandan is 956 MVA. MPC owns equipment at Mandan. WAPA owns the line. BEPC owns Ward.	\$0
Fox Tail-Tatanka North 230 kV	MDU	Replace Breakers. New Rating: 478/526 MVA	\$2,000,000
Merricourt-Tatanka North 230 kV	MDU	Rebuild line to ACSS. New Rating: 478/478 MVA	\$1,000,000
Wishek-J302 POI 230 kV	MDU	J302 POI-Wishek 230 kV Rebuild	\$4,750,000 ²
Wabaco-Rochester 161 kV	DPC	Rebuild line with 795 ACSS conductor. \$11M. MTEP Appendix A project	\$0 ³
Wabaco-Alma 161 kV	DPC	Rebuild line with 795 ACSS conductor	\$6,300,000

Note 1: This Network Upgrade is also identified in summer peak scenario.

Note 2: This Network Upgrade was required in DPP 2017 Aug. Ph2 study. The cost will not be allocated to DPP 2018 Apr. Ph1 projects.

Note 3: This is approximate \$11,000,000 Appendix A project in MTEP that is being disputed.

Section

5

Local Planning Criteria Analysis

Local Planning Criteria (LPC) analyses were performed to identify additional constraints per Transmission Owning Companies' LPC.

5.1 GRE Local Planning Criteria Analysis

Great River Energy (GRE) determined with provided rationale that the GRE LPC should be applied to the projects listed in Table 5-1. The GRE LPC analysis consisted of steady-state contingency analysis for summer shoulder condition.

Table 5-1: DPP Projects with GRE LPC Applicable

MISO Project	Pmax (MW)	Fuel type	POI	Rationale
J1106	414	Wind	Lyon County - Cedar Mountain 345 kV	GRE is the maintainer for this CapX-owned facility and responsible for the compliance with the NERC standards (FAC-002, TPL-001) associated with new interconnections
J1140	100	Solar	Platte River 115 kV	GRE is the Transmission owner of the transmission line
J1187	151.8	Wind	Stanton 230 kV	GRE is the owner of the substation

Based on geographic locations of the projects' Point of Interconnection (POI), the projects were separated into the following three groups for GRE LPC study:

- J1106 GRE LPC group: J1106
- J1140 GRE LPC group: J1140
- CCS GRE LPC group: J1187

Siemens PTI performed the GRE local planning criteria analysis based on GRE's LPC. The J1106 GRE local planning criteria analysis details can be found in Appendix F.1.1, the J1140 GRE local planning criteria analysis details can be found in Appendix F.1.2, and the CCS (J1187) GRE local planning criteria analysis details can be found in Appendix F.1.3.

5.1.1 Additional Network Upgrades Identified in J1106 GRE LPC Analysis

No additional Network Upgrades are required in the J1106 GRE LPC study.

5.1.2 Additional Network Upgrades Identified in J1140 GRE LPC Analysis

No additional Network Upgrades are required in the J1140 GRE LPC study.

5.1.3 Additional Network Upgrades Identified in CCS GRE LPC Analysis

Voltage collapse and voltage constraints were identified in the benchmark case under 12 CUDC related contingencies. Since all projects in DPP 2017 August West Phase 1 study were modeled and dispatched in this CCS GRE LPC study, network upgrades required in the GRE LPC in DPP 2017 August West Phase 1 study were tested for mitigation. It is concluded that the following Network Upgrades (Table 5-2) are required for mitigating voltage collapse and voltage constraints identified in the CCS GRE LPC benchmark case. Study projects in the DPP 2018 April cycle are not responsible for these Network Upgrades required in the benchmark case.

Table 5-2: Network Upgrades Required in the CCS GRE LPC Benchmark Case

Network Upgrades	Owner	Cost (\$)
Build Center-Ellendale 345 kV line (200 miles)	GRE MPC MDU	\$406,750,000
Build Alexandria -Quarry 345 kV line (77 miles)	GRE MRES XEL	TBD
2×75 Mvar switched cap bank at Riverview 345 kV (615664)	GRE	\$8,000,000
3×40 Mvar switched cap bank at Audubon 230 kV (620336)	OTP	\$6,000,000
100 Mvar SVC at Audubon 230 kV (620336)	OTP	\$15,000,000
4×75 Mvar switched cap bank at Jamestown 345 kV (620369)	OTP	\$16,000,000
150 Mvar SVC at Jamestown 345 kV (620369)	OTP	\$24,370,000
150 Mvar SVC at Prairie 230 kV (657755)	MPC	\$25,000,000
1×40 Mvar switched cap bank at Winger 230 kV (657758)	MPC	\$2,000,000
3×50 MVAR MSC at Alexandria 345 kV (658201)	MRES	\$10,000,000
150 MVAR SVC at Alexandria 345 kV (658201)	MRES	\$25,000,000

The Network Upgrades (Table 5-2) required in the benchmark case were added to the CCS GRE LPC study case. No voltage collapse or voltage constraints were identified in the CCS GRE LPC study case under the 12 CUDC related contingencies. Therefore, no additional voltage collapse mitigations were required in the study case.

Additional Network Upgrades required in the CCS GRE LPC study are listed in Table 5-3.

Table 5-3: Additional Network Upgrades for Constraints Identified in CCS GRE LPC Analysis

Constraint	Network Upgrades	Owner	Cost (\$)
Voltage collapse in Benchmark Case under CUDC contingencies ¹	Build Center-Ellendale 345 kV line (200 miles)	GRE MPC MDU	\$406,750,000
Voltage collapse in Benchmark Case under CUDC contingencies ¹	Build Alexandria -Quarry 345 kV line (77 miles)	GRE MRES XEL	TBD
Voltage collapse in Benchmark Case under CUDC contingencies ¹	2×75 Mvar switched cap bank at Riverview 345 kV (615664)	GRE	\$8,000,000
Voltage collapse in Benchmark Case under CUDC contingencies ¹	3×40 Mvar switched cap bank at Audubon 230 kV (620336)	OTP	\$6,000,000
Voltage collapse in Benchmark Case under CUDC contingencies ¹	100 Mvar SVC at Audubon 230 kV (620336)	OTP	\$15,000,000
Voltage collapse in Benchmark Case under CUDC contingencies ¹	4×75 Mvar switched cap bank at Jamestown 345 kV (620369)	OTP	\$16,000,000
Voltage collapse in Benchmark Case under CUDC contingencies ¹	150 Mvar SVC at Jamestown 345 kV (620369)	OTP	\$24,370,000
Voltage collapse in Benchmark Case under CUDC contingencies ¹	150 Mvar SVC at Prairie 230 kV (657755)	MPC	\$25,000,000
Voltage collapse in Benchmark Case under CUDC contingencies ¹	1×40 Mvar switched cap bank at Winger 230 kV (657758)	MPC	\$2,000,000
Voltage collapse in Benchmark Case under CUDC contingencies ¹	3×50 MVAR MSC at Alexandria 345 kV (658201)	MRES	\$10,000,000
Voltage collapse in Benchmark Case under CUDC contingencies ¹	150 MVAR SVC at Alexandria 345 kV (658201)	MRES	\$25,000,000
Benton 230 kV bus bar	Upgrade jumpers for two breakers on the 230 kV bus	GRE	\$100,000

Note 1: Study projects in the DPP 2018 Apr. cycle are not responsible for these Network Upgrades required in the benchmark case.

5.2 OTP Local Planning Criteria Analysis

J951, J975, and J1038 are to be interconnected in Otter Tail Power (OTP) transmission system. In addition to MISO's standard DPP analysis, OTP determined that J951, J975, and J1038 are required for OTP LPC study. The OTP LPC analysis consisted of steady-state contingency analysis for summer shoulder, summer peak, and Light Load No Wind conditions.

Three additional scenarios were analyzed in the OTP LPC study:

1. OTP LPC Summer Peak (SPK)
2. OTP LPC Summer Shoulder (SH)
3. OTP LPC Light Load No Wind (LLNW)

Siemens PTI performed the local planning criteria analysis based on OTP's LPC. The OTP local planning criteria analysis details can be found in Appendix F.2.

5.2.1 Additional Network Upgrades Identified in OTP LPC Analysis

No thermal or voltage constraints were identified in the summer peak or summer shoulder scenario.

In the light load no wind scenario, high voltage violation was identified at J933 POI 345 kV bus for the loss of J933 POI – Twin Brooks 345 kV line. The high voltage violation should be mitigated by J933 in DPP 2017 Aug. study cycle. Projects in DPP 2018 April cycle are not responsible for this violation.

No additional Network Upgrades are required in the OTP LPC study.

5.3 MDU Local Planning Criteria Analysis

Montana-Dakota Utilities Company (MDU) determined that J997 has more than 20% Distribution Factor (DF) on the outlets of J436/J437/J488, J302/J503, G359 (Merricourt), J580, J929, J933, J1040, and J1040 has more than 20% DF on the outlets of Tatanka, Foxtail, J436/J437/J488, J302/J503, G359 (Merricourt), J580, J929, J933, J997. In addition to MISO's standard DPP analysis, MDU determined that J997 and J1040 are required for MDU LPC study per Section 3.2 of the MDU Local Planning Criteria. The MDU LPC analysis consisted of steady-state contingency analysis for summer shoulder condition.

One additional scenario was analyzed in the MDU LPC study:

- a. MDU LPC Summer Shoulder (SH)

Siemens PTI performed the local planning criteria analysis based on MDU's LPC. The MDU local planning criteria analysis details can be found in Appendix F.3.

5.3.1 Additional Network Upgrades Identified in MDU LPC Analysis

Additional Network Upgrades required in the MDU LPC study are listed in Table 5-4.

Table 5-4: Additional Network Upgrades for Constraints Identified in MDU LPC Analysis

Constraint	Owner	Mitigation	Cost (\$)
J580 POI-Mandan 230 kV	MDU	Reconductor Mandan-J580 POI 230 kV line w/ 954 ACSS conductor (~40 miles, including river crossing) (rating: 687/687 MVA [N/E]).	\$12,000,000
Ellendale-Aberdeen Jct 115 kV	NWPS MDU	Substation upgrades at Ellendale 115. NWPS owns the line.	\$250,000
East Bismark-Bismark 115 kV	MDU WAPA	MDU equipment can be rated up to 199/249 MVA [N/E]. WAPA terminal is limiting.	\$0
Stein-NE Bismark 115 kV	MDU	Minor substation upgrades at E Bismarck 115 (new rating: 120/149 MVA [N/E]).	\$250,000
Heskett-Mandan 230 kV	MDU	Replace conductor w/ 954 ACSS and minor substation upgrades at Heskett 230(new rating: 478/526 MVA [N/E]).	\$250,000
Mandan-Ward 230 kV	MDU BEPC WAPA MPC	MDU's rating at Mandan is 956 MVA. MPC owns equipment at Mandan. WAPA owns the line. BEPC owns Ward.	\$0
Fox Tail-Tatanka North 230 kV	MDU	Major substation upgrades at Tatanka North 230 (new rating: 610/610 MVA [N/E]).	\$1,500,000 ¹
Fox Tail-Ellendale 230 kV	MDU	Rebuild Foxtail-Ellendale 230 kV line w/ 1272 ACSS (new rating: 769/769 MVA [N/E]).	\$20,000,000
Merricourt-Ellendale 230 kV	MDU	Rebuild Merricourt-Ellendale 230 kV line w/ 1272 ACSS (new rating: 776/776 MVA [N/E]).	\$30,000,000

Constraint	Owner	Mitigation	Cost (\$)
Ellendale 345-230-34.5 xfmr	MDU	Add a 2nd Ellendale345 345/230/34.5 kV transformer.	\$7,000,000

Note 1: This MDU LPC NU is in addition to the MISO NU.

5.4 MRES Local Planning Criteria Analysis

J1108 is to be interconnected in Missouri River Energy Services (MRES) transmission system. In addition to MISO's standard DPP analysis, MRES determined that J1108 is required for MRES LPC study. The MRES LPC analysis consisted of steady-state contingency analysis for summer shoulder.

Siemens PTI performed the local planning criteria analysis based on MRES's LPC. The MRES local planning criteria analysis details can be found in Appendix F.4.

5.4.1 Additional Network Upgrades Identified in MRES LPC Analysis

Additional Network Upgrades required in the MRES LPC study are listed in Table 5-5.

Table 5-5: Additional Network Upgrades for Constraints Identified in MRES LPC Analysis

Constraint	Owner	Mitigation	Cost (\$)
Alexandria MRES-Alexandria 115 kV	MRES OTP	Replace line jumpers	\$30,000

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MWEX Voltage Stability Study

ATC performed steady state voltage stability analysis. Voltage stability analysis is required to determine if the initial conditions of the DPP system models under study are in a stable state as defined by Power-Voltage (PV) curves of the Minnesota Wisconsin Export Interface (MWEX) for the worst contingency.

As shown in Table 6-1, the Pre-DPP scenario in the 2024SH case is not voltage stable. The Pre-DPP scenario does not converge with the worst contingency and therefore is in violation of ATC Planning Criteria.

The Post-DPP scenario in the 2024SH case is voltage stable but is also in violation of ATC Planning Criteria because the voltage stability margin is less than 10% with the worst contingency.

Because the Post-DPP scenario is not aggravating the criteria violations, Network Upgrades related to voltage stability will not be assigned to the Interconnection Customers, based on the assumptions used in this analysis.

The MWEX voltage stability study details can be found in Appendix G.

Table 6-1: MWEX Margins to Collapse in the 2024SH Cases

	Real Power Flow (MW)						
	AHD-SLK ¹	MWEX			Margin to Nose ²		
Case	N-0 Initial Condition	N-0 I.C. ³	N-1 I.C. ³	N-1 Nose	(MW)	(%)	Notes
Pre-DPP	542.1	1343.4	N/A – Contingency does not converge ⁶				Voltage Unstable
Post-DPP	540.7	1352.0	969.7	1016.9	47.2	4.6	Voltage Stable ^{4b,5} Insufficient Margin ^{4a}

Notes:

- As described in the active MWEX Operating Guide, the AHD-SLK interface is a single element PTDF interface measured at the Minnesota Power 230 kV side of the Arrowhead 230 kV phase shifter.
- Margin to Nose is defined as:
 - "Margin to Nose (MW)" = "MWEX N-1 Nose" – "N-1 Initial Condition After Phase Shift"
 - "Margin to Nose (%)" = "Margin to Nose (MW)" / "MWEX N-1 Nose"
- Initial Condition flows were measured in the base cases with an intact system and the worst contingency, plus operation of various control systems as needed with all transformer taps, switched shunts, and PARs locked.
- ATC Planning Criteria requires:
 - A 10% voltage stability margin.

- b. $V_{nose} < V_{min}$.
- 5. Xcel Planning Criteria requires:
 - a. $V_{nose} < V_{min}$. The voltage is measured at the Xcel Eau Claire 345 kV bus. Per Xcel's Planning Criteria, the post-contingent minimum voltage is 0.92 p.u. at the Eau Claire 345 kV bus.
- 6. N-1 measurements cannot be made because the worst contingency does not converge.

Deliverability Study

7.1 Project Description

Interconnection requests requesting Network Resource Interconnection Services (NRIS) were considered for deliverability analysis.

7.2 Introduction

Generator interconnection projects have to pass Generator Deliverability Study to be granted Network Resource Interconnection Services (NRIS).

If the generator is determined as not fully deliverable, the customer can choose either to change his project to an Energy Resource (ER) project or proceed with the system upgrades that will make the generator fully deliverable.

Generator Deliverability Study ensures that the Network Resources, on an aggregate basis, can meet the MISO aggregate load requirements during system peak condition without getting bottlenecked. The wind generators are tested at 100 % of their maximum output level which then can be used to meet Resource Adequacy obligations, under Module E, of the MISO Transmission and Energy Market Tariff (TEMT).

7.3 Study Methodology

MISO Generator Deliverability Study whitepaper describing the algorithm can be found at https://cdn.misoenergy.org/Generator_Deliverability_Study_Methodology108139.pdf.

7.4 Determining the MW restriction

If one facility is overloaded based on the assessed “severe yet credible dispatch” scenario described in the study methodology, and the generator under study is in the “Top 30 DF List” (see white paper for detail), part or all of its output is not deliverable. The restricted MW is calculated as following:

$$(\text{MW restricted}) = (\text{worst loading} - \text{MW rating}) / (\text{generator sensitivity factor})$$

If the result is larger than the maximum output of the generator, 100% of this generator’s output is not deliverable.

The generator is also responsible for any NEW base case (pre-shift) overload or NEW “severe yet credible dispatch overload” where the generator is not in the “Top 30 DF List”, if the generator’s DF is greater than 5%. Please see white paper for detail. The formula above also applies to these situations.

7.5 2024 Deliverability Study Result

7.5.1 J953

J953 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	1.83 MW (100%)
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7.5.2 J954

J954 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	1.4 MW (100%)
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7.5.3 J958

J958 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	100 MW (100%)
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7.5.4 J959

J959 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	150 MW (100%)
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7.5.5 J963

J963 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	10.2 MW (100%)
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7.5.6 J966

J966 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	400 MW (100%)
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7.5.7 J967

J967 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	150 MW (100%)
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7.5.8 J971

J971 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	125.71 MW (50.3%)						
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
Upgrade Fenton 115/69 kV Transformer	250	0.0779	No		J971, J1045	\$1,386,121	\$1,500,000

7.5.9 J975

J975 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	0 MW (0%)						
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
2nd Coyote 345/115 kV Transformer	0.00	0.0501	No		J975, J1187	\$2,611,349	\$7,000,000
New Square Butte-Mandan 230 kV Line	25.71	0.0767	No		J975, J1109, J1170, J1187	\$5,190,143	\$31,000,000
Leland Olds-Stanton 230 kV Rebuild	80.73	0.0750	No		J975, J1109, J1170, J1187	\$300,887	\$2,400,000
Maple River TR 2 Replacement	82.56	0.2597	Yes	J975, J1109, J1170	J975, J1109, J1170, J1187	\$949,680	\$4,600,000
Maple River TR 1 Replacement	150.00	0.2582	Yes	J975, J1109, J1170	J975, J1109, J1170, J1187	\$949,870	\$4,600,000

7.5.10 J977

J977 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIIS and IC upgrades and case assumptions)	300 MW (100%)
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7.5.11 J981

J981 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIIS and IC upgrades and case assumptions)	200 MW (100%)
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7.5.12 J982

J982 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	300 MW (100%)
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7.5.13 J997

J997 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	0 MW (0%)						
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
East Bismark Terminal Upgrades	200	0.0564	No		J997, J1187	\$54,675	\$100,000

7.5.14 J998

J998 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	0 MW (0%)						
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
Council Bluffs-S3456 345 kV Terminal Equipment Upgrade	50	0.0900	No		J998, J1024, J1050, J1122, J1132	\$5,403	\$200,000

7.5.15 J999

J999 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	100 MW (100%)
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7.5.16 J1001

J1001 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	40 MW (100%)
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7.5.17 J1006

J1006 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	100 MW (100%)
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7.5.18 J1008

J1008 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	50 MW (100%)
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7.5.19 J1024

J1024 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	0 MW (0%)						
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
Reconductor MCKSBRG-WNTRST	0.00	0.1684	No		J1024, J1132	\$6,370,342	\$10,000,000
Council Bluffs-S3456 345 kV Terminal Equipment Upgrade	104.71	0.1295	No		J998, J1024, J1050, J1122, J1132	\$31,099	\$200,000
WNTRST-NORWLK Structure Replacements	134.64	0.1418	No		J1024, J1132	\$189,678	\$300,000
Adams-Creston Structure Replacements	200.00	0.2840	No		J1024	\$800,000	\$800,000

7.5.20 J1032

J1032 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	50 MW (100%)
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7.5.21 J1040

J1040 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	250 MW (100%)
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7.5.22 J1041

J1041 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	20 MW (100%)
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7.5.23 J1045

J1045 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	10.6 MW (53.0%)						
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
Upgrade Fenton 115/69 kV Transformer	20	0.0800	No		J971, J1045	\$113,879	\$1,500,000

7.5.24 J1050

J1050 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	0 MW (0%)						
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
Council Bluffs-S3456 345 kV Terminal Equipment Upgrade	225	0.0665	No		J998, J1024, J1050, J1122, J1132	\$17,966	\$200,000

7.5.25 J1054

J1054 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	30 MW (100%)
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7.5.26 J1057

J1057 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	125 MW (100%)
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7.5.27 J1061

J1061 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	300 MW (100%)
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7.5.28 J1072

J1072 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	150 MW (100%)
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7.5.29 J1084

J1084 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	150 MW (100%)
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7.5.30 J1086

J1086 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	120 MW (100%)
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7.5.31 J1092

J1092 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	100 MW (100%)
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7.5.32 J1098

J1098 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	0 MW (0%)						
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
Second Webster 345/115 kV Transformer	40	0.0587	No		J1098, J1114	\$1,405,315	\$10,000,000

7.5.33 J1105

J1105 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	200 MW (100%)
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7.5.34 J1106

J1106 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	414 MW (100%)
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7.5.35 J1108

J1108 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	150 MW (100%)
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7.5.36 J1109

J1109 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	0 MW (0%)						
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Constrain t in ERIS Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
New Square Butte-Mandan 230 kV Line	41.96	0.0618	No		J975, J1109, J1170, J1187	\$5,771,006	\$31,000,000
Leland Olds-Stanton 230 kV Rebuild	131.76	0.0639	No		J975, J1109, J1170, J1187	\$353,771	\$2,400,000
Maple River TR 2 Replacement	134.74	0.3492	Yes	J975, J1109, J1170	J975, J1109, J1170, J1187	\$1,762,116	\$4,600,000
Maple River TR 1 Replacement	207.00	0.3471	Yes	J975, J1109, J1170	J975, J1109, J1170, J1187	\$1,762,487	\$4,600,000

7.5.37 J1110

J1110 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	100 MW (100%)
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7.5.38 J1114

J1114 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	0 MW (0%)						
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
Second Webster 345/115 kV Transformer	200	0.0718	No		J1098, J1114	\$8,594,685	\$10,000,000

7.5.39 J1122

J1122 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	0 MW (0%)						
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
Council Bluffs-S3456 345 kV Terminal Equipment Upgrade	200	0.5835	No		J998, J1024, J1050, J1122, J1132	\$140,128	\$200,000

7.5.40 J1124

J1124 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	100 MW (100%)
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7.5.41 J1128

J1128 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	147.25 MW (98.2%)						
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
Emery Jumper Replacement	150	0.0547			J1128, J1174, J1175	\$6,150	\$50,000

7.5.42 J1131

J1131 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	100 MW (100%)
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7.5.43 J1132

J1132 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	0 MW (0%)						
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
Reconductor MCKSBRG-WNTRST	0.00	0.3838	No		J1024, J1132	\$3,629,658	\$10,000,000
Council Bluffs-S3456 345 kV Terminal Equipment Upgrade	26.18	0.0900	No		J998, J1024, J1050, J1122, J1132	\$5,403	\$200,000
WNTRST-NORWLK Structure Replacements	50.00	0.3299	No		J1024, J1132	\$110,322	\$300,000

7.5.44 J1135

J1135 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	50 MW (100%)
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7.5.45 J1140

J1140 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	100 MW (100%)
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7.5.46 J1143

J1143 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	100 MW (100%)
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7.5.47 J1149

J1149 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	200 MW (100%)
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7.5.48 J1164

J1164 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	200 MW (100%)
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7.5.49 J1169

J1169 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	50 MW (100%)
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7.5.50 J1170

J1170 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	0 MW (0%)						
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
New Square Butte-Mandan 230 kV Line	41.14	0.0618	No		J975, J1109, J1170, J1187	\$5,575,851	31,000,000.00
Leland Olds-Stanton 230 kV Rebuild	131.76	0.0639	No		J975, J1109, J1170, J1187	\$341,808	2,400,000.00
Maple River TR 2 Replacement	134.74	0.3492	Yes	J975, J1109, J1170	J975, J1109, J1170, J1187	\$1,701,753	4,600,000.00
Maple River TR 1 Replacement	200.00	0.3471	Yes	J975, J1109, J1170	J975, J1109, J1170, J1187	\$1,702,110	4,600,000.00

7.5.51 J1174

J1174 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	144.5 MW (48.2%)						
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
Emery Jumper Replacement	300	0.0975	No		J1128, J1174, J1175	\$21,925	\$50,000

7.5.52 J1175

J1175 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	144.5 MW (48.2%)						
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
Emery Jumper Replacement	300	0.0975	No		J1128, J1174, J1175	\$21,925	\$50,000

7.5.53 J1177

J1177 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	300 MW (100%)
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7.5.54 J1179

J1179 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	300 MW (100%)
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7.5.55 J1181

J1181 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	200 MW (100%)
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7.5.56 J1182

J1182 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	250 MW (100%)
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7.5.57 J1187

J1187 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	0 MW (0%)						
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
2nd Coyote 345/115 kV Transformer	0.00	0.0832	No		J975, J1187	\$4,388,651	\$7,000,000
New Square Butte-Mandan 230 kV Line	0.00	0.2112	No		J975, J1109, J1170, J1187	\$14,463,000	\$31,000,000
East Bismark Terminal Upgrades	31.22	0.0616	No		J997, J1187	\$45,325	\$100,000
Leland Olds-Stanton 230 kV Rebuild	98.04	0.3457	No		J975, J1109, J1170, J1187	\$1,403,533	\$2,400,000
Maple River TR 2 Replacement	100.26	0.0634	Yes	J975, J1109, J1170	J975, J1109, J1170, J1187	\$186,451	\$4,600,000
Maple River TR 1 Replacement	151.80	0.0630	Yes	J975, J1109, J1170	J975, J1109, J1170, J1187	\$185,533	\$4,600,000

Shared Network Upgrades Analysis

Shared Network Upgrade (SNU) test for Network Upgrades driven by higher queued interconnection project was performed for this System Impact Study.

No SNUs were identified in this study.

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Cost Allocation

The cost allocation of Network Upgrades for the study group reflects responsibilities for mitigating system impacts based on Interconnection Customer-elected level of Network Resource Interconnection service as of the draft System Impact Study report date.

9.1 Cost Assumptions for Network Upgrades

The cost estimate for each network upgrade was provided by the corresponding transmission owning company.

9.2 ERIS Network Upgrades Proposed for DPP West Area Projects

Network upgrades for Energy Resource Interconnection Service (ERIS) were identified in the MISO ERIS analysis and the affected system analysis. The ERIS network upgrades include voltage support network upgrades and thermal network upgrades identified in the MISO steady-state analysis, network upgrades identified in the Local Planning Criteria analysis and affected system analysis, voltage support network upgrades identified in the MWEX voltage stability analysis, stability network upgrades identified in the MISO transient stability analysis, and short circuit network upgrades identified in the MISO short circuit analysis. The total costs of ERIS network upgrades for the 2024 scenario are summarized in Table 9-1.

Table 9-1: Summary of ERIS Network Upgrades

Category of Network Upgrades	Cost (\$)
Base Case Network Upgrades	\$210,829,263
Network Upgrades Identified in MWEX Voltage Stability analysis	\$0
Additional Thermal Network Upgrades Identified in MISO Steady-State Analysis	\$477,503,000
Additional Reactive Power Network Upgrades for Voltage Constraints	\$0
Network Upgrades Identified in GRE LPC Analysis	\$100,000
Network Upgrades Identified in OTP LPC Analysis	\$0
Network Upgrades Identified in MDU LPC Analysis	\$71,250,000
Network Upgrades Identified in MRES LPC Analysis	\$30,000
Shared Network Upgrades	\$0
Total	\$759,712,263

ERIS network upgrades are listed below.

Table 9-2: Base Case Network Upgrades

NUs	Miles	Cost (\$)
Hazel Creek-Scott County 345 kV	115	\$210,829,263
Total		\$210,829,263

Table 9-3: Network Upgrades Required for MWEX Voltage Stability

NUs	Miles	Cost (\$)
No MWEX NUs		\$0

Table 9-4: Additional Thermal Network Upgrades in MISO Steady-State Analysis

Constraint	Owner	Mitigation	Cost (\$)
J1008 POI-Blue Earth 69 kV	ITCM	Rebuild 5.25 miles	\$3,800,000
J1008 POI-Blue Earth Tap 69 kV	ITCM	Rebuild 0.5 miles	\$363,000
J1024 POI-J611 POI 161 kV	MEC	MTEP 18885 to reconductor line in 2020	\$0
J1024 POI-Clarinda 161 kV	MEC	Substation terminal equipment upgrades	\$700,000
J1057 POI-Heron Lake 161 kV	ITCM	MTEP 15 Project 8106 and rebuild 14.7 miles	\$16,200,000
J1086 POI-Elk 161 kV	ITCM	Replace jumpers and rebuild approx 0.7 miles	\$770,000
J1143 POI-Blanch Rd. 115 kV	MP	Rebuild Blanchard - J1143 POI - Bellevue - Little Falls 115 kV (8.34 mi)	\$4,170,000
J1181 POI-Hazleton 345 kV	ITCM MEC	MEC: Structure replacements. New MEC Only rating expected to be 1154/1154 MVA. \$750K ITCM: Structure replacements. New ITCM rating 1151 SN/SE. \$840K	\$1,590,000
J580 POI-J302 POI 230 kV	MDU	Rebuild line to ACSS New Rating: 610 / 610 MVA	\$2,300,000 ¹
J580 POI-Heskett 230 kV	MDU	J580 POI-Heskett 230 kV Rebuild	\$4,500,000 ¹
J611 POI-Maryville 161 kV	MEC GMO	MEC: Existing MEC only rating expected to be 410 MVA after DPP 2016 AUG West line reconductor network upgrade. GMO: NU is not required unless it is identified as constraint in affected system study.	\$0
J748 POI-Raun 345 kV	MEC	Structure replacements. New MEC Only rating expected to be 952/952 MVA.	\$300,000
Adair-Thomas Hill 161 kV	AMRN AECI	Rebuild to 3000 A	\$28,600,000

Constraint	Owner	Mitigation	Cost (\$)
Montgomery-Spencer Crk 345 kV	AMRN	Rebuild Montgomery-Spencer Creek-5503 345 kV line	\$35,000,000
Herleman-Marblehead N 138 kV	AMRN	2nd Maywood-Herleman 345 kV line	\$21,000,000
Marblehead N 161-138 kV xfmr	AMRN	2nd Maywood-Herleman 345 kV line	Incl.
Marblehead N-Palmyra 161 kV	AMRN AECI	2nd Maywood-Herleman 345 kV line	Incl.
Wilmarth-Sheas Lake 345 kV	XEL	Wilmarth-Sheas Lake 345 kV Rebuild	\$49,400,000 ¹
Split Rock-White 345 kV	XEL WAPA	XEL: Limiter is on WAPA facility. \$0 WAPA: NU is not required unless it is identified as constraint in affected system study.	\$0
Blue Lake-Scott Co 345 kV	XEL	Blue Lake-Scott County 345 kV Rebuild	\$24,000,000 ¹
Lakefield Tap-Lakefield 345 kV	XEL ITCM	Rebuild 18.8 mi to 2x954 ACSS	\$37,600,000
Field S-Crandal 345 kV	XEL	Crandal-Fieldon 345 kV Rebuild	\$40,000,000 ¹
Briggs Rd 345-161-34.5 kV xfmr #9	XEL	Add 2nd 448 MVA TR10	\$4,000,000
Helena-Sheas Lk 345 kV	XEL	Helena-Sheas Lake 345 kV Rebuild	\$15,000,000 ¹
Hazel Crk 345-230-13.8 kV xfmr #9	XEL	Upgrade to 672 MVA transformer	\$3,500,000
Split Rock-Rock Co. 161 kV	XEL	Uprate with 795 ACSS	\$21,900,000
Sheyenne-Lake Park 230 kV	XEL OTP	XEL: Uprate 3 miles of line owned by Xcel to 954 ACSR. \$3M OTP: line clearance mitigations. \$150K	\$3,150,000
Sheyenne-Maple River 230 kV	XEL MPC	XEL: Rebuild 8.0 mi to 954 ACSR. \$14.4M MPC: Replace Bus Jumpers at Maple River. \$300K	\$14,400,000
Minn Valley Tap-Granite Falls 230 kV	XEL WAPA	XEL: Summer normal and emergency ratings are 413 MVA WAPA: NU is not required unless it is identified as constraint in affected system study.	\$0
Rock Co.-Magnolia 161 kV	XEL ITCM	XEL: Uprate with 795 ACSS ITCM: Rebuild 11.4 miles, \$12.54M	\$24,600,000
Split Rock 161-115-36.2 kV xfmr #6	XEL	Upgrade with 336 MVA	\$2,000,000
Sheyenne-Mapleton 115 kV	XEL OTP	Sheyenne-Mapleton 115 kV Rebuild	\$22,200,000 ¹
Douglas Co 115-69 kV xfmr #2	XEL	Replace TR1 to match TR2 with 80 MVA	\$1,200,000
Douglas Co-Hudson 115 kV	XEL GRE	Rebuild 6.8 mi to 795 ACSS	\$10,390,000
Douglas Co-Little Sauk 115 kV	XEL GRE	Rebuild 9.9 mi to 795 ACSS	\$14,850,000
Quarry 345-115-34.5 kV xfmr #9	XEL	Add 2nd 448 MVA TR10	\$4,000,000

Constraint	Owner	Mitigation	Cost (\$)
Douglas Co-West Union 69 kV	XEL GRE	Rebuild 5.4 mi to 795 ACSR	\$3,240,000
Blackoak-Grove 69 kV	XEL GRE	Rebuild 0.2 mi to 795 ACSR	\$120,000
Riverton-Mud Lake 230 kV	MP GRE	MP: No MP limiting equipment, 414.0/414.0 SN/SE. \$0 GRE: Rebuild 8.6 total miles of line. \$4.1M	\$4,100,000
Pepin Lake-Long Prairie 115 kV	MP	Rebuild Pepin Lake-Long Prairie 115 kV (12.77 miles)	6,385,000
Pepin Lake-Blanchard 115 kV	MP	Rebuild Blanchard - Pepin Lake 115 kV (14.23 miles)	7,115,000
Austin-Murphy 161 kV	SMPMPA	Austin-Murphy Creek 161 kV Rebuild	\$1,875,000 ¹
Murphy-J1128 POI 161 kV	SMPMPA	Murphy Creek-J1128 POI 161 kV Rebuild	\$5,812,500 ¹
Rutland-Fox Lake 161 kV	ITCM	Rebuild 8.9 miles	\$9,800,000
Rutland-Huntley 161 kV	ITCM	Rebuild 0.73 miles	\$800,000
Inman-Wing River 230 kV	GRE	Rebuild line, replace jumpers, CT, hookstick switches	\$13,100,000
Inman-Henning 230 kV	GRE	Rebuild line, replace jumpers at terminals	\$2,800,000
Hubbard-Erie Jct 230 kV	GRE OTP MP	GRE: GRE equipment at Hubbard is rated 522.6 MVA. \$0 OTP: Sufficient for flows seen in study. \$0 MP: Reconductor on MP's segment. \$2.265M	\$2,265,000
West St. Cloud-Westwood 115 kV	GRE XEL	Rebuild 0.6 mi to 2x795 ACSS	\$900,000
West St. Cloud-Little Sauk Tap 115 kV	GRE XEL	GRE: XEL owns the circuit and equipmnet. \$0 XEL: Uprate line to 795 ACSS. \$2.1M	\$2,100,000
Wing River-Riverton 230 kV	GRE MP	GRE: Increase conductor clearance to 75C on GRE portion (13.3 miles). 327.0/359.0 SN/SE. \$4M MP: Increase conductor clearance to 75C on MP portion of Riverton-Wing River 230 kV (35.97 miles). 327.0/359.0 SN/SE. \$9.72M	\$13,720,000
Silver Lake-Henning 230 kV	GRE MRES OTP	GRE: Replace jumpers at Henning. \$50K OTP: line clearance mitigations. \$650k MRES: Phase raisers. \$1M	\$1,700,000
Silver Lake-Fergus Falls 230 kV	GRE OTP MRES	GRE: GRE equipment at Silver Lake is rated 796.7 MVA. \$0 OTP: Line clearance mitigations. \$400K MRES: Phase raisers. \$60K	\$460,000
Chub Lake 345-115-34.5 kV xfmr	GRE	Add second 345/115 kV transformer at Chub Lake	\$11,400,000
Dewing-Little Falls 115 kV	GRE MP	Increase conductor clearance on Little Falls - Dewing - Mud Lake 115 kV (28.23 mi)	\$1,600,000
Dewing-Mud Lake 115 kV	GRE	Increase conductor clearance on Little Falls - Dewing - Mud Lake	Incl.

Constraint	Owner	Mitigation	Cost (\$)
	MP	115 kV (28.23 mi)	
J1143 POI-Bellevu 115 kV	GRE MP	Rebuild Blanchard - J1143 POI - Bellevue - Little Falls 115 kV (8.34 mi)	Incl.
Bellevu-Little Falls 115 kV	GRE MP	Rebuild Blanchard - J1143 POI - Bellevue - Little Falls 115 kV (8.34 mi)	Incl.
Rushmore-Wolf Tap 69 kV	GRE XEL	GRE: Rebuild the Rushmore-Wolf Tap-Rushmore 69 kV line, replace three 3-way line switches.	\$3,300,000
Little Sauk Tap-Long Prairie 115 kV	GRE MP	GRE: GRE tap switch (Little Sauk) is rated 239 MVA. \$0 MP: Rebuild MP segment of Long Prairie-GRE Little Sauk 115 kV (9.34 miles). 185.0/203.0 SN/SE. \$4.67M XEL: Don't own anything on that section of line	\$4,670,000
Lesauk Tap-Fishill 115 kV	GRE XEL	GRE: XEL owns equipment and line. \$0 XEL: Uprate line to 795 ACSS. \$4.5M	\$4,500,000
St. Stephens-Fishill 115 kV	GRE XEL	GRE: XEL owns equipment and line. \$0 XEL: Uprate line to 795 ACSS. \$5.1M	\$5,100,000
West Union-Sauk Center Muni 69 kV	GRE XEL	Rebuild 8.0 mi to 795 ACSR	\$4,800,000
Kandota Tap-Sauk Center Muni 69 kV	GRE XEL	Rebuild 0.9 mi to 795 ACSR	\$540,000
Kandota Tap-Grove 69 kV	GRE XEL	GRE: Replace Kandota 3-way switch (1200 amp). \$385K XEL: Rebuild to 795 ACSR, \$2.5M	\$2,885,000
Aldrich-Verndale 115 kV	GRE MP	Increase conductor clearance to 65C on Dog Lake - Verndale 115 kV (16.99 miles)	\$1,020,000
CSLTNET-Mapleton 115 kV	OTP	Rebuild of ~12 miles of 115 kV line and terminal equipment replacement.	\$7,000,000
CSLTNET-Buffalo 115 kV	OTP	Rebuild of ~16 miles of 115 kV line and terminal equipment replacement.	\$9,900,000
Hoot Lake-J1038 POI 115 kV	OTP	Rebuild of ~2 miles of 115 kV line and terminal equipment replacement.	\$1,250,000
Donalds-J951 POI 115 kV	OTP	Line clearance mitigations and terminal equipment replacement.	\$500,000
Erie Jct-Audubon 230 kV	OTP XEL	OTP: Clearance mitigations. \$150K XEL: Clearance mitigations. \$300K	\$450,000 ¹
Hankinson-Forman 230 kV	OTP	Clearance mitigations.	\$500,000
Wahpeton-Fergus Falls 230 kV	OTP	Wahpeton-Fergus Falls 230 kV Uprate	\$850,000 ¹
Audubon-Lake Park 230 kV	OTP	Clearance mitigations.	\$200,000
Maple River 345-230-13.8 kV xfmr #1	OTP MPC	Replace the #1 transformer with a larger transformer.	\$4,600,000
Maple River 345-230-13.8 kV xfmr #2	OTP MPC	Replace the #2 transformer with a larger transformer.	\$4,600,000

Constraint	Owner	Mitigation	Cost (\$)
Ellendale-County Line 69 kV	ITCM	Rebuild 5.79 miles	\$4,200,000
Huntley-Blue Earth Tap 69 kV	ITCM	Rebuild 7.34 miles	\$5,300,000
Elmore-Blue Earth Tap 69 kV	ITCM	Rebuild 5.24 miles	\$3,800,000
Elmore-Lakota CBPC 69 kV	ITCM CBPC	Rebuild 7.58 miles	\$5,500,000
Rushmore Tap-Wolf Tap 69 kV	GRE XEL	Rebuild the Rushmore-wolf tap-Rushmore 69 kV line, replace three 3-way line switches.	Incl.
Hayward-County Line 69 kV	ITCM	Rebuild 13.32 miles	\$9,700,000
Lakota-Lakota CBPC 69 kV	ITCM CBPC	Rebuild 2.02 miles	\$1,500,000
Osceola-Osceola REC 69 kV	ITCM CIPCO	ITCM: ITCM rating 42/44 MVA SN/SE CIPCO: NU is not required unless identified in affected system study	\$0
Magnolia-Elk 161 kV	ITCM	Replace structures to mitigate sag limits, new rating 210 MVA SN/SE	\$2,000,000
Heron Lake-Lakefield 161 kV	ITCM	MTEP 15 Project 8106 and replace wavetrap	\$20,000
Lime Crk-Colby 161 kV	ITCM	Rebuild 4.39 miles	\$4,800,000
Ottumwa 345-161 kV xfmr	ITCM	Add second 345/161 kV transformer	\$8,000,000
Adams-Hayward 161 kV	ITCM	Adams-Hayward 161 kV Uprate	\$480,000 ¹
Adams-Beaver Creek 161 kV	ITCM DPC	Rebuild line with 795 ACSS conductor	\$15,100,000
Hazleton-Arnold 345 kV	ITCM MEC	MEC: Structure replacements. New MEC Only rating expected to be 1229/1229 MVA. \$750K ITCM: Structure replacements. New ITCM rating 1285 MVA/SN/SE. \$480K	\$1,230,000
Colby 345-161 kV xfmr	ITCM	Add second 345/161 kV transformer	\$8,000,000
Webster-Wright 161 kV	MEC	Reconductor line and substation terminal equipment upgrades. New rating predicted to be 315/335 MVA.	\$8,000,000
Wilton-Winger 230 kV	MPC OTP	OTP: Sufficient for the flows of 395.2 MVA. \$0 MPC: Raise structures to 90 C. \$350K	\$0
Ellendale-Aberdeen 115 kV	MDU NWE	MDU's rating for Ellendale is 142 MVA. NWPS owns line and equipment at Aberdeen.	\$0
Mandan-Ward 230 kV	MDU BEPC	MDU's rating at Mandan is 956 MVA. MPC owns equipment at Mandan. WAPA owns the line. BEPC owns Ward.	\$0

Constraint	Owner	Mitigation	Cost (\$)
Fox Tail-Tatanka North 230 kV	MDU	Replace Breakers. New Rating: 478/526 MVA	\$2,000,000
Merricourt-Tatanka North 230 kV	MDU	Rebuild line to ACSS. New Rating: 478/478 MVA	\$1,000,000
Wishek-J302 POI 230 kV	MDU	J302 POI-Wishek 230 kV Rebuild	\$4,750,000 ¹
Wabaco-Rochester 161 kV	DPC	Rebuild line with 795 ACSS conductor. \$11M. MTEP Appendix A project	\$0 ²
Wabaco-Alma 161 kV	DPC	Rebuild line with 795 ACSS conductor	\$6,300,000

Note 1: This Network Upgrade was required in DPP 2017 Aug. Ph2 study. The cost will not be allocated to DPP 2018 Apr. Ph1 projects.

Note 2: This is approximate \$11,000,000 Appendix A project in MTEP that is being disputed.

Table 9-5: Additional Reactive Power NUs Required for Voltage Constraints

Network Upgrades	Owner	Cost (\$)
No reactive power NUs		\$0

Table 9-6: J1106 GRE Local Planning Criteria Network Upgrades

Constraint	Owner	Mitigation	Cost (\$)
No J1106 GRE LPC NUs			\$0

Table 9-7: J1140 GRE Local Planning Criteria Network Upgrades

Constraint	Owner	Mitigation	Cost (\$)
No J1140 GRE LPC NUs			\$0

Table 9-8: CCS (J1187) GRE Local Planning Criteria Network Upgrades

Constraint	Network Upgrades	Owner	Cost (\$)
Voltage collapse in Benchmark Case under CUDC contingencies ¹	Build Center-Ellendale 345 kV line (200 miles)	GRE MPC MDU	\$406,750,000
Voltage collapse in Benchmark Case under CUDC contingencies ¹	Build Alexandria -Quarry 345 kV line (77 miles)	GRE MRES XEL	TBD

Constraint	Network Upgrades	Owner	Cost (\$)
Voltage collapse in Benchmark Case under CUDC contingencies ¹	2×75 Mvar switched cap bank at Riverview 345 kV (615664)	GRE	\$8,000,000
Voltage collapse in Benchmark Case under CUDC contingencies ¹	3×40 Mvar switched cap bank at Audubon 230 kV (620336)	OTP	\$6,000,000
Voltage collapse in Benchmark Case under CUDC contingencies ¹	100 Mvar SVC at Audubon 230 kV (620336)	OTP	\$15,000,000
Voltage collapse in Benchmark Case under CUDC contingencies ¹	4×75 Mvar switched cap bank at Jamestown 345 kV (620369)	OTP	\$16,000,000
Voltage collapse in Benchmark Case under CUDC contingencies ¹	150 Mvar SVC at Jamestown 345 kV (620369)	OTP	\$24,370,000
Voltage collapse in Benchmark Case under CUDC contingencies ¹	150 Mvar SVC at Prairie 230 kV (657755)	MPC	\$25,000,000
Voltage collapse in Benchmark Case under CUDC contingencies ¹	1×40 Mvar switched cap bank at Winger 230 kV (657758)	MPC	\$2,000,000
Voltage collapse in Benchmark Case under CUDC contingencies ¹	3×50 MVAR MSC at Alexandria 345 kV (658201)	MRES	\$10,000,000
Voltage collapse in Benchmark Case under CUDC contingencies ¹	150 MVAR SVC at Alexandria 345 kV (658201)	MRES	\$25,000,000
Benton 230 kV bus bar	Upgrade jumpers for two breakers on the 230 kV bus	GRE	\$100,000

Note 1: Study projects in the DPP 2018 Apr. cycle are not responsible for these Network Upgrades required in the benchmark case.

Table 9-9: OTP Local Planning Criteria Network Upgrades

Network Upgrades	Owner	Cost (\$)
No OTP LPC NUs		\$0

Table 9-10: MDU Local Planning Criteria Network Upgrades

Constraint	Owner	Mitigation	Cost (\$)
J580 POI-Mandan 230 kV	MDU	Reconductor Mandan-J580 POI 230 kV line w/ 954 ACSS conductor (~40 miles, including river crossing) (rating: 687/687 MVA [N/E]).	\$12,000,000
Ellendale-Aberdeen Jct 115 kV	NWPS MDU	Substation upgrades at Ellendale 115. NWPS owns the line.	\$250,000
East Bismark-Bismark 115 kV	MDU WAPA	MDU equipment can be rated up to 199/249 MVA [N/E]. WAPA terminal is limiting.	\$0
Stein-NE Bismark 115 kV	MDU	Minor substation upgrades at E Bismarck 115 (new	\$250,000

Constraint	Owner	Mitigation	Cost (\$)
		rating: 120/149 MVA [N/E].	
Heskett-Mandan 230 kV	MDU	Replace conductor w/ 954 ACSS and minor substation upgrades at Heskett 230(new rating: 478/526 MVA [N/E].	\$250,000
Mandan-Ward 230 kV	MDU BEPC WAPA MPC	MDU's rating at Mandan is 956 MVA. MPC owns equipment at Mandan. WAPA owns the line. BEPC owns Ward.	\$0
Fox Tail-Tatanka North 230 kV	MDU	Major substation upgrades at Tatanka North 230 (new rating: 610/610 MVA [N/E]).	\$1,500,000 ¹
Fox Tail-Ellendale 230 kV	MDU	Rebuild Foxtail-Ellendale345 230 kV line w/ 1272 ACSS (new rating: 769/769 MVA [N/E]).	\$20,000,000
Merricourt-Ellendale 230 kV	MDU	Rebuild Merricourt-Ellendale345 230 kV line w/ 1272 ACSS (new rating: 776/776 MVA [N/E]).	\$30,000,000
Ellendale 345-230-34.5 xfmr	MDU	Add a 2nd Ellendale345 345/230/34.5 kV transformer.	\$7,000,000

Note 1: This MDU LPC NU is in addition to the MISO NU.

Table 9-11: MRES Local Planning Criteria Network Upgrades

Constraint	Owner	Mitigation	Cost (\$)
Alexandria MRES-Alexandria 115 kV	MRES OTP	Replace line jumpers	\$30,000

Table 9-12: Shared Network Upgrades

Network Upgrades	Project Study Cycle	Projects sharing cost	MW Contribution	Total Network Upgrade Cost (\$)	Cost Responsibility
No SNUs					\$0

9.3 Cost Allocation Methodology

The costs of Network Upgrades (NU) for a set of generation projects (one or more sub-groups or entire group with identified NU) are allocated based on the MW impact from each project on the constrained facilities in the Post Case. For constraints identified in the shoulder peak scenario, the MW impact is calculated using the shoulder peak post-DPP case. The MW impact on constraints identified in the summer peak scenario is calculated using the summer peak post-DPP case. With all Group Study generation projects dispatched in the Post Case, all thermal and voltage constraints will be identified and a distribution factor from each project on each constraint will be obtained.

Constraints which are mitigated by one or a subset of NU are identified. The MW contribution on these constraints from each generating facility is calculated in the Post Case without any network upgrades. Then the cost of each NU is allocated based on the pro rata share of the MW contribution from each generating facility on the constraints mitigated or partly mitigated by this NU. The methodology to determine the cost allocation of NU is:

$$\text{Project A cost portion of NU} = \text{Cost of NU} \times \left(\frac{\text{Max}(\text{Proj.A MW contribution on constraint})}{\sum_i \text{Max}(\text{Proj.i MW contribution on constraint})} \right)$$

9.4 Cost Allocation

The cost allocation of Network Upgrades for the study group reflects responsibilities for mitigating system impacts based on Interconnection Customer-elected level of Network Resource Interconnection service as of the draft System Impact Study report date.

The Distribution Factor (DF) from each generating facility is calculated on the constraints identified in the steady-state analysis in the Post Case without any network upgrades. For a reactive power network upgrade required for mitigating voltage constraints identified in the steady-state AC contingency analysis and stability analysis, DFs are calculated under the most critical contingency on all branches (proxy branches for reactive power network upgrade) connecting at the constraint bus. For a reactive power network upgrade required for mitigating MWEX voltage stability constraints identified in the voltage stability analysis, DFs are calculated under the most critical contingency on all branches (proxy branches) connecting to the high voltage side of the transformer, where the voltage collapse occurs.

For each thermal constraint, the maximum MW contribution (increasing flow) from each generating facility is calculated. MW contribution from one generating facility is set as zero if the MW contribution is less than 1 MW, or the constraint is not categorized as MISO ERS constraint or affected system constraint for that specific generating facility.

For reactive power network upgrades, or MWEX network upgrades and other voltage stability network upgrades, generators with positive net MW impact (harming the constraint) on all branches connected at the constraint bus will be responsible for mitigating these constraints.

Additional NRIS Network Upgrades are allocated to the impacting NRIS projects. ERS Network Upgrades will be allocated to the impacting projects only based on the ERS results.

Transient stability Network Upgrades are allocated based on projects causing instability. If multiple projects are causing instability, cost allocation will be based on pro rata share of total MW of all projects causing instability.

The calculated DF results and the MW contribution on each constraint are in Appendix H.1 for the 2024 scenario.

Finally, the cost allocation for each NU is calculated based on the MW contribution of each generating facility, as detailed in Appendix H.2 for the 2024 scenario.

Assuming all generating facilities in the DPP 2018 August West Area group advance, a summary of the costs for total NUs (NUs for ERS, NRIS, and Interconnection Facilities) allocated to each generating facility is listed in Table 9-13.

Table 9-13: Summary of Total NU Costs Allocated to Each Generation Project

Project	Max Output (MW)	Total Cost of NU per Project (\$)	\$/MW	Share %
J951	80	\$4,000,000	\$50,000	0.37%
J952	54	\$6,164,349	\$114,155	0.57%
J953	1.83	\$0	\$0	0.00%
J954	1.4	\$0	\$0	0.00%
J958	100	\$11,264,609	\$112,646	1.03%
J959	150	\$4,112,985	\$27,420	0.38%
J963	10.2	\$0	\$0	0.00%
J966	400	\$60,327,888	\$150,820	5.53%
J967	150	\$7,119,266	\$47,462	0.65%
J971	250	\$40,266,134	\$161,065	3.69%
J975	150	\$21,587,859	\$143,919	1.98%
J977	300	\$54,872,349	\$182,908	5.03%
J981	200	\$5,000,000	\$25,000	0.46%
J982	300	\$35,545,311	\$118,484	3.26%
J997	200	\$7,598,412	\$37,992	0.70%
J998	50	\$8,061,709	\$161,234	0.74%
J999	100	\$1,500,000	\$15,000	0.14%
J1001	40	\$5,052,922	\$126,323	0.46%
J1006	100	\$12,473,881	\$124,739	1.14%
J1008	50	\$24,753,232	\$495,065	2.27%
J1024	200	\$23,100,890	\$115,504	2.12%
J1032	50	\$12,437,403	\$248,748	1.14%
J1038	32.5	\$13,726,635	\$422,358	1.26%
J1040	250	\$77,238,741	\$308,955	7.08%
J1041	20	\$8,143,961	\$407,198	0.75%
J1045	20	\$2,003,334	\$100,167	0.18%
J1050	225	\$6,374,828	\$28,333	0.58%
J1054	30	\$6,018,524	\$200,617	0.55%
J1057	125	\$29,803,154	\$238,425	2.73%
J1061	300	\$23,008,089	\$76,694	2.11%
J1072	150	\$6,809,633	\$45,398	0.62%
J1084	150	\$2,000,000	\$13,333	0.18%

Project	Max Output (MW)	Total Cost of NU per Project (\$)	\$/MW	Share %
J1086	120	\$31,105,265	\$259,211	2.85%
J1092	100	\$3,500,000	\$35,000	0.32%
J1098	40	\$3,855,668	\$96,392	0.35%
J1105	200	\$15,381,476	\$76,907	1.41%
J1106	414	\$59,701,678	\$144,207	5.47%
J1108	150	\$90,905,665	\$606,038	8.33%
J1109	207	\$33,868,847	\$163,618	3.11%
J1110	100	\$3,412,763	\$34,128	0.31%
J1114	200	\$24,261,116	\$121,306	2.22%
J1122	200	\$21,502,682	\$107,513	1.97%
J1124	100	\$3,603,735	\$36,037	0.33%
J1128	150	\$19,382,041	\$129,214	1.78%
J1131	100	\$2,500,000	\$25,000	0.23%
J1132	50	\$5,305,112	\$106,102	0.49%
J1135	50	\$500,000	\$10,000	0.05%
J1140	100	\$13,699,332	\$136,993	1.26%
J1143	100	\$15,062,842	\$150,628	1.38%
J1149	200	\$14,242,456	\$71,212	1.31%
J1164	200	\$46,748,785	\$233,744	4.29%
J1169	50	\$4,930,338	\$98,607	0.45%
J1170	200	\$24,381,651	\$121,908	2.24%
J1174	300	\$21,965,005	\$73,217	2.01%
J1175	300	\$25,908,086	\$86,360	2.38%
J1177	300	\$30,018,068	\$100,060	2.75%
J1179	300	\$11,000,000	\$36,667	1.01%
J1181	200	\$14,290,234	\$71,451	1.31%
J1182	250	\$7,015,013	\$28,060	0.64%
J1187	151.8	\$22,348,307	\$147,222	2.05%
Total/Average	9072.7	\$1,090,762,263	\$126,779	100.00%

Model Development

A.1 DPP 2018 April Generation Projects

Table A-1: DPP 2018 April West Area Projects

MISO Project Num	State	County	Trans. Owner	Point of Interconnection	ERIS Output	NRIS Output	Fuel Type	Service Type
J951	MN	Kittson	OTP	Donaldson - Warsaw 115 kV	80	0	Wind	ERIS
J952	SD	Corson	MDU	McIntosh Junction 115 kV	54	0	Wind	ERIS
J953	IA	Johnson	ITCM	AMIL.IOW_AFRYT	1.83	1.83	Diesel	External NRIS
J954	IA	Johnson	ITCM	AMIL.IOW_AFRYT	1.4	1.4	Solar	External NRIS
J958	MN	Lincoln	Xcel	Buffalo Ridge 115 kV	100	100	Wind	NRIS
J959	IA	Fayette	SMMPA	Windsor 161 kV	150	150	Wind	NRIS
J963	IA	Cedar	ITCM	Bennett - Graham 69 kV	10.2	10.2	Diesel	NRIS
J966	MO	Putnam, Sullivan	ATXI	Zachary 345 kV	400	400	Wind	NRIS
J967	MN	Mower	Xcel	Adams 345 kV	150	150	Wind	NRIS
J971	MN	Nobles	Xcel	Nobles Co - Fenton 115 Kv	250	250	Wind	NRIS
J975	ND	Cass	OTP	Buffalo 115 kV	150	150	Wind	NRIS
J977	MN	Jackson, Martin	ITCM	Jackson North 161 kV	300	300	Wind	NRIS
J981	IA	Washington	MEC	Sub T 345 kV	200	200	Wind	NRIS
J982	IA	Dickinson, Emmet	MEC	Obrien County - Kossuth 345 kV (33 mi from Obrien)	300	300	Wind	NRIS
J997	ND	LaMoure	MDU	Ellendale345 230 kV	200	200	Solar	NRIS
J998	IA	Dallas	MEC	Adel - Waukee 69 kV	50	50	Solar	NRIS
J999	IA	Linn	ITCM	Fairfax 161 kV	100	100	Solar	NRIS
J1001	MN	Lincoln	Xcel	Buffalo Ridge 115 kV	40	40	Solar	NRIS
J1006	SD	Brookings	Xcel	Brookings - Yankee 115 kV	100	100	Solar	NRIS
J1008	MN	Faribault	ITCM	Blue Earth IPL - Blue Earth 69 kV	50	50	Solar	NRIS
J1024	MO	Nodaway	MEC	J611 - Clarinda 161 kV	200	200	Wind	NRIS

MISO Project Num	State	County	Trans. Owner	Point of Interconnection	ERIS Output	NRIS Output	Fuel Type	Service Type
J1032	MN	Wabasha	Xcel	Briggs Rd. - N. Rochester 345 kV	50	50	Battery	NRIS
J1038	MN	Grant	OTP	Grant Co. - E. Fergus Falls - Hoot Lake 115 kV	32.5	0	Solar	ERIS
J1040	ND	McIntosh	MDU	Wishek Junction 230 kV	250	250	Wind	NRIS
J1041	MN	Jackson	GRE	Brewster-Heron Lake 161 kV	20	20	Battery	NRIS
J1045	MN	Murray	Xcel	Fenton - Chanarambie 115 kV	20	20	Battery	NRIS
J1050	IA	Boone, Hamilton	ITCM	Doud Tap 161 kV	225	225	Wind	NRIS
J1054	MN	Goodhue	Xcel	Cannon Falls 115 kV	30	30	Battery	NRIS
J1057	MN	Jackson	GRE	Brewster-Heron Lake 161 kV (0.2 mi from Brewster)	125	125	Solar	NRIS
J1061	MN	Wabasha	Xcel	Briggs Rd - N. Rochester 345 kV	300	300	Solar	NRIS
J1072	MN	Mower	Xcel	Adams 345 kV	150	150	Solar	NRIS
J1084	IA	Clinton	ITCM	Rock Creek 345 kV	150	150	Solar	NRIS
J1086	MN	Nobles	ITCM	Elk - Brewster 161 kV	120	120	Solar	NRIS
J1092	WI	Saint Croix	Xcel	Three Lakes 115 kV	100	100	Solar	NRIS
J1098	MN	Jackson	Xcel	Lakefield 345 kV	40	40	Solar	NRIS
J1105	MN	Dakota	Xcel	Hampton Corners 345 kV	200	200	Solar	NRIS
J1106	MN	Redwood	GRE	Lyon County - Cedar Mountain 345 kV (12 mi from Lyon Co)	414	414	Wind	NRIS
J1108	MN	Douglas	MRES	Alexandria 345 kV	150	150	Solar	NRIS
J1109	ND	Cass	Xcel	Bison 345 kV	207	207	Wind	NRIS
J1110	MN	Mower	SMMPA	North Austin 161 kV	100	100	Solar	NRIS
J1114	IA	Plymouth	MEC	J506 (Raun - Highland 345 kV)	200	200	Wind	NRIS
J1122	IA	Pottawattamie	MEC	Council Bluffs - Fallow Avenue 345 kV (23.56 mi from Overland)	200	200	Wind	NRIS
J1124	MN	Olmsted	SMMPA	Byron 345 kV	100	100	Solar	NRIS
J1128	MN	Freeborn	SMMPA	Hayward - Murphy Creek 161 kV	150	150	Solar	NRIS
J1131	IA	Scott	MEC	Sub 56 161 kV	100	100	Solar	NRIS
J1132	IA	Union	ITCM	Creston East 69 kV	50	50	Solar	NRIS
J1135	IA	Des Moines	ITCM	Agency Street 69 kV	50	50	Solar	NRIS
J1140	MN	Benton	MP	Platte River 115 kV	100	100	Solar	NRIS
J1143	MN	Morrison	MP	Blanchard - Bellevu 115 kV (1000 feet from Blanchard)	100	100	Solar	NRIS
J1149	MN	Yellow	Xcel	Hazel Creek 230 kV	200	200	Solar	NRIS

MISO Project Num	State	County	Trans. Owner	Point of Interconnection	ERIS Output	NRIS Output	Fuel Type	Service Type
		Medicine						
J1164	MN	Rock	ITCM	Magnolia 161 kV	200	200	Solar	NRIS
J1169	SD	McCook	Xcel	Grant 115 kV	50	50	Solar	NRIS
J1170	ND	Cass	Xcel	Bison 345 kV	200	200	Solar	NRIS
J1174	IA	Worth	ITCM	Bison - Colby 345 kV (16 mi from Colby)	300	300	Solar	NRIS
J1175	IA	Worth	ITCM	Bison - Colby 345 kV (16 mi from Colby)	300	300	Wind	NRIS
J1177	MO	Schuyler	ATXI	Mark Twain 345 kV	300	300	Wind	NRIS
J1179	MN	Chisago	Xcel	Rush City - Red Rock 230 kV	300	300	Solar	NRIS
J1181	IA	Chickasaw	ITCM	Hazleton - Mitchell county 345 kV	200	200	Wind	NRIS
J1182	MO	Adair	ATXI	Zachary 345 kV	250	250	Solar	NRIS
J1187	ND	Mercer	GRE	Stanton 230 kV	151.8	151.8	Wind	NRIS

Table A-2: Dynamic Modeling, Collector System and Shunt Compensation Modeling for DPP West Area Projects

MISO Project #	Turbine / Inverter	Shunt Compensation	Generator Modeling	Generator Reactive Power Capability	Collector System
J951	32 GE 2.5 MW -116	None	One 80 MW unit	Qmin = - 0.8217 MVAR Qmax = +0.8217 MVAR per turbine	R = 0.01324 pu X = 0.01924 pu B = 0.0348 pu
J952	15 Vestas V136 3.6 MW	7.2 MVAR	One 54 MW unit	Qmin = -1.423 MVAR Qmax = +1.609 MVAR per turbine	R = 0.00287 X = 0.00426 B = 0.00906
J958	40 GE 2.5 MW -116	2x6 MVAR	One 100 MW unit	Qmin = - 0.822 MVAR Qmax = +0.822 MVAR per turbine	R = 0.0074 X = 0.0112 B = 0.0292
J959	60 GE 2.5 MW -116	2x9 MVAR	One 150 MW unit	Qmin = - 0.8217 MVAR Qmax = +0.8217 MVAR per turbine	R = 0.007 X = 0.0123 B = 0.04553
J963	3 CAT Diesel 2 MW	None	One 4.2 MW unit One 6 MW unit	PF: 0.95 (For existing generation) Qmin = - 1.38047 MVAR Qmax = +1.3804 MVAR PF: 0.8 (For CAT Diesel) Qmin = -1.5 MVAR Qmax = +1.5 MVAR	Not Applicable
J966	160 GE 2.5 MW -116	2x14 MVAR	One 400 MW unit	Qmin = - 0.8217 MVAR Qmax = +0.8217 MVAR per turbine	R = 0.0039 X = 0.0064 B = 0.0507
J967	60 GE 2.5 MW -116	2x14 MVAR	One 150 MW unit	Qmin = - 0.8217 MVAR Qmax = +0.8217 MVAR per turbine	R = 0.007 X = 0.0123 B = 0.04553
J971	64 Vestas V136 - 3.6 MW and 10 Vestas V110 - 2.0 MW	1x16 MVAR	One 230 MW unit One 20 MW unit	Vestas V136 Qmin = -1.4228 Qmax = 1.6086 per turbine Vestas V110 Qmin = -0.665 Qmax = 0.660 per turbine	Collector1 (For Vestas V136) R = 0.00409 X = 0.00653 B = 0.10606 Collector2 (For Vestas V110) R = 0.00665 X = 0.01062 B = 0.012
J975	60 GE 2.5 MW -116	1x6 MVAR	One 150 MW unit	Qmin = - 1.2108 MVAR Qmax = +1.2108 MVAR per turbine	R = 0.00340 X = 0.00490 B = 0.01250

MISO Project #	Turbine / Inverter	Shunt Compensation	Generator Modeling	Generator Reactive Power Capability	Collector System
J977	150 Vestas V110 2 MW	2x26 MVAR	Two 150 MW units	Qmin = - 0.65736 MVAR Qmax = +0.65736 MVAR per turbine	Collector 1: R = 0.01156 X = 0.01097 B = 0.05129 Collector 2: R = 0.00984 X = 0.00903 B = 0.05059
J981	100 Vestas V110 2 MW	2x13.5 MVAR	One 92 MW unit One 108 MW unit	Qmin = - 0.65736 MVAR Qmax = +0.65736 MVAR per turbine	Collector 1: R = 0.03013 X = 0.02732 B = 0.05867 Collector 2: R = 0.04341 X = 0.03548 B = 0.08374
J982	150 Vestas V110 2 MW	2x25.5 MVAR	Two 150 MW units	Qmin = - 0.65736 MVAR Qmax = +0.65736 MVAR per turbine	Collector 1: R = 0.02885 X = 0.02659 B = 0.12103 Collector 2: R = 0.02800 X = 0.02534 B = 0.11000
J997	54 TMEIC 3.71 MW	2x14 MVAR	One 200 MW unit	Qmin = - 1.2194 MVAR Qmax = 1.2194 MVAR per inverter	R = 0.00037 X = 0.00068 B = 0.00394
J998	47 GE 1.064 MW	1x7 MVAR	One 50 MW unit	Qmin = - 0.3497 MVAR Qmax = 0.3497 MVAR per inverter	R = 0.0018 X = 0.0011 B = 0.0012
J999	27 TMEIC 3.71 MW	2x7 MVAR	One 100 MW unit	Qmin = - 1.2194 MVAR Qmax = 1.2194 MVAR per inverter	R = 0.0013 X = 0.0012 B = 0.0025
J1001	11 TMEIC 3.71 MW	1x6 MVAR	One 40 MW unit	Qmin = - 1.2194 MVAR Qmax = 1.2194 MVAR per inverter	R = 0.0018 X = 0.0011 B = 0.0012
J1006	27 TMEIC 3.71 MW	2x7 MVAR	One 100 MW unit	Qmin = - 1.2194 MVAR Qmax = 1.2194 MVAR per inverter	R = 0.0013 X = 0.0012 B = 0.0025
J1008	14 TMEIC 3.71 MW	1x7 MVAR	One 50 MW unit	Qmin = - 1.2194 MVAR Qmax = 1.2194 MVAR per inverter	R = 0.0018 X = 0.0011 B = 0.0012

MISO Project #	Turbine / Inverter	Shunt Compensation	Generator Modeling	Generator Reactive Power Capability	Collector System
J1024	94 Vestas V120 2.2 MW	1x30 MVAR	One 200 MW unit	Qmin = - 0.7230 MVAR Qmax = 0.7230 MVAR per turbine	R = 0.01094 X = 0.00953 B = 0.09425
J1032	14 4.2 MVA TMEIC Ninja 840KW (5*0.84 MVA) EES	None	One 50 MW unit	Qmin = -30.94 MVAR Qmax = +30.94 MVAR	Not Available
J1038	14 2.5 MW SMA 250-EV-US	1*5 MVAR	One 32.5 MW unit	Qmin = - 21 MVAR Qmax = 21 MVAR	R = 0.000003 pu X = 0.000001 pu B = 0.00585 pu
J1040	100 GE 2.52 MW -127	1*17 MVAR	Two 125 MW units	Qmin = - 122.05 MVAR Qmax = 122.05 MVAR	Collector 1: R = 0.00547 pu X = 0.00639 pu B = 0.02249 pu Collector 2: R = 0.0102 pu X = 0.01436 pu B = 0.03942 pu
J1041	6 4.2 MVA TMEIC Ninja 840KW (5*0.84 MVA) EES	None	One 20 MW unit	Qmin = - 15.33 MVAR Qmax = + 15.33 MVAR	Not Available
J1045	6 4.2 MVA TMEIC Ninja 840KW (5*0.84 MVA) EES	None	One 20 MW unit	Qmin = - 15.33 MVAR Qmax = + 15.33 MVAR	Not Available
J1050	90 GE 2.5 MW -116	4*15 MVAR	Two 112.5 MW units	For each unit Qmin = - 54.49 MVAR Qmax = + 54.49 MVAR	Collector 1: R = 0.0054307 pu X = 0.005201 pu B = 0.0505 pu Collector 2: R = 0.009021 pu X = 0.0009061 pu B = 0.0764 pu
J1054	8 4.2 MVA TMEIC Ninja 840KW (5*0.84 MVA) EES	None	One 30 MW unit	Qmin = -15.13 MVAR Qmax = +15.13 MVAR	R = 0.01929 pu X = 0.0201 pu B = 0.01 pu
J1057	34 4.2 MVA TMEIC Ninja 840 KW (5*0.84 kVA) PV	1*4 MVAR	One 125 MW unit	Qmin = -60.54 MVAR Qmax = +60.54 MVAR	R = 0.0046 pu X = 0.0069 pu B = 0.01250 pu
J1061	80 4.2 MVA TMEIC Ninja 840 KW (5*0.84 kVA) PV	4*2 MVAR	One 300 MW unit	Qmin = -146.4 MVAR Qmax = +146.4 MVAR	R = 0.0046 pu X = 0.0069 pu B = 0.01250 pu
J1072	41 TMEIC 4.05 MVA	2*12 MVAR	One 150 MW unit	Qmin = -49.3 MVAR Qmax = +49.3 MVAR	R = 0.0011 pu X = 0.0009 pu B = 0.0054 pu
J1084	41 TMEIC 4.05 MVA	2*12 MVAR	One 150 MW unit	Qmin = -49.3 MVAR Qmax = +49.3 MVAR	R = 0.0011 pu X = 0.0009 pu

MISO Project #	Turbine / Inverter	Shunt Compensation	Generator Modeling	Generator Reactive Power Capability	Collector System
					B = 0.0054 pu
J1086	41 3 MW HEM PV	None	One 120 MW unit	Qmin = - 39.4421 MVAR Qmax = + 39.4421 MVAR	R = 0.00414 pu X = 0.003779 pu B = 0.00917 pu
J1092	152 Schneider XC680-NA	None	One 100 MW unit	Qmax = +48.4 MVAR Qmin = - 48.4 MVAR	R = 0.001858 pu X = 0.001644 pu B = 0.000363 pu
J1098	69 Schneider XC680-NA	None	One 40 MW unit	Qmin = - 19.37 MVAR Qmax = +19.37 MVAR	R = 0.02037 pu X = 0.02274 pu B = 0.02335 pu
J1105	54 TMEIC 4.05 MVA	2*14 MVAR	One 200 MW unit	Qmin = - 65.73 MVAR Qmax = +65.73 MVAR	R = 0.001 pu X = 0.0008 pu B = 0.00820 pu
J1106	120 Vestas V126 3.45 MW	None	One 414 MW unit	Qmin = - 176.36 MVAR Qmax = + 200.51 MVAR	R = 0.004 pu X = 0.013 pu B = 0.094 pu
J1108	60 2.5 MW TMEIC PVH-L2700GR	1 x 20 MVAR capacitor	One 150 MW unit	Qmin = -61.188 MVAR Qmax = +61.188 MVAR	R = 0.003 pu X = 0.0029 pu B = 0.0237 pu
J1109	60 Vestas V136, 3.45 MW, 3.75 MVA	None	One 207 MW unit	Qmin = -88.1816 MVAR Qmax = 100.2547 MVAR	R = 0.007 pu X = 0.025 pu B = 0.047 pu
J1110	34 Power Electronics HEMK FS3150KU 3.15 MVA	None	One 100 MW unit	Qmin = -48.55 MVAR Qmax = 48.55 MVAR	R = 0.00622 pu X = 0.00846 pu B = 0.02999 pu
J1114	100 Vestas V110 2 MW	2x 27MVAR	Two 100 MW units	For each unit Qmin = -32.87 MVAR Qmax = 32.87 MVAR	Collector 1: R = 0.026793 pu X = 0.030726 pu B = 0.02195 pu Collector 2: R = 0.026793 pu X = 0.030726 pu B = 0.02195 pu
J1122	91 Vestas V110 2.2 MW	1 x 60 MVAR capacitor	One 200 MW unit	Qmin = -65.7368 MVAR Qmax = +65.7368 MVAR	R = 0.006 pu X = 0.0074 pu B = 0.1385 pu
J1124	34 Power Electronics HEMK FS3150KU 3.15MVA, 3.276 MVA derated	None	One 100 MW unit	Qmin = -48.55 MVAR Qmax = 48.55 MVAR	R = 0.00622 pu X = 0.00846 pu B = 0.0299 pu
J1128	51 Power Electronics FS3000MU	None	One 150 MW unit	Qmin = -49.30 MVAR Qmax = 49.30 MVAR	R = 0.003403 pu X = 0.003126 pu B = 0.011686 pu

MISO Project #	Turbine / Inverter	Shunt Compensation	Generator Modeling	Generator Reactive Power Capability	Collector System
J1131	34 Power Electronics FS3000CU15	None	One 100 MW unit	Qmin = -59.55 MVAR Qmax = 59.55 MVAR	R = 0.00776 pu X = 0.00951 pu B = 0.02263 pu
J1132	17 Power Electronics FS3000CU15	None	One 50 MW unit	Qmin = -29.7 MVAR Qmax = 29.7 MVAR	R = 0.01593 pu X = 0.01958 pu B = 0.01154 pu
J1135	17 Power Electronics FS3000CU15	None	One 50 MW unit	Qmin = -29.7 MVAR Qmax = 29.7 MVAR	R = 0.01593 pu X = 0.01958 pu B = 0.01154 pu
J1140	36 Power Electronics FS3001CU15	4X8 MVAR	One 100 MW unit	Qmin = -32.87 MVAR Qmax = 32.87 MVAR	R = 0.00070 pu X = 0.00112 pu B = 0.00046 pu
J1143	36 Power Electronics FS3001CU15	4X8 MVAR	One 100 MW unit	Qmin = -32.87 MVAR Qmax = 32.87 MVAR	R = 0.00070 pu X = 0.00112 pu B = 0.00046 pu
J1149	80 SMA SC2750	1X36 MVAR	One 200 MW unit	Qmin = -85.2 MVAR Qmax = 85.2 MVAR	R = 0.001028 pu X = 0.00213 pu B = 0.00458 pu
J1164	72 Power Electronics FS3001CU15	Two 4X8 MVAR	One 200 MW unit	Qmin = - 65.737 MVAR Qmax = 65.737 MVAR	R = 0.0007 pu X = 0.00112 pu B = 0.00046 pu
J1169	18 FS3001CU15	4X4 MVAR	One 50 MW unit	Qmin = - 16.86 MVAR Qmax = 16.86 MVAR	R = 0.005008 pu X = 0.006485 pu B = 0.005161 pu
J1170	72 FS3001CU15	Two 4X8 MVAR	One 200 MW unit	Qmin = - 65.737 MVAR Qmax = 65.737 MVAR	R = 0.007 pu X = 0.025 pu B = 0.047 pu
J1174	80 4.2 MVA TMEIC Ninja 840 KW (5*0.84 kVA) PV	2x4 MVAR	One 300 MW unit	Qmin = - 151.3 MVAR Qmax = 151.3 MVAR	R = 0.0046 pu X = 0.0069 pu B = 0.0125 pu
J1175	120 GE 2.5 MW -116	2X9 MVAR	One 300 MW unit	Qmin = - 145.3 MVAR Qmax = 145.3 MVAR	R = 0.0034 pu X = 0.0049 pu B = 0.0125 pu
J1177	140 Vestas V116 2.2 MW	2x8 MVAR	One 300 MW unit	Qmin = - 0.3135 MVAR Qmax = +0.35824 MVAR per turbine	R = 0.005195 pu X = 0.006491 pu B = 0.21041 pu
J1179	80 4.2 MVA TMEIC Ninja 840 KW (5*0.84 kVA) PV	2x4 MVAR	One 300 MW unit	Qmin = - 151.31 MVAR Qmax = +151.31 MVAR	R = 0.00460 pu X = 0.00690 pu B = 0.01250 pu
J1181	80 GE 2.5 MW -116	2x6 MVAR	One 200 MW unit	Qmin = - 1.2108 MVAR Qmax = +1.2108 MVAR per turbine	R = 0.00340 pu X = 0.0049 pu B = 0.0125 pu

MISO Project #	Turbine / Inverter	Shunt Compensation	Generator Modeling	Generator Reactive Power Capability	Collector System
J1182	103 SMASC 2750-EV 2.4475 MW	None	One 250 MW unit	Qmin = - 1.2539 MVAR Qmax = +1.2539 MVAR per inverter	R = 0.00113140 pu X = 0.00130881 pu B = 0.022690 pu
J1187	69 Vestas V120 2.2 MW	None	One 151.8 MW unit	Qmin = - 0.723 MVAR Qmax = +0.723 MVAR per turbine	R = 0.00510 pu X = 0.00530 pu B = 0.02994 pu

Table A-3: DPP 2018 April Central Area Projects

MISO Project Num	State	County	Trans. Owner	Point of Interconnection	ERIS Output	NRIS Output	Fuel Type	Service Type
J955	IL	Sangamon	ATXI	Austin 345 kV	1040 sum/ 1165 win	1040 sum/ 1165 win	CC	NRIS
J956	MO	Ralls	UEC	Spencer Creek 345 kV	200	200	Solar	NRIS
J968	IN	Jasper, White	NIPS	Reynolds 345 kV	200	200	Wind	NRIS
J974	IL	Fulton, Peoria	AIC	Mapleridge 345 kV	250	250	Wind	NRIS
J976	MO	Warren	UEC	Montgomery - Enon 345 kV	300	300	Solar	NRIS
J979	IL	Christian	ATXI	Pana 345 kV	200	200	Wind	NRIS
J980	IL	Christian	ATXI	Pana 345 kV	200	200	Wind	NRIS
J985	MO	Scott	UEC	Miner - Kelso 161 kV	120	120	Solar	NRIS
J987	MO	Montgomery	UEC	Montgomery 161 kV	100	100	Solar	NRIS
J991	IL	Clay	AIC	Xenia 345 kV	200	200	Solar	NRIS
J992	IN	Cass	DEI	Walton 230 kV	200	200	Solar	NRIS
J993	IN	Boone	IPL	Hortonville - Whitestown 345 kV	200	200	Solar	NRIS
J994	MO	Callaway	UEC	Guthrie 161 kV	100	100	Solar	NRIS
J995	IL	Brown, Jersey	AIC	NW Jerseyville Northwest 138 kV	100	100	Solar	NRIS
J1016	IN	Gibson	SIGE	Francisco - Toyota Z75 138 kV	50	50	Solar	NRIS
J1021	IL	Piatt	AIC	Goose Creek 345 Kv	200	200	Wind	NRIS
J1022	IL	McLean	AIC	Weedman 138 kV	150	150	Wind	NRIS
J1025	MO	Knox	ATXI	Zachary - Maywood 345 kV	300	300	Wind	NRIS
J1026	MO	Audrain, Ralls	UEC	Maywood - Spencer Creek 345 kV	400	400	Wind	NRIS
J1027	IN	Pike	HE	Ratts 161 kV	150	150	Solar	NRIS
J1028	IN	Pike	HE	Ratts - Victory 161 kV	150	150	Solar	NRIS
J1031	IN	Henry	DEI	Fall Creek - New Castle 138 kV	150	150	Wind	NRIS
J1033	MO	Stoddard	UEC	Stoddard - Morely 161 kV	50	50	Battery	NRIS
J1034	MO	Stoddard	UEC	Stoddard - Morley 161 kV	225	225	Solar	NRIS
J1039	MO	Warren	UEC	Enon - Montgomery 345 kV	50	50	Battery	NRIS
J1055	IL	Mason	AIC	Mason 138 kV	144	144	Wind	NRIS
J1058	IN	Lake	NIPS	Schahfer - St. John 345 kV	300	300	Solar	NRIS

MISO Project Num	State	County	Trans. Owner	Point of Interconnection	ERIS Output	NRIS Output	Fuel Type	Service Type
J1059	MO	Cape Girardeau	UEC	Kelso - Lutesville 345 kV	300	300	Solar	NRIS
J1063	IN	Clinton	DEI	New London - Frankfort 230 kV	300	300	Solar	NRIS
J1067	IN	Jasper, Pulaski	NIPS	Reynolds - Burr Oak 345 kV	240	240	Solar	NRIS
J1069	IN	Jasper, Pulaski	NIPS	Reynolds 345 kV	200	200	Wind	NRIS
J1074	IN	Gibson	SIGE	Francisco 138 kV	200	200	Solar	NRIS
J1087	MO	Scott	UEC	Miner - Kelso 161 kV	200	200	Solar	NRIS
J1094	IL	Washington	AIC	Prest 138 kV	150	150	Solar	NRIS
J1096	IL	Saline	AIC	Norris City North - Muddy 138 kV	150	150	Solar	NRIS
J1102	IL	Logan	AIC	Fogarty 138 kV	70	70	Solar	NRIS
J1107	MO	Cape Girardeau	UEC	Kelso - Lutesville 345 kV	200	200	Solar	NRIS
J1111	IL	Jackson	SIPC	Campbell Hill - Jackson 161 kV	150	150	Solar	NRIS
J1112	IN	Kosciusko	NIPS	Leesburg 138 kV	150	150	Solar	NRIS
J1115	IL	Macon	AIC	Latham - Oreana 345 kV	200	200	Wind	NRIS
J1123	MO	Audrain	UEC	Spencer Creek 345 kV	150	150	Wind	NRIS
J1139	IL	Champaign	AIC	Sidney 138 kV	150	150	Solar	NRIS
J1145	MO	Callaway	UEC	Overton - (McCrede) - Montgomery 345 kV	250	250	Solar	NRIS
J1148	IN	Posey	SIGE	AB Brown 138 kV	54	54	CT	NRIS
J1152	IN	Hancock, Shelby	IPL	Gwynneville - Sunnyside 345 kV	200	200	Solar	NRIS
J1161	MO	Scott	UEC	Miner - Kelso 161 kV	100.23	100.23	Solar	NRIS
J1180	IL	Clark	AIC	Casey West - Sullivan 345 kV	75	75	Solar	NRIS
J1189	IN	Brown, Martin	DEI	Crane Solar 69 kV	5	5	Battery	NRIS

Table A-4: DPP 2018 April Michigan Area Projects

MISO Project Num	State	County	Trans. Owner	Point of Interconnection	ERIS Output	NRIS Output	Fuel Type	Service Type
J978	MI	Montcalm	METC	Eureka - Vestaburg 138 kV	100	100	Wind	NRIS
J984	MI	Gratiot	METC	Nelson Rd. 345 kV	200	200	Wind	NRIS
J989	MI	Oakland	METC	Halsey 138 kV	80	80	Solar	NRIS
J996	MI	Lenawee	METC	Beecher - Moraco 138 kV	80	80	Solar	NRIS
J1005	MI	Midland, Saginaw	METC	Murphy 345 kV	200	200	Solar	NRIS
J1013	MI	Branch	METC	Batavia 138 kV	80	80	Solar	NRIS
J1014	MI	Jackson	METC	Tompkins - Blackstone 138 kV	80	80	Solar	NRIS
J1043	MI	Montcalm	METC	Vergennes - Nelson Rd 345 kV	374.4	374.4	Wind	NRIS
J1046	MI	Washtenaw	ITCT	Majestic - Lemoyne 345 kV	50	50	Battery	NRIS
J1056	MI	Monroe	ITCT	Lulu - Monroe 345 kV	200	200	Wind	NRIS
J1062	MI	Washtenaw	ITCT	Majestic - Lemoyne 345 kV	200	200	Solar	NRIS
J1064	MI	Ionia	METC	Marquette - Vergennes 138 kV	200	200	Wind	NRIS
J1065	MI	Gratiot	METC	Alma - Vestaburg 138 kV	200	200	Wind	NRIS
J1068	MI	Washtenaw	ITCT	Dorset 120 kV	200	200	Wind	NRIS
J1071	MI	Oceana	METC	Donaldson 138 kV	100	100	Solar	NRIS
J1083	MI	Eaton	METC	Oneida138 kV	70	70	Solar	NRIS
J1088	MI	Shiawassee	METC	Cornell - Layton 138 kV	150	150	Solar	NRIS
J1089	MI	Shiawassee	METC	Cornell - Bell Rd 138 kV	170	170	Solar	NRIS
J1090	MI	Ingham	METC	Tompkins - Churchill Jct 138 kV	90	90	Solar	NRIS
J1103	MI	Tuscola	ITCT	Kirk 345 kV	20	20	Battery	NRIS
J1117	MI	Ingham	METC	Green 138 kV	80	80	Solar	NRIS
J1119	MI	Jackson	METC	Tompkins 345 kV	600	600	CC	NRIS
J1172	MI	Genesee	METC	Dort - Garfield 138 kV	50	50	Solar	NRIS
J1173	MI	Lenawee	CE	Raisin - METC Tap 138 kV	80	80	Solar	NRIS
J1178	MI	Eaton	METC	Oneida 138 kV	65	65	Solar	NRIS

Table A-5: DPP 2018 April ATC Area Projects

MISO Project Num	State	County	Trans. Owner	Point of Interconnection	ERIS Output	NRIS Output	Fuel Type	Service Type
J986	WI	Wood	ATC	Port Edwards - Sand Lake 138kV	149.76	149.76	Solar	NRIS
J1000	WI	Grant	ATC	Nelson Dewey 138 kV	50	50	Solar	NRIS
J1002	WI	Waushara	ATC	Wautoma 138 kV	99	99	Solar	NRIS
J1003	WI	Dodge	ATC	North Beaver Dam 69 Kv	50	50	Solar	NRIS
J1009	WI	Kenosha, Racine	ATC	Pleasant Prairie - Racine 345 kV	400	400	Solar	NRIS
J1010	WI	Kenosha, Racine	ATC	Arcadian - Zion 345 kV	400	400	Solar	NRIS
J1011	WI	Kenosha, Racine	ATC	Arcadian - Zion 345 kV	400	400	Solar	NRIS
J1042	WI	Walworth	ATC	North Lake Geneva 138 kV	200	200	Solar	NRIS
J1051	WI	Jefferson	ATC	Concord 138 kV	50	50	Battery	NRIS
J1053	WI	Jefferson	ATC	Concord 138 kV	300	300	Solar	NRIS
J1085	WI	Marathon, Taylor, Wood	ATC	Gardner Park - Stone Lake 345 kV	300	300	Wind	NRIS
J1101	WI	Manitowoc	ATC	Kewaunee 138 kV	20	20	Battery	NRIS
J1121	WI	Marathon	ATC	Gardner Park - Stone Lake 345 kV	200	200	Solar	NRIS
J1153	WI	Sheboygan	ATC	Holland 138 kV	150	150	Solar	NRIS
J1154	WI	Jefferson	ATC	Jefferson 138 kV	75	75	Solar	NRIS
J1171	WI	Dodge	ATC	Butternut 138 kV	100	100	Solar	NRIS
J1183	MI	Delta	ATC	Heritage Garden	1.35	0	Solar	ERIS
J1188	WI	Rock	ATC	Sheepskin 69 kV	50	50	Solar	NRIS

A.2 DPP 2017 August West Area Phase 1 Network Upgrades

Table A-6: DPP 2017 August West Phase 1 NUs

Network Upgrades	Owner	Cost (\$)
Build Twin Brooks-Alexandria 345 kV new line	OTP MRES	\$242,400,000
Add a Mandan-J580 POI 230 kV line	MDU	\$28,000,000
Rebuild Merricourt-Wishek 230 kV line	MDU	\$5,800,000
Increase clearance on Merricourt-Tatanka North 230 kV line	MDU	\$100,000
Coon Creek Substation Reconfiguration	XEL	\$15,000,000
Sheyenne-Lake Park 230 kV Uprate	XEL OTP MPC	\$1,300,000
Sheyenne-Mapleton 115 kV Rebuild	XEL OTP	\$17,600,000
Magic City-Souris 115 kV Rebuild	XEL	\$7,000,000
Granite City-Crossroads 115 kV Rebuild	XEL	\$3,700,000
Crossroads-MEI International 115 kV Rebuild	XEL	\$1,300,000
MEI International-Westwood 115kV Rebuild	XEL	\$2,000,000
West St. Cloud-Westwood 115 kV Rebuild	XEL	\$600,000
Buffalo 345/115 kV Transformer #1 Upgrade	OTP	\$3,000,000
Buffalo 345/115 kV Transformer #2 Upgrade	OTP	\$3,000,000
Mapleton-CSLTNET 115 kV Uprate	OTP	\$25,000
Mahnomn-Ulrich 115 kV Uprate	OTP	\$50,000
Winger-Fertile 115 kV Uprate	OTP	\$100,000
Wilton-Bemidji 115 kV Uprate	OTP	\$50,000
Crookston-Fertile 115 kV Uprate	OTP	\$250,000
Crookston-Falconer 115 kV Uprate	OTP	\$25,000
Devil's Lake-Ramsey 115 kV Rebuild	OTP	\$450,000
Badura-Hubbard 230 kV Uprate	MP GRE	\$600,000
Riverton-Mud Lake 230 kV Uprate	GRE	\$882,000
McHenry 230/115 kV Transformer #1 Upgrade	GRE	\$3,000,000
Hubbard-Audubon 230 kV Uprate	MP GRE XEL OTP	\$2,000,000
Wanda-Sundown 69 kV Rebuild	GRE	\$3,150,000

Network Upgrades	Owner	Cost (\$)
Wanda-Waterbury 69 kV Rebuild	GRE	\$1,500,000
Windms-Wilder 69 kV Uprate	GRE	\$100,000
Heron Lake 161/69 kV Transformer #1 Upgrade	ITCM	\$2,520,000
Heron Lake 161/69 kV Transformer #2 Upgrade	ITCM	\$2,520,000
Walters-Alden 69 kV Rebuild	ITCM	\$4,700,000
Alden-Aleawst 69 kV Rebuild	ITCM	\$5,100,000
Cottonwood-Waterbury 69 kV Rebuild	ITCM	\$4,400,000
Cottonwood-Highway Tap 69 kV Rebuild	ITCM	\$725,000
Highway Tap-Lambrtn 69 kV Rebuild	ITCM	\$334,000
Jefferson Tap-N Storden 69 kV Rebuild	ITCM	\$2,900,000
Jefferson Tap-Storden 69 kV Rebuild	ITCM	\$900,000
N Storden-Lambrtn 69 kV Rebuild	ITCM	\$5,900,000
S Storden-Heron Lake 69 kV Rebuild	ITCM	\$8,400,000
S Storden-Storden 69 kV Rebuild	ITCM	\$4,000,000
Springfield-Dotson Corners 69 kV Uprate	ITCM GRE	\$60,000
Brookville-Sundown 69 kV Rebuild	ITCM	\$2,700,000
Brookville-Cobden Tap 69 kV Rebuild	GRE ITCM	\$2,570,000
Springfield-Cobden Tap 69 kV Rebuild	ITCM	\$396,000
New J628 POI– Prairie 230 kV 2nd Circuit	GRE	\$22,360,000
J628 POI–Prairie 230 kV Rebuild	GRE	\$16,770,000
J628 POI– J897 POI 230 kV Rebuild	GRE	\$28,545,000
J889 POI–Ramsey 230 kV Rebuild	GRE	\$46,290,000
Alexandria four 50 MVar Capacitor Banks and 345 kV bus expansion	MRES	\$26,000,000
Morgan Vallley-Tiffin 345 kV Uprate	MEC	\$1,500,000
Hills-Tiffin 345 kV Uprate	MEC	\$400,000

A.3 Model Review Comments

Table A-7: Model Review Comments

Company	Python/ Idev File Name	2024 SH Study	2024 SH Benchmark	2024 SPK Study	2024 SPK Benchmark
ICs	J951 Comments.py	X	X	X	X
ICs	J1038 Comments.py	X	X	X	X
ICs	J1050 Comments.py	X	X	X	X
ICs	J1086 Comments.py	X	X	X	X
ICs	J1086 Comments_SH.py	X	X		
ICs	J1086 Comments_PK.py			X	X
ICs	J1092 Comments.py	X	X	X	X
ICs	J1098 Comments.py	X	X	X	X
ICs	J1106 Comments.py	X	X	X	X
ICs	J1108 Comments.py	X	X	X	X
ICs	J1109 Comments_SH.py	X	X		
ICs	J1109 Comments_PK.py			X	X
ICs	J1110 Comments.idv	X	X	X	X
ICs	J1114_Update_SH.idv	X	X		
ICs	J1114_Update_PK.idv			X	X
ICs	J1122 Comments.py	X	X	X	X
ICs	J1124 Comments.py	X	X	X	X
ICs	J1128 Comments.py	X	X	X	X
ICs	J1140 Comments.py	X	X	X	X
ICs	J1143 Comments.py	X	X	X	X
ICs	J1164 Comments.py	X	X	X	X
ICs	J1169 Comments.py	X	X	X	X
ICs	J1170 Comments.py	X	X	X	X
ICs	J1187 Comments.py	X	X	X	X
ATC	ATC_SH_Study.py	X			
ATC	ATC_SH_Bench.py		X		
ATC	ATC_PK_Study.py			X	
ATC	ATC_PK_Bench.py				X
CIPCO	CIPCO IR23_V33_SH.idv	X	X		
CIPCO	CIPCO IR23_V33_PK.idv			X	X
CIPCO	CIPCO IR24-IR34_V33_SH.IDV	X	X		

Company	Python/ Idev File Name	2024 SH Study	2024 SH Benchmark	2024 SPK Study	2024 SPK Benchmark
CIPCO	CIPCO IR24-IR34_V33_PK.IDV			X	X
MDU	MDU_Updates-DPP_2018_APR_West_Phase1_Models_190924.idv	X	X	X	X
MDU	J929_SH_Study.py	X			
MDU	J929_SH_Bench.py		X		
MDU	J929_PK_Study.py			X	
MDU	J929_PK_Bench.py				X
MEC	MEC_DPP_2018_APR_West_Ph1_SH_Updates.py	X	X		
MEC	MEC_DPP_2018_APR_West_Ph1_SUM_Updates.py			X	X
MEC	Turn off retirements.py	X	X	X	X
MEC	Correct V Control.py	X	X	X	X
MEC	Correct X 631144-41814-631139.py	X	X	X	X
MEC	Gen Correction_SH_Study.idv	X			
MEC	Gen Correction_SH_Bench.idv		X		
MEC	Gen Correction_PK_Study.idv			X	
MEC	Gen Correction_PK_Bench.idv				X
MP	J1143 POI Chng.py	X	X	X	X
MPC	MPC-fixrtngs-APR18_West_DPP-SH_study_190918.sav.idv	X	X	X	X
ICs	J1032 Chng.py	X	X	X	X
ICs	J1041 Chng.py	X	X	X	X
ICs	J1045 Chng.py	X	X	X	X
ICs	J1054 Chng.py	X	X	X	X
ICs	J1057 Chng.py	X	X	X	X
ICs	J1061 Chng.py	X	X	X	X
ICs	J1174 Chng.py	X	X	X	X
ICs	J1179 Chng.py	X	X	X	X

A.4 MISO Classic as the Study Sink

Table A-8: MISO Classic as the Study Sink

Area #	Area Name	Area #	Area Name
207	HE	608	MP
208	DEI	613	SMMPA
210	SIGE	615	GRE
216	IPL	620	OTP
217	NIPS	627	ALTW
218	METC	633	MPW
219	ITC	635	MEC
295	WEC	661	MDU
296	MIUP	663	BEPC-MISO
314	BREC	680	DPC
333	CWLD	694	ALTE
356	AMMO	696	WPS
357	AMIL	697	MGE
360	CWLP	698	UPPC
361	SIPC	701	MISO Prior
600	Xcel		

A.5 PJM Market as PJM Projects Sink

Table A-9: PJM Market as PJM Projects Sink

Area #	Area Name	Area #	Area Name
201	AP	230	PECO
202	ATSI	231	PSE&G
205	AEP	232	BGE
209	DAY	233	PEPCO
212	DEO&K	234	AE
215	DLCO	235	DP&L
222	CE	236	UGI
225	PJM	237	RECO
226	PENELEC	320	EKPC
227	METED	345	DVP
228	JCP&L	363	LGEE
229	PPL	703	PJM Prior

A.6 SPP Market as SPP Projects Sink

Table A-10: SPP Market as SPP Projects Sink

Area #	Area Name	Area #	Area Name
515	SWPA	541	KCPL
520	AEPW	542	KACY
523	GRDA	544	EMDE
524	OKGE	545	INDN
525	WFEC	546	SPRM
526	SPS	640	NPPD
527	OMPA	645	OPPD
531	MIDW	650	LES
534	SUNC	652	WAPA
536	WERE	659	BEPC-SPP
540	GMO	702	SPP Prior

A.7 Contingency Files used in Steady-State Analysis

Table A-11: List of Contingencies used in Steady-State Analysis

Contingency File Name	Description	Shoulder	Peak
Automatic single element contingencies	Single element outages at buses 60 kV and above in the study region	x	x
CC Bipole Events.con	Specified category P1, P7 contingencies in GRE Coal Creek	x	x
HVDC_Red_2024SH.con	Contingencies with HVDC reduction	x	
HVDC_Red_2024PK.con	Contingencies with HVDC reduction		x
MEC-DPP2018APRWESTPH1-Cat P1-2019.10.08.con	Specified category P1 contingencies in MEC	x	x
MEC-DPP2018APRWESTPH1-Cat P2-2019.10.08.con	Specified category P2 contingencies in MEC	x	x
MEC-DPP2018APRWESTPH1-Cat P5-2019.10.08.con	Specified category P5 contingencies in MEC	x	x
MEC-DPP2018APRWESTPH1-Cat P7-2019.10.08.con	Specified category P7 contingencies in MEC	x	x
MISO19_2024_SUM_TA_P1_ATC.con	Specified category P1 contingencies in ATC	x	x
MISO19_2024_SUM_TA_P1_IOWA.con	Specified category P1 contingencies in Iowa	x	x
MISO19_2024_SUM_TA_P1_MINN-DAKS.con	Specified category P1 contingencies in Minnesota, Dakotas	x	x
MISO19_2024_SUM_TA_P1_P2_P4_P5_NoLoadLoss.con	Specified category P1, P2, P4, P5 contingencies in MISO cLASSIC	x	x
MISO19_2024_SUM_TA_P2_P4_P5_P7_LoadLoss.con	Specified category P2, P4, P5, P7 contingencies in MISO Classic	x	x
MISO19_2024_SUM_TA_P1_AMEREN.con	Specified category P1 contingencies in Ameren	x	x
2020_RTEP_Single_2017Aug-Updated-3-4-19.con	Specified single contingencies in PJM	x	x
2020_RTEP_Bus_AC2-updated-3-4-19.con	Specified bus contingencies in PJM	x	x
2020_RTEP_Line_FB_2017Aug-updated-3-4-19.con	Specified breaker failure contingencies in PJM	x	x
2020_RTEP_Tower_AC2-updated-3-4-19.con	Specified common structure contingencies in PJM	x	x
160303-KACY_P1.con	Specified category P1 contingencies in KACY	x	x
160303-KACY_P2.con	Specified category P2 contingencies in KACY	x	x
AECI-AMMO.CON	Specified contingencies between AECI and AMMO	x	x
KCPL_P1.con	Specified category P1 contingencies in KCPL	x	x
KCPL_P2.con	Specified category P2 contingencies in KCPL	x	x
KCPL_P4.con	Specified category P4 contingencies in KCPL	x	x

Contingency File Name	Description	Shoulder	Peak
KCPL_P5.con	Specified category P5 contingencies in KCPL	x	x
KCPL_P7.con	Specified category P7 contingencies in KCPL	x	x

Model Data

B.1 Power Flow Model Data

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B.2 2024 Slider Diagrams

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Appendix

C

Reactive Power Requirement Analysis Results (FERC Order 827)

Table C-1: Reactive Power Requirement Analysis Results											
Project #	HV Side Bus #	Lagging Power Factor Results				Leading Power Factor Results				Turbine Inherent Power Factor	Shunt Compensation
		MW from plant to HV side (P)	MVAR from plant to HV side (Q)	Lagging Power Factor at HV Side	Meet Lagging Power Factor Req.?	MW from plant to HV side (P)	MVAR from plant to HV side (Q)	Leading Power Factor at HV Side	Meet Leading Power Factor Req.?		
J951	89513	78.28	12.56	0.9874	No	78.21	-40.41	-0.888	Yes	± 0.95	None
J952	89523	53.22	18.46	0.9448	Yes	53.13	-36.36	-0.825	Yes	+0.913 / -0.93	1×7.2 MVAR cap bank on 34.5 kV system
J958	89583	98.19	28.95	0.9592	No	98.12	-49.67	-0.892	Yes	± 0.95	2×6 MVAR cap bank on 34.5 kV system
J959	89593	146.81	40.58	0.9639	No	146.65	-75.89	-0.888	Yes	± 0.95	2×9 MVAR cap bank on 34.5 kV system
J966	89663	389.39	80.93	0.9791	No	388.55	-218.19	-0.872	Yes	± 0.95	2×14 MVAR cap bank on 34.5 kV system
J967	89673	146.83	50.76	0.9451	Yes	146.60	-77.38	-0.884	Yes	± 0.95	2×14 MVAR cap bank on 34.5 kV system
J971	89713	245.23	74.49	0.9568	No	244.71	-153.13	-0.848	Yes	+0.913/-0.93 for Vestas V136 ± 0.95 for Vestas V110	1×16 MVAR cap bank on 34.5 kV system
J975	89753	147.50	49.66	0.9477	Yes	147.05	-108.29	-0.805	Yes	± 0.90	1×6 MVAR cap bank on 34.5 kV system
J977	89773	291.53	96.02	0.9498	Yes	290.82	-159.99	-0.876	Yes	± 0.95	1×26 MVAR cap bank on each of the two 34.5 kV system
J981	89813	189.82	53.36	0.9627	No	189.06	-105.02	-0.874	Yes	± 0.95	1×13.5 MVAR cap bank on each of the two 34.5 kV system
J982	89823	283.89	94.58	0.9487	Yes	281.87	-161.69	-0.867	Yes	± 0.95	1×25.5 MVAR cap bank on each of the two 34.5 kV system
J997	89973	197.54	53.83	0.9648	No	197.43	-106.12	-0.881	Yes	± 0.95	2×14 MVAR cap bank on 34.5 kV system
J998	89983	49.38	14.48	0.9596	No	49.29	-26.66	-0.880	Yes	± 0.95	1×7 MVAR cap bank on 34.5 kV system
J999	89993	98.74	29.38	0.9585	No	98.65	-51.54	-0.886	Yes	± 0.95	2×7 MVAR cap bank on 34.5 kV system
J1001	40013	39.63	15.86	0.9284	Yes	39.59	-16.91	-0.920	Yes	± 0.95	1×6 MVAR cap bank on 34.5 kV system
J1006	40063	98.74	29.38	0.9585	No	98.65	-51.54	-0.886	Yes	± 0.95	2×7 MVAR cap bank on 34.5 kV system
J1008	40083	49.42	17.21	0.9444	Yes	49.35	-23.27	-0.904	Yes	± 0.95	1×7 MVAR cap bank on 34.5 kV system
J1024	40243	193.46	59.45	0.9559	No	192.84	-107.61	-0.873	Yes	± 0.95	1×30 MVAR cap bank on 34.5 kV system
J1032	40613	49.52	20.02	0.9271	Yes	49.43	-44.40	-0.744	Yes	± 0.857	None
J1038	40383	31.93	18.52	0.8650	Yes	31.84	-30.57	-0.721	Yes	± 0.857	1×5 MVAR cap bank on 34.5 kV system
J1040	40403	244.54	86.93	0.9422	Yes	243.53	-186.97	-0.793	Yes	± 0.90	1×17 MVAR cap bank on 34.5 kV system
J1041	40573	19.86	13.53	0.8265	Yes	19.83	-17.52	-0.749	Yes	± 0.794	None
J1045	88742	19.86	13.61	0.8248	Yes	19.85	-17.17	-0.756	Yes	± 0.794	None
J1050	40503	220.03	129.56	0.8617	Yes	219.41	-158.28	-0.811	Yes	± 0.90	2×15 MVAR cap bank on each of the two 34.5 kV system
J1054	40543	29.54	10.71	0.9401	Yes	29.48	-20.24	-0.824	Yes	± 0.90	None
J1057	40573	123.14	42.31	0.9457	Yes	122.85	-86.40	-0.818	Yes	± 0.90	1×4 MVAR cap bank on 34.5 kV system
J1061	40613	292.60	89.66	0.9561	No	291.29	-225.58	-0.791	Yes	± 0.90	4×2 MVAR cap bank on 34.5 kV system
J1072	40723	148.12	46.10	0.9548	No	147.95	-78.84	-0.883	Yes	± 0.95	2×12 MVAR cap bank on 34.5 kV system
J1084	40842	148.14	46.56	0.9540	No	147.98	-78.13	-0.884	Yes	± 0.95	2×12 MVAR cap bank on 34.5 kV system
J1086	40863	117.89	16.93	0.9899	No	117.64	-64.64	-0.876	Yes	± 0.95	None
J1092	40923	99.03	26.74	0.9654	No	98.87	-74.33	-0.799	Yes	± 0.90	None
J1098	601068	39.35	16.52	0.9220	Yes	39.27	-23.04	-0.863	Yes	± 0.90	None
J1105	40923	197.52	59.72	0.9572	No	197.15	-107.40	-0.878	Yes	± 0.95	2×14 MVAR cap bank on 34.5 kV system
J1106	41063	402.78	119.53	0.9587	No	401.25	-274.02	-0.826	Yes	+0.90 / -0.92	None
J1108	41083	147.06	52.62	0.9415	Yes	146.75	-95.37	-0.838	Yes	± 0.926	1×20 MVAR cap bank on 34.5 kV system
J1109	41093	201.93	59.99	0.9586	No	201.39	-134.08	-0.832	Yes	+0.90 / -0.92	None
J1110	41103	98.50	31.86	0.9515	No	98.26	-68.47	-0.820	Yes	± 0.90	None
J1114	41143	192.81	77.09	0.9285	Yes	189.85	-127.64	-0.830	Yes	± 0.95	1×27 MVAR cap bank on each of the two 34.5 kV system
J1122	41223	194.96	90.01	0.9079	Yes	194.46	-103.99	-0.882	Yes	± 0.95	1×60 MVAR cap bank on 34.5 kV system
J1124	41243	98.51	32.10	0.9508	No	98.20	-70.69	-0.812	Yes	± 0.90	None
J1128	41283	147.48	20.59	0.9904	No	147.16	-82.07	-0.873	Yes	± 0.95	None
J1131	41313	97.99	37.36	0.9344	Yes	97.45	-88.43	-0.741	Yes	± 0.86	None
J1132	41323	48.99	20.15	0.9248	Yes	48.77	-41.58	-0.761	Yes	± 0.86	None
J1135	41353	48.99	20.15	0.9248	Yes	48.77	-41.58	-0.761	Yes	± 0.86	None
J1140	41403	98.98	45.89	0.9072	Yes	98.95	-54.32	-0.877	Yes	± 0.95	4×8 MVAR cap bank on 34.5 kV system
J1143	41433	99.17	52.78	0.8828	Yes	98.78	-43.10	-0.917	Yes	± 0.95	4×8 MVAR cap bank on 34.5 kV system
J1149	41493	197.60	79.67	0.9275	Yes	197.33	-130.07	-0.835	Yes	± 0.92	1×36 MVAR cap bank on 34.5 kV system
J1164	41643	198.20	105.13	0.8834	Yes	198.00	-97.00	-0.898	Yes	± 0.95	Two 4×8 MVAR cap bank on 34.5 kV system
J1169	41693	49.42	26.70	0.8798	Yes	49.42	-24.23	-0.898	Yes	± 0.95	4×4 MVAR cap bank on 34.5 kV system
J1170	41703	195.66	93.33	0.9026	Yes	195.09	-103.04	-0.884	Yes	± 0.95	Two 4×8 MVAR cap bank on 34.5 kV system
J1174	41743	292.77	93.34	0.9528	No	290.82	-234.98	-0.778	Yes	± 0.90	2×4 MVAR cap bank on 34.5 kV system
J1175	41753	292.75	97.48	0.9488	Yes	291.96	-222.22	-0.796	Yes	± 0.90	2×9 MVAR cap bank on 34.5 kV system
J1177	41773	292.10	16.76	0.9984	No	290.91	-105.84	-0.940	Yes	+0.987 / -0.99	2×8 MVAR cap bank on 34.5 kV system
J1179	41793	292.52	88.94	0.9568	No	290.93	-235.98	-0.777	Yes	± 0.90	2×4 MVAR cap bank on 34.5 kV system
J1181	41813	196.30	69.84	0.9422	Yes	195.64	-144.37	-0.805	Yes	± 0.90	2×6 MVAR cap bank on 34.5 kV system
J1182	41823	246.81	78.82	0.9526	No	246.29	-189.83	-0.792	Yes	± 0.89	None
J1187	41873	148.71	17.11	0.9934	No	148.08	-90.16	-0.854	Yes	± 0.95	None

2024 Summer Peak Contingency Analysis Results

D.1 Constraints in 2024 Summer Peak (SPK) Discharging Condition

Table D-1: 2024 SPK Discharging System Intact Thermal Constraints

Table D-2: 2024 SPK Discharging System Intact Voltage Constraints

Table D-3: 2024 SPK Discharging Category P1 Thermal Constraints

Table D-4: 2024 SPK Discharging Category P1 Voltage Constraints

Table D-5: 2024 SPK Discharging Category P2-P7 Thermal Constraints

Table D-6: 2024 SPK Discharging Category P2-P7 Voltage Constraints

Table D-7: 2024 SPK Discharging Non-Converged Contingencies

Table D-8: 2024 SPK Discharging Non-Converged Contingencies DCCC Results

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D.2 Constraints in 2024 Summer Peak (SPK) Charging Condition

Table D-9: 2024 SPK Charging System Intact Thermal Constraints

Table D-10: 2024 SPK Charging System Intact Voltage Constraints

Table D-11: 2024 SPK Charging Category P1 Thermal Constraints

Table D-12: 2024 SPK Charging Category P1 Voltage Constraints

Table D-13: 2024 SPK Charging Category P2-P7 Thermal Constraints

Table D-14: 2024 SPK Charging Category P2-P7 Voltage Constraints

Table D-15: 2024 SPK Charging Non-Converged Contingencies

Table D-16: 2024 SPK Charging Non-Converged Contingencies DCCC Results

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2024 Summer Shoulder Contingency Analysis Results

E.1 Stage-1 Contingency Analysis Results

Table E-1: Voltage Violations in 2024 SH Discharging Scenario

Table E-2: Thermal Violations in 2024 SH Discharging Scenario

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E.2 Base Case Network Upgrades Justification Results

Table E-3: Voltage Justification Results in 2024 SH Discharging Scenario

Table E-4: Thermal Justification Results in 2024 SH Discharging Scenario

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E.3 Stage-2 Contingency Analysis with Base Case NUs

E.3.1 Constraints in Stage-2 Summer Shoulder (SH) Discharging Condition

Table E-5: Stage-2 SH Discharging System Intact Thermal Constraints

Table E-6: Stage-2 SH Discharging System Intact Voltage Constraints

Table E-7: Stage-2 SH Discharging Category P1 Thermal Constraints

Table E-8: Stage-2 SH Discharging Category P1 Voltage Constraints

Table E-9: Stage-2 SH Discharging Category P2-P7 Thermal Constraints

Table E-10: Stage-2 SH Discharging Category P2-P7 Voltage Constraints

Table E-11: Stage-2 SH Discharging Non-Converged Contingencies

Table E-12: Stage-2 SH Discharging Non-Converged Contingencies DCCC Results

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E.3.2 Constraints in Stage-2 Summer Shoulder (SH) Charging Condition

Table E-13: Stage-2 SH Charging System Intact Thermal Constraints

Table E-14: Stage-2 SH Charging System Intact Voltage Constraints

Table E-15: Stage-2 SH Charging Category P1 Thermal Constraints

Table E-16: Stage-2 SH Charging Category P1 Voltage Constraints

Table E-17: Stage-2 SH Charging Category P2-P7 Thermal Constraints

Table E-18: Stage-2 SH Charging Category P2-P7 Voltage Constraints

Table E-19: Stage-2 SH Charging Non-Converged Contingencies

Table E-20: Stage-2 SH Charging Non-Converged Contingencies DCCC Results

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Local Planning Criteria Analysis Results

F.1 GRE Local Planning Criteria Analysis Results

F.1.1 J1106 GRE Local Planning Criteria Analysis

Below is the J1106 GRE local planning criteria analysis report.

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F.1.2 J1140 GRE Local Planning Criteria Analysis

Below is the J1140 GRE local planning criteria analysis report.

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F.1.3 CCS (J1187) GRE Local Planning Criteria Analysis

Below is the CCS GRE local planning criteria analysis report.

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F.2 OTP LPC Analysis

Below is the OTP local planning criteria analysis report.

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F.3 MDU LPC Analysis

Below is the MDU local planning criteria analysis report.

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F.4 MRES LPC Analysis

Below is the MRES local planning criteria analysis report.

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MWEX Voltage Study Details

Below is the MWEX voltage stability study report provided by ATC.

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2024 Cost Allocation Results

H.1 Distribution Factor (DF) and MW Contribution Results for Cost Allocation in 2024

Table H-1: Distribution Factor and MW Contribution on Constraints for Hazel Creek-Scott County 345 kV Base Case NU Cost Allocation

Table H-2: Distribution Factor and MW Contribution on Constraints for Other Thermal NU Cost Allocation

Table H-3: Distribution Factor and MW Contribution on MISO Voltage Constraints for MISO Reactive Power NU Cost Allocation

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H.2 Cost Allocation Details

Table H-4: Network Upgrades Cost Allocation in 2024

Project Summary		Key Dates		2020		2021		2022		2023		2024		2025		2026		2027		2028		2029		2030		2031		2032		2033		2034		2035		2036		2037		2038		2039		2040		2041		2042		2043		2044		2045		2046		2047		2048		2049		2050		2051		2052		2053		2054		2055		2056		2057		2058		2059		2060		2061		2062		2063		2064		2065		2066		2067		2068		2069		2070		2071		2072		2073		2074		2075		2076		2077		2078		2079		2080		2081		2082		2083		2084		2085		2086		2087		2088		2089		2090		2091		2092		2093		2094		2095		2096		2097		2098		2099		2100		2101		2102		2103		2104		2105		2106		2107		2108		2109		2110		2111		2112		2113		2114		2115		2116		2117		2118		2119		2120		2121		2122		2123		2124		2125		2126		2127		2128		2129		2130		2131		2132		2133		2134		2135		2136		2137		2138		2139		2140		2141		2142		2143		2144		2145		2146		2147		2148		2149		2150		2151		2152		2153		2154		2155		2156		2157		2158		2159		2160		2161		2162		2163		2164		2165		2166		2167		2168		2169		2170		2171		2172		2173		2174		2175		2176		2177		2178		2179		2180		2181		2182		2183		2184		2185		2186		2187		2188		2189		2190		2191		2192		2193		2194		2195		2196		2197		2198		2199		2200		2201		2202		2203		2204		2205		2206		2207		2208		2209		2210		2211		2212		2213		2214		2215		2216		2217		2218		2219		2220		2221		2222		2223		2224		2225		2226		2227		2228		2229		2230		2231		2232		2233		2234		2235		2236		2237		2238		2239		2240		2241		2242		2243		2244		2245		2246		2247		2248		2249		2250		2251		2252		2253		2254		2255		2256		2257		2258		2259		2260		2261		2262		2263		2264		2265		2266		2267		2268		2269		2270		2271		2272		2273		2274		2275		2276		2277		2278		2279		2280		2281		2282		2283		2284		2285		2286		2287		2288		2289		2290		2291		2292		2293		2294		2295		2296		2297		2298		2299		2300		2301		2302		2303		2304		2305		2306		2307		2308		2309		2310		2311		2312		2313		2314		2315		2316		2317		2318		2319		2320		2321		2322		2323		2324		2325		2326		2327		2328		2329		2330		2331		2332		2333		2334		2335		2336		2337		2338		2339		2340		2341		2342		2343		2344		2345		2346		2347		2348		2349		2350		2351		2352		2353		2354		2355		2356		2357		2358		2359		2360		2361		2362		2363		2364		2365		2366		2367		2368		2369		2370		2371		2372		2373		2374		2375		2376		2377		2378		2379		2380		2381		2382		2383		2384		2385		2386		2387		2388		2389		2390		2391		2392		2393		2394		2395		2396		2397		2398		2399		2400		2401		2402		2403		2404		2405		2406		2407		2408		2409		2410		2411		2412		2413		2414		2415		2416		2417		2418		2419		2420		2421		2422		2423		2424		2425		2426		2427		2428		2429		2430		2431		2432		2433		2434		2435		2436		2437		2438		2439		2440		2441		2442		2443		2444		2445		2446		2447		2448		2449		2450		2451		2452		2453		2454		2455		2456		2457		2458		2459		2460		2461		2462		2463		2464		2465		2466		2467		2468		2469		2470		2471		2472		2473		2474		2475		2476		2477		2478		2479		2480		2481		2482		2483		2484		2485		2486		2487		2488		2489		2490		2491		2492		2493		2494		2495		2496		2497		2498		2499		2500		2501		2502		2503		2504		2505		2506		2507		2508		2509		2510		2511		2512		2513		2514		2515		2516		2517		2518		2519		2520		2521		2522		2523		2524		2525		2526		2527		2528		2529		2530		2531		2532		2533		2534		2535		2536		2537		2538		2539		2540		2541		2542		2543		2544		2545		2546		2547		2548		2549		2550		2551		2552		2553		2554		2555		2556		2557		2558		2559		2560		2561		2562		2563		2564		2565		2566		2567		2568		2569		2570		2571		2572		2573		2574		2575		2576		2577		2578		2579		2580		2581		2582		2583		2584		2585		2586		2587		2588		2589		2590		2591		2592		2593		2594		2595		2596		2597		2598		2599		2600		2601		2602		2603		2604		2605		2606		2607		2608		2609		2610		2611		2612		2613		2614		2615		2616		2617		2618		2619		2620		2621		2622		2623		2624		2625		2626		2627		2628		2629		2630		2631		2632		2633		2634		2635		2636		2637		2638		2639		2640		2641		2642		2643		2644		2645		2646		2647		2648		2649		2650		2651		2652		2653		2654		2655		2656		2657		2658		2659		2660		2661		2662		2663		2664		2665		2666		2667		2668		2669		2670		2671		2672		2673		2674		2675		2676		2677		2678		2679		2680		2681		2682		2683		2684		2685		2686		2687		2688		2689		2690		2691		2692		2693		2694		2695		2696		2697		2698		2699		2700		2701		2702		2703		2704		2705		2706		2707		2708		2709		2710		2711		2712		2713		2714		2715		2716		2717		2718		2719		2720		2721		2722		2723		2724		2725		2726		2727		2728		2729		2730		2731		2732		2733		2734		2735		2736		2737		2738		2739		2740		2741		2742		2743		2744		2745		2746		2747		2748		2749		2750		2751		2752		2753		2754		2755		2756		2757		2758		2759		2760		2761		2762		2763		2764		2765		2766		2767		2768		2769		2770		2771		2772		2773		2774		2775		2776		2777		2778		2779		2780		2781		2782		2783		2784		2785		2786		2787		2788		2789		2790		2791		2792		2793		2794		2795		2796		2797		2798		2799		2800		2801		2802		2803		2804		2805		2806		2807		2808		2809		2810		2811		2812		2813		2814		2815		2816		2817		2818		2819		2820		2821		2822		2823		2824		2825		2826		2827		2828		2829		2830		2831		2832		2833		2834		2835		2836		2837		2838		2839		2840		2841		2842		2843		2844		2845		2846		2847		2848		2849		2850		2851		2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Project Name	Project ID	Start Date	End Date	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035	2036	2037	2038	2039	2040	2041	2042	2043	2044	2045	2046	2047	2048	2049	2050	2051	2052	2053	2054	2055	2056	2057	2058	2059	2060	2061	2062	2063	2064	2065	2066	2067	2068	2069	2070	2071	2072	2073	2074	2075	2076	2077	2078	2079	2080	2081</																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																												

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